PMC

# Master Training Manual

## v3 · Sovereign Edition · 150-page complete

End-to-end operator handbook · every screen · every field · every workflow

PMC MANAGERS LIMITED · CONFIDENTIAL · V46M · SEPTEMBER 2026

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# 1 · Platform overview & architecture

The PMC platform is a multi-role tender lifecycle engine covering the full path from notice publication → registration → bidding → EMD → evaluation → award → e-signature → invoicing → post-contract audit. Delivered as a React SPA backed by FastAPI + MongoDB with same-origin cookie sessions and CSRF double-submit tokens.

**Frontend** — React SPA served from S3 + CloudFront. Sovereign navy/gold editorial design system with Libre Baskerville + IBM Plex fonts. Routing via React Router; state via Context (i18n, auth) + local component state.

**Backend** — FastAPI on AWS Elastic Beanstalk (Docker running on 64-bit Amazon Linux 2). All routes prefixed `/api/v1/…`. Auth realm split into admin, staff and vendor with per-realm login endpoints and cookie names.

**Data** — MongoDB via Motor (async driver). Collections include tenants, notices, tender_templates, registrations, vendors, bids, rubrics, scores, evaluations, awards, escrow_transactions, auctions, corrigenda, audits, ratings, engagements, packages, categories, countries, dpr_jobs.

**AI** — Managed AI Gateway (universal LLM key) routed through the integration proxy. Used for DPR ingestion (map-reduce), AI Fill All, anomaly detection and the admin copilot. v46l lifts the 800 KB cap and adds an async worker.

**Storage** — S3 for licences, tender attachments, signed PDFs, archived auction rooms. CloudFront in front for global CDN.

**Notifications** — Resend (email), planned Firebase/APNs (mobile push).

**Payments** — Stripe (test + live keys separated); Kaspi + PayPal on the roadmap.

**E-signature** — DocuSign envelope templates + status webhooks.

**Observability** — Structured logs via Python logging, CloudWatch tail, audit_log() collection for every mutation.

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# 2 · Roles at a glance

| Role | Portal | Scope |
|---|---|---|
| Admin | `/admin` | Full control of every module. |
| Management | `/staff` | Executive dashboards, committee outcomes, escrow financials. |
| Partner | `/staff` | GOC / Bank / Regulator — audit hooks and co-procurement dashboards. |
| Agent | `/staff` | Field-ops — draft tenders, review dossiers, publish corrigenda. |
| Reviewer | `/admin` | Read-only + evaluation scoring. |
| Escrow agent | `/staff` | EMD desk, forfeitures, refunds. |
| Certifier | `/staff` | Tender packet sign-off. |
| Legal counsel | `/staff` | Contract review, DocuSign envelope owner. |
| Accounting | `/staff` | Fee invoicing, refunds, reconciliation. |
| Committee member | `/staff` | Evaluation committee voting. |
| Auditor | `/admin (read)` | Full audit trail with entity filters. |
| Support | `/staff` | Vendor helpdesk, activation resends. |
| Vendor | `/vendor` | Bid on notices, manage profile & invoices. |

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# 3 · Security, audit & data-protection

**Authentication** — Three cookie realms (admin_session, staff_session, vendor_session). HttpOnly, SameSite=Lax, Secure in production. CSRF double-submit on every mutating request. Brute-force lockout after 5 failed attempts within 15 minutes.

**Authorisation** — `require_role(*roles)` guards every mutating endpoint. Read endpoints use realm-specific `current_admin` / `current_staff` / `current_vendor` dependencies.

**Audit trail** — Every mutation calls `audit_log()` which writes an immutable row (actor id, timestamp, entity, action, JSON payload) to the audits collection. Auditors filter by entity + action + actor + date and export CSV.

**Data protection** — Licences and legal docs sit in a private S3 bucket behind pre-signed URLs. PII is redacted from Auditor-role CSV exports.

**Compliance** — GDPR consent captured at registration; ESG questionnaire lives under the vendor profile.

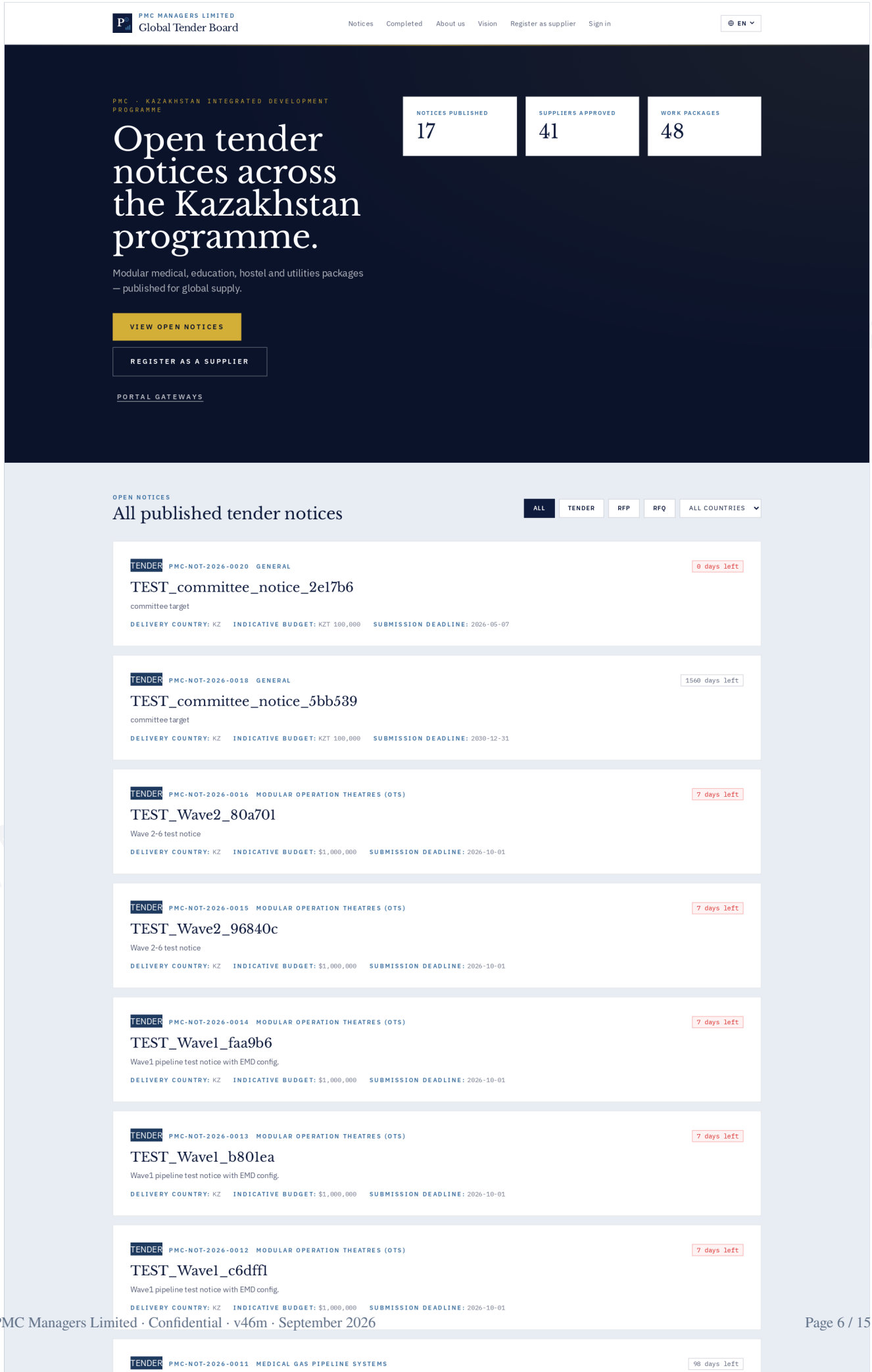
DENTIAL — INTERNAL USE

# 4 · Public visitor journey

## 1 · Sovereign home — hero + live KPIs

*The public landing page. Editorial navy/gold hero with a Libre Baskerville headline, three live KPI cards (Notices Published, Suppliers Approved, Work Packages), and three primary CTAs — View Open Notices, Register as Supplier, and the Portal Gateways link.*

DENTIAL — INTERNAL USE


PMC MANAGERS LIMITED
Global Tender Board
Notices
Completed
About us
Vision
Register as supplier
Sign in
EN
PMC · KAZAKHSTAN INTEGRATED DEVELOPMENT PROGRAMME
Open tender notices across the Kazakhstan programme.
Modular medical, education, hostel and utilities packages — published for global supply.
VIEW OPEN NOTICES
REGISTER AS A SUPPLIER
PORTAL GATEWAYS
NOTICES PUBLISHED
17
SUPPLIERS APPROVED
41
WORK PACKAGES
48
OPEN NOTICES
All published tender notices
ALL
TENDER
RFP
RFQ
ALL COUNTRIES
TENDER PMC-NOT-2026-0020 GENERAL
0 days left
TEST_committee_notice_2e17b6
committee target
DELIVERY COUNTRY: KZ INDICATIVE BUDGET: KZT 100,000 SUBMISSION DEADLINE: 2026-05-07
TENDER PMC-NOT-2026-0018 GENERAL
1560 days left
TEST_committee_notice_5bb539
committee target
DELIVERY COUNTRY: KZ INDICATIVE BUDGET: KZT 100,000 SUBMISSION DEADLINE: 2030-12-31
TENDER PMC-NOT-2026-0016 MODULAR OPERATION THEATRES (OTS)
7 days left
TEST_Wave2_80a701
Wave 2-6 test notice
DELIVERY COUNTRY: KZ INDICATIVE BUDGET: $1,000,000 SUBMISSION DEADLINE: 2026-10-01
TENDER PMC-NOT-2026-0015 MODULAR OPERATION THEATRES (OTS)
7 days left
TEST_Wave2_96840c
Wave 2-6 test notice
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7 days left
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Wave1 pipeline test notice with EMD config.
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Wave1 pipeline test notice with EMD config.
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Wave1 pipeline test notice with EMD config.
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TENDER PMC-NOT-2026-0011 MEDICAL GAS PIPELINE SYSTEMS
98 days left

*Sovereign home — hero + live KPIs*

### Purpose

The sovereign public entry. Establishes brand authority, states current scope (Kazakhstan Integrated Development), and exposes three live counters and three primary CTAs.

### How the workflow unfolds

1) A prospective bidder lands from a Google search or a Snapchat post. 2) The live KPIs prove the platform has real, current activity. 3) They tap 'View Open Notices' to browse tenders or 'Portal Gateways' to sign in to their existing account.

### Fields

- **Kicker line** — 'PMC Managers · Sovereign Gateway' — anchors the brand in an editorial serif register.
- **Sovereign hero** — Libre Baskerville headline, italic gold accent, IBM Plex Mono kicker — deliberately different from generic SaaS.
- **KPI cards** — Notices Published / Suppliers Approved / Work Packages — pulled live from /api/v1/meta/public/stats.
- **Primary CTAs** — View Open Notices (gold button), Register as Supplier (ghost button), Portal Gateways (underline link).

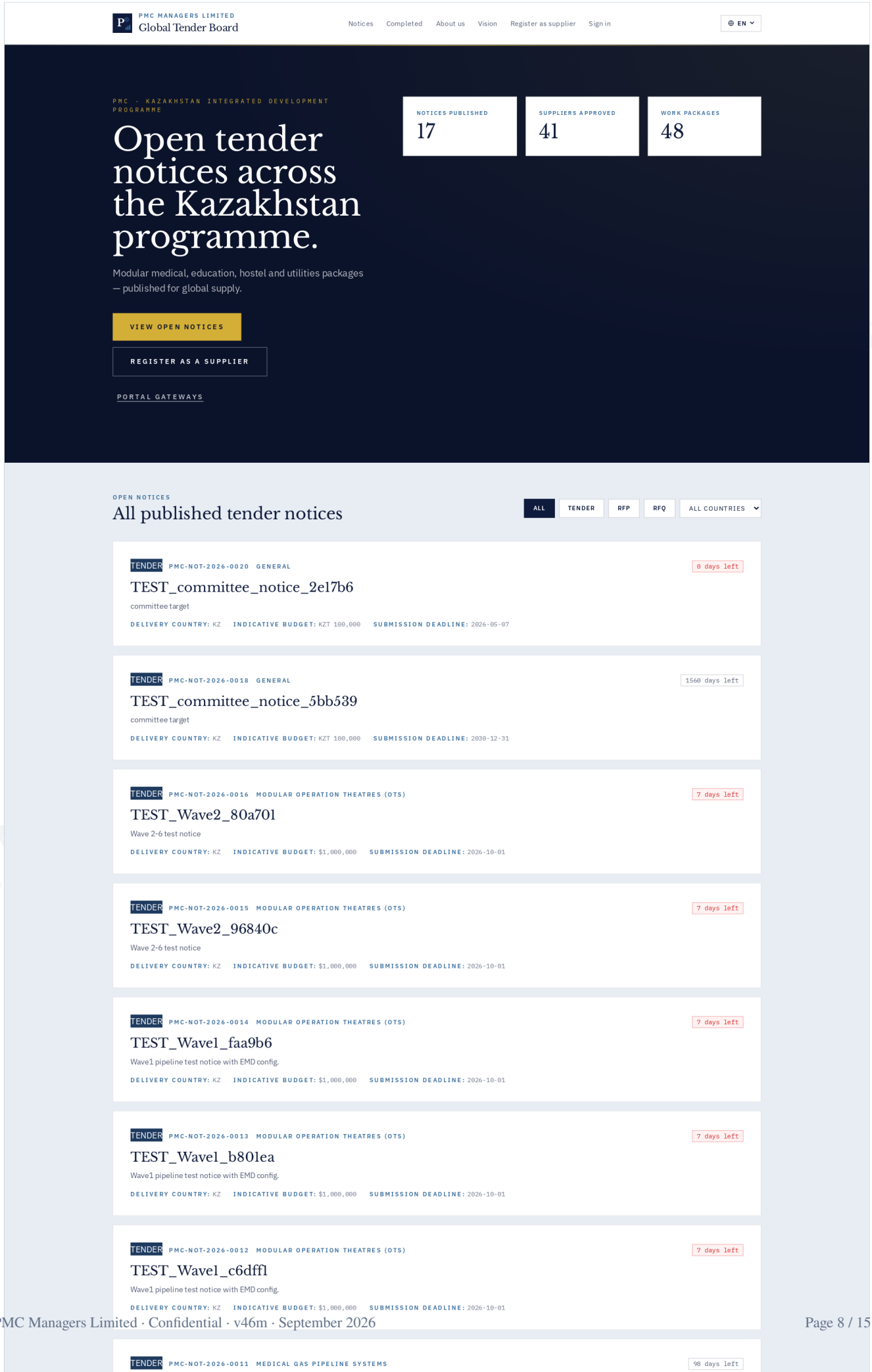
### Known gotchas

- Counters cache in CloudFront for 60 s — a fresh notice may take up to a minute to appear.
- The hero background is a radial gold gradient at 8% opacity — do not brighten it or the serif headline loses contrast.

## 2 · Public notice board

*The reverse-chronological list of every published notice. Fuzzy search, country/category filters, deadline chips and per-row EMD hints. This is the primary discovery surface for prospective bidders.*

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PMC MANAGERS LIMITED
Global Tender Board
Notices
Completed
About us
Vision
Register as supplier
Sign in
EN
PMC · KAZAKHSTAN INTEGRATED DEVELOPMENT PROGRAMME
Open tender notices across the Kazakhstan programme.
Modular medical, education, hostel and utilities packages — published for global supply.
VIEW OPEN NOTICES
REGISTER AS A SUPPLIER
PORTAL GATEWAYS
NOTICES PUBLISHED
17
SUPPLIERS APPROVED
41
WORK PACKAGES
48
OPEN NOTICES
All published tender notices
ALL
TENDER
RFP
RFQ
ALL COUNTRIES
TENDER PMC-NOT-2026-0020 GENERAL
0 days left
TEST_committee_notice_2e17b6
committee target
DELIVERY COUNTRY: KZ INDICATIVE BUDGET: KZT 100,000 SUBMISSION DEADLINE: 2026-05-07
TENDER PMC-NOT-2026-0018 GENERAL
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Wave 2-6 test notice
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TENDER PMC-NOT-2026-0011 MEDICAL GAS PIPELINE SYSTEMS
98 days left

*Public notice board*

### Purpose

The reverse-chronological list of every published notice. Primary discovery surface for bidders and researchers.

### How the workflow unfolds

1) Enter a keyword or filter by country / category. 2) Review the card headline, deadline and EMD hint. 3) Click a row to open the full public detail page.

### Fields

- **Search box** — Fuzzy match on title, description, country and category — server side.
- **Country / Category filters** — Backed by the same masters used in the admin console.
- **Deadline chip** — Turns red at T-24 h to create urgency.
- **EMD hint** — Shows the EMD amount or percentage inline so bidders can pre-arrange the deposit.

### Known gotchas

- Deadlines display in the visitor's local timezone; the backend stores UTC ISO strings.

## 3 · Live tenders listing

*All tenders currently open for bidding. Same card structure as the notice board with a countdown chip on each card driving urgency.*

DENTIAL — INTERNAL USE

PMC MANAGERS LIMITED
Global Tender Board

Notices · Completed · About us · Vision · Register as supplier · Sign in · EN

PUBLIC NOTICE BOARD

# Live tenders

Open for bidding right now — click any card for full documents

LIVE | COMPLETED | ALL NOTICES

Search title, reference, category...

COUNTRY (KZ, IN, GB...)

PMC-NOT-2026-0018 · TENDER · ● LIVE · KZ

## TEST_committee_notice_5bb539

committee target

BUDGET: KZT 100,000 · DEADLINE: 2030-12-31 · CATEGORY: general

PMC-NOT-2026-0016 · TENDER · ● LIVE · KZ

## TEST_Wave2_80a701

Wave 2-6 test notice

BUDGET: USD 1,000,000 · DEADLINE: 2026-10-01 · CATEGORY: Modular operation theatres (OTs)

PMC-NOT-2026-0015 · TENDER · ● LIVE · KZ

## TEST_Wave2_96840c

Wave 2-6 test notice

BUDGET: USD 1,000,000 · DEADLINE: 2026-10-01 · CATEGORY: Modular operation theatres (OTs)

PMC-NOT-2026-0014 · TENDER · ● LIVE · KZ

## TEST_Wave1_faa9b6

Wave1 pipeline test notice with EMD config.

BUDGET: USD 1,000,000 · DEADLINE: 2026-10-01 · CATEGORY: Modular operation theatres (OTs)

PMC-NOT-2026-0013 · TENDER · ● LIVE · KZ

## TEST_Wave1_b801ea

Wave1 pipeline test notice with EMD config.

BUDGET: USD 1,000,000 · DEADLINE: 2026-10-01 · CATEGORY: Modular operation theatres (OTs)

PMC-NOT-2026-0012 · TENDER · ● LIVE · KZ

## TEST_Wave1_c6dff1

Wave1 pipeline test notice with EMD config.

BUDGET: USD 1,000,000 · DEADLINE: 2026-10-01 · CATEGORY: Modular operation theatres (OTs)

PMC-NOT-2026-0011 · TENDER · ● LIVE · KZ

## TEST_Notice_4ec2ff

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0010 · TENDER · ● LIVE · KZ

## TEST_Notice_6c5ebf

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0009 · TENDER · ● LIVE · KZ

## TEST_Notice_1f2591

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0008 · TENDER · ● LIVE · KZ

## TEST_Notice_4db920

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0006 · TENDER · ● LIVE · KZ

## TEST_Notice_d29602

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0005 · TENDER · ● LIVE · KZ

## TEST_Notice_aa0373

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0004 · TENDER · ● LIVE · KZ

## TEST_Notice_18efc5

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC Managers
SOVEREIGN TENDER BOARD

The Big-4 procurement authority for cross-border public works, operating since 2018 across 11 jurisdictions.

EXPLORE
Live tenders
Completed tenders
All notices
Register as supplier

COMPANY
About us
Vision & Mission
Sign in

FOLLOW

LEGAL
Terms of use
Privacy & GDPR
Cookie preferences

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*Live tenders listing*

### Purpose

Filter of the notice board scoped to tenders currently open for bidding.

### How the workflow unfolds

Bidders check this daily; the URL is optimised for RSS and Snapchat spotlight posts.

### Fields

- **Countdown chip** — Live H:MM:SS to submission deadline.
- **Awarding authority** — Small navy tag on each card.
- **EMD hint** — Percentage or absolute value inline.

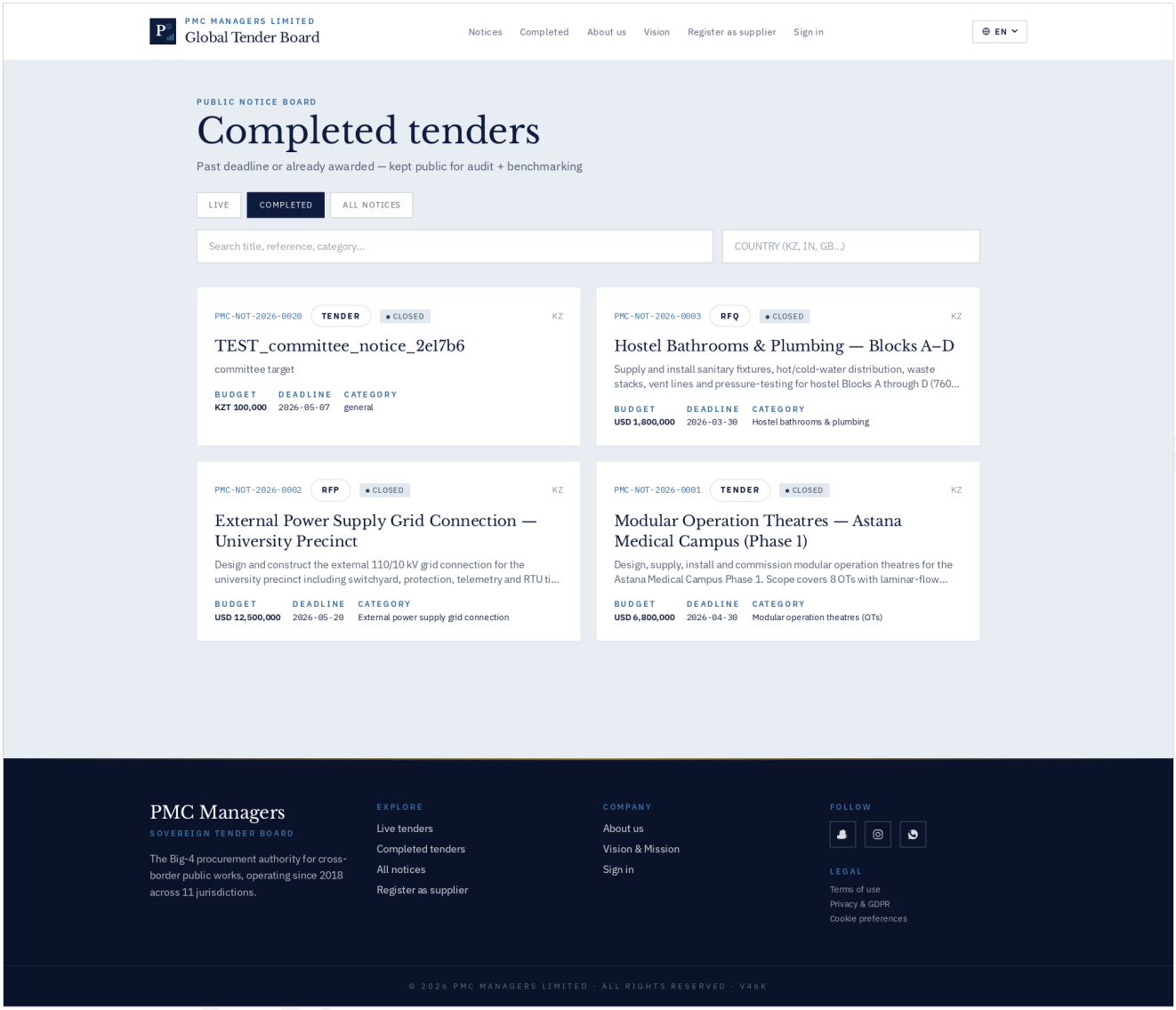
### Known gotchas

- If the counter reads 0 despite live data, force a CloudFront invalidation.

## 4 · Completed tenders

*Public archive of every tender that has passed its submission deadline. Awards, final prices and winning bidders are shown once the award is finalised and cleared for publication.*

DENTIAL — INTERNAL USE




*Completed tenders*

### Purpose

Public archive of every tender past its submission deadline.

### How the workflow unfolds

Journalists, auditors and inbound partners use this to audit PMC's throughput.

### Fields

- **Award column** — Winning bidder + final price once award is finalised and cleared for publication.
- **Signed award PDF** — Downloadable letter of award.

### Known gotchas

- A tender only appears here after the award is confirmed and legal has flipped the publish flag.

## 5 · All tenders — unified archive

*The combined live + completed archive with a status filter. Used by researchers, auditors, and inbound partners doing due diligence on PMC.*

PMC MANAGERS LIMITED
Global Tender Board

Notices · Completed · About us · Vision · Register as supplier · Sign in · EN

PUBLIC NOTICE BOARD

# All published notices

Everything that has ever been published

LIVE | COMPLETED | ALL NOTICES

Search title, reference, category... | COUNTRY (KZ, IN, GB...)

PMC-NOT-2026-0020 · TENDER · CLOSED · KZ

## TEST_committee_notice_2e17b6

committee target

BUDGET: KZT 100,000 · DEADLINE: 2026-05-07 · CATEGORY: general

PMC-NOT-2026-0018 · TENDER · LIVE · KZ

## TEST_committee_notice_5bb539

committee target

BUDGET: KZT 100,000 · DEADLINE: 2030-12-31 · CATEGORY: general

PMC-NOT-2026-0016 · TENDER · LIVE · KZ

## TEST_Wave2_80a701

Wave 2-6 test notice

BUDGET: USD 1,000,000 · DEADLINE: 2026-10-01 · CATEGORY: Modular operation theatres (OTs)

PMC-NOT-2026-0015 · TENDER · LIVE · KZ

## TEST_Wave2_96840c

Wave 2-6 test notice

BUDGET: USD 1,000,000 · DEADLINE: 2026-10-01 · CATEGORY: Modular operation theatres (OTs)

PMC-NOT-2026-0014 · TENDER · LIVE · KZ

## TEST_Wave1_faa9b6

Wave1 pipeline test notice with EMD config.

BUDGET: USD 1,000,000 · DEADLINE: 2026-10-01 · CATEGORY: Modular operation theatres (OTs)

PMC-NOT-2026-0013 · TENDER · LIVE · KZ

## TEST_Wave1_b801ea

Wave1 pipeline test notice with EMD config.

BUDGET: USD 1,000,000 · DEADLINE: 2026-10-01 · CATEGORY: Modular operation theatres (OTs)

PMC-NOT-2026-0012 · TENDER · LIVE · KZ

## TEST_Wave1_c6dff1

Wave1 pipeline test notice with EMD config.

BUDGET: USD 1,000,000 · DEADLINE: 2026-10-01 · CATEGORY: Modular operation theatres (OTs)

PMC-NOT-2026-0011 · TENDER · LIVE · KZ

## TEST_Notice_4ec2ff

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0010 · TENDER · LIVE · KZ

## TEST_Notice_6c5ebf

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0009 · TENDER · LIVE · KZ

## TEST_Notice_1f2591

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0008 · TENDER · LIVE · KZ

## TEST_Notice_4db920

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0006 · TENDER · LIVE · KZ

## TEST_Notice_d29602

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0005 · TENDER · LIVE · KZ

## TEST_Notice_aa0373

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0004 · TENDER · LIVE · KZ

## TEST_Notice_18efc5

TEST description for auto-license attach behaviour on KZ delivery.

DEADLINE: 2026-12-31 · CATEGORY: Medical gas pipeline systems

PMC-NOT-2026-0003 · RFQ · CLOSED · KZ

## Hostel Bathrooms & Plumbing — Blocks A–D

Supply and install sanitary fixtures, hot/cold-water distribution, waste stacks, vent lines and pressure-testing for hostel Blocks A through D (760...

BUDGET: USD 1,800,000 · DEADLINE: 2026-03-30 · CATEGORY: Hostel bathrooms & plumbing

PMC-NOT-2026-0002 · RFP · CLOSED · KZ

## External Power Supply Grid Connection — University Precinct

Design and construct the external 110/10 kV grid connection for the university precinct including switchyard, protection, telemetry and RTU ti...

BUDGET: USD 12,500,000 · DEADLINE: 2026-05-20 · CATEGORY: External power supply grid connection

PMC-NOT-2026-0001 · TENDER · CLOSED · KZ

## Modular Operation Theatres — Astana Medical Campus (Phase 1)

Design, supply, install and commission modular operation theatres for the Astana Medical Campus Phase 1. Scope covers 8 OTs with laminar-flow...

BUDGET: USD 6,800,000 · DEADLINE: 2026-04-30 · CATEGORY: Modular operation theatres (OTs)

*All tenders — unified archive*

### Purpose

The combined live+completed archive with a status filter.

### How the workflow unfolds

Used by regulators, auditors and inbound partners doing due diligence on PMC.

### Fields

- **Status filter** — Live / Completed / Cancelled / Withdrawn.
- **Column sort** — Deadline, budget, publish date, awarding authority.

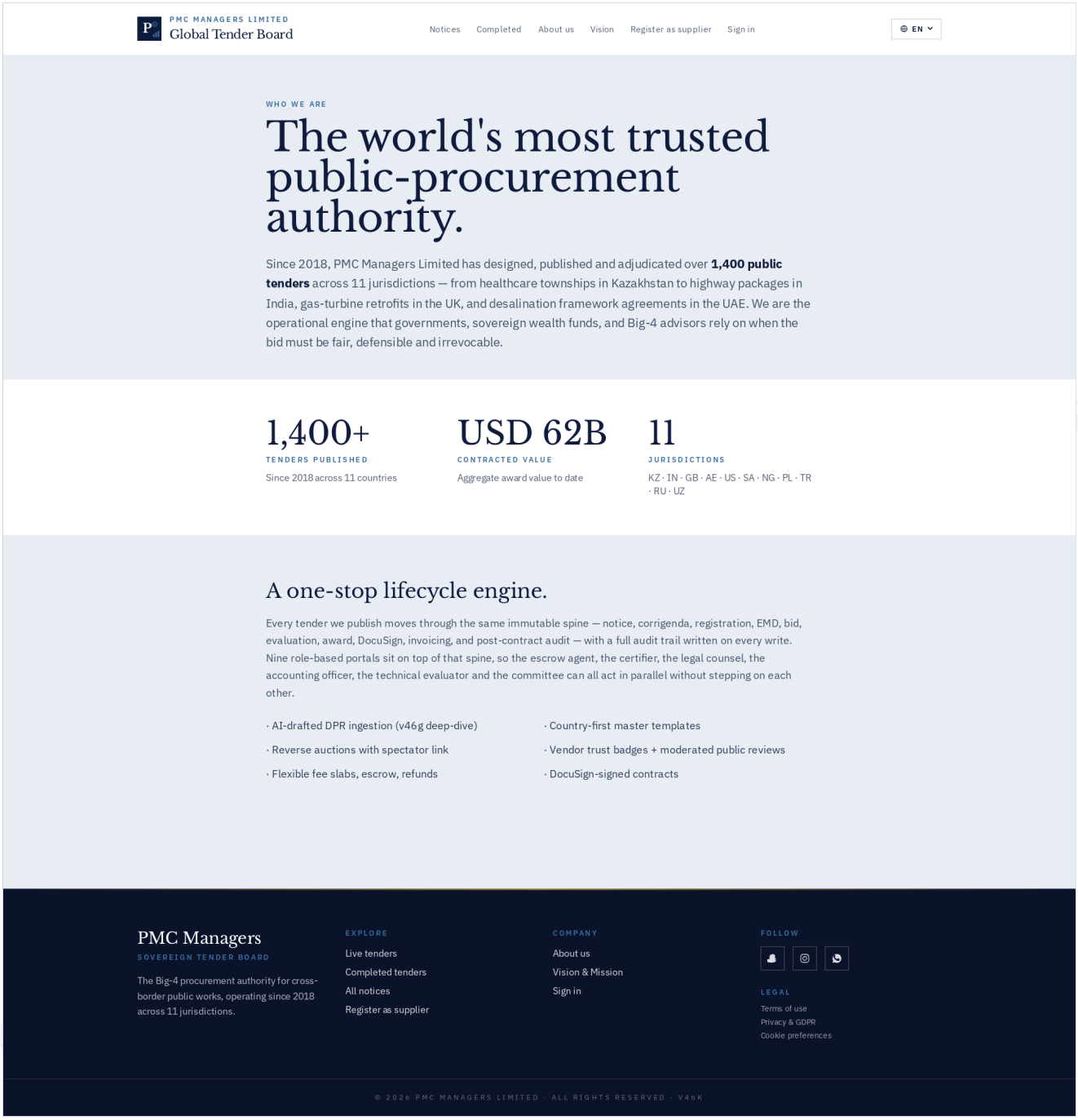
### Known gotchas

- Cancelled tenders keep their audit trail — never delete a notice, always withdraw it.

## 6 · About Us — company narrative

*Editorial About page. Charter, milestones, leadership grid, sector coverage, and press quotes. All copy is content-managed via the notice-board CMS.*

DENTIAL — INTERNAL USE




*About Us — company narrative*

### Purpose

Editorial About Us — company narrative, milestones, leadership grid, sector coverage, press quotes.

### How the workflow unfolds

Marketing and BD share this URL with prospective GOC/Bank partners; it also drives Snapchat / Instagram profile clicks.

### Fields

- **Charter** — One-paragraph mission statement — content-managed via CMS.
- **Milestones** — Chronological chips — feed this monthly during releases.

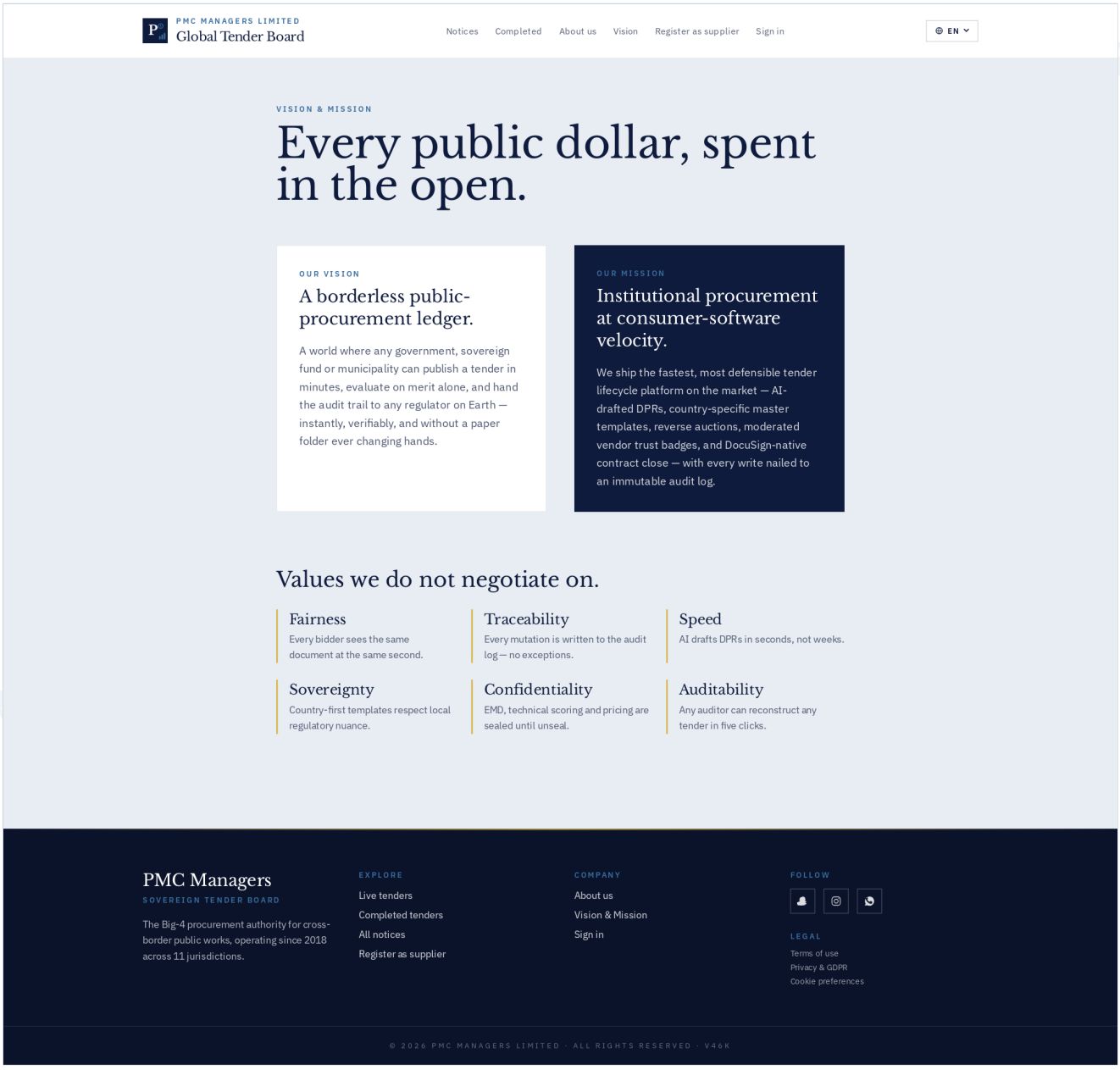
- **Leadership grid** — Portrait cards with role and short bio.
- **Press quotes** — Curated third-party validators.

### Known gotchas

- Portraits should be square 512×512 PNGs sized ≤ 120 KB — larger images push the fold below the CTA.

## 7 · Vision & Mission

*The strategic doctrine page — Vision, Mission, Values, and the North-Star KPIs the platform is optimised for. Used in tender-committee briefings.*




*Vision & Mission*

### Purpose

The strategic doctrine page — Vision, Mission, Values, North-Star KPIs the platform is optimised for.

### How the workflow unfolds

Referenced by the tender committee in every quarterly briefing. Also cited in ESG questionnaires.

### Fields

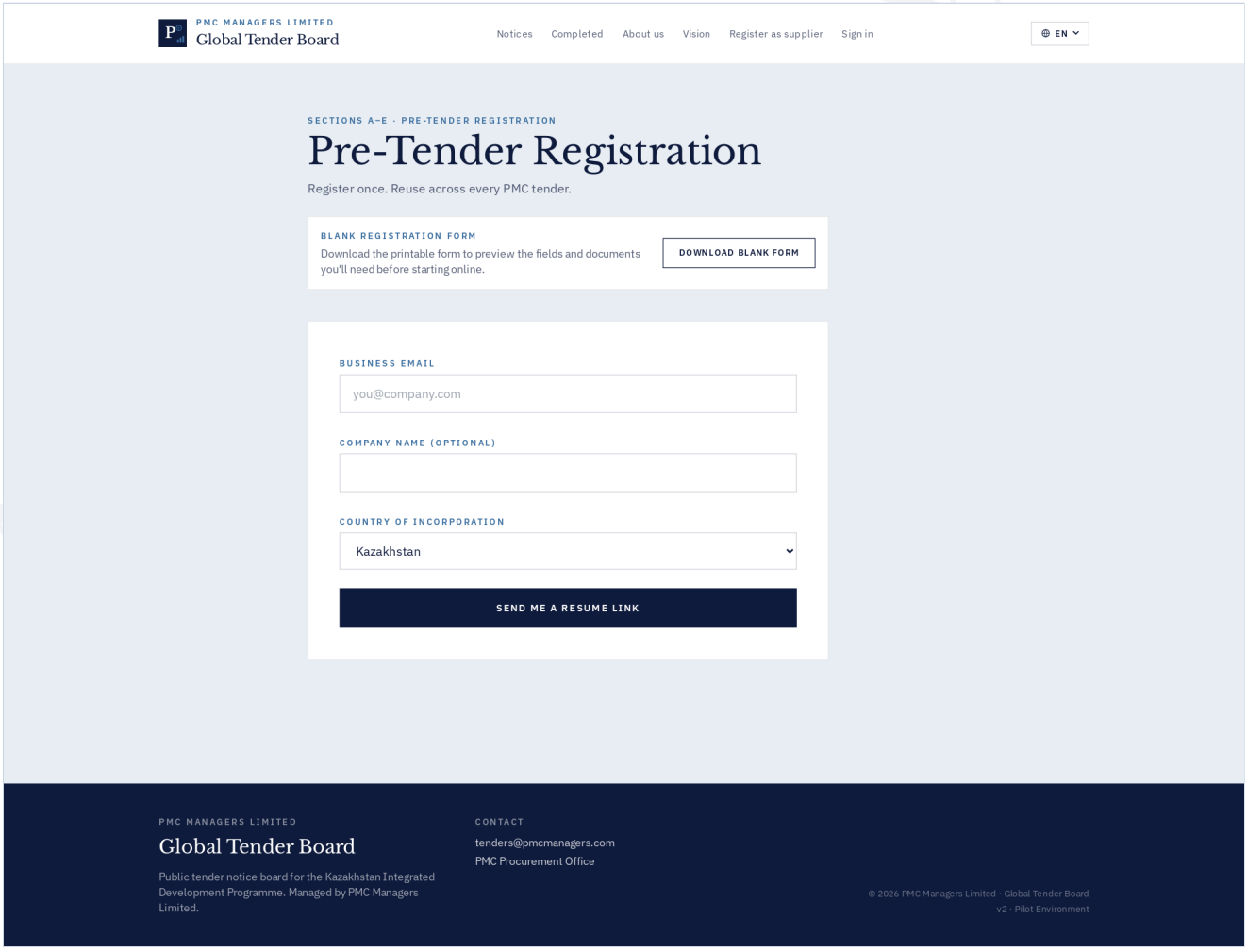
- **Vision statement** — 10-year direction.
- **Mission** — 12-month operational focus.
- **Values** — Three principles used to arbitrate roadmap trade-offs.
- **North-star KPIs** — Cycle time, dispute rate, EMD-refund SLA, vendor NPS.

### Known gotchas

- Whenever the North-Star KPIs change, re-open every tender committee meeting with the delta explained.

## 8 · Supplier registration — start

*Public entry point for vendor self-onboarding. Email, company name, country. A magic link is dispatched via Resend and the bidder resumes the deep intake wizard from that link.*




*Supplier registration — start*

### Purpose

Public entry point for vendor self-onboarding.

### How the workflow unfolds

1) Bidder submits email + company + country. 2) A Resend magic link is emailed. 3) Bidder clicks the link and lands on RegistrationResume where the 5-step wizard begins.

### Fields

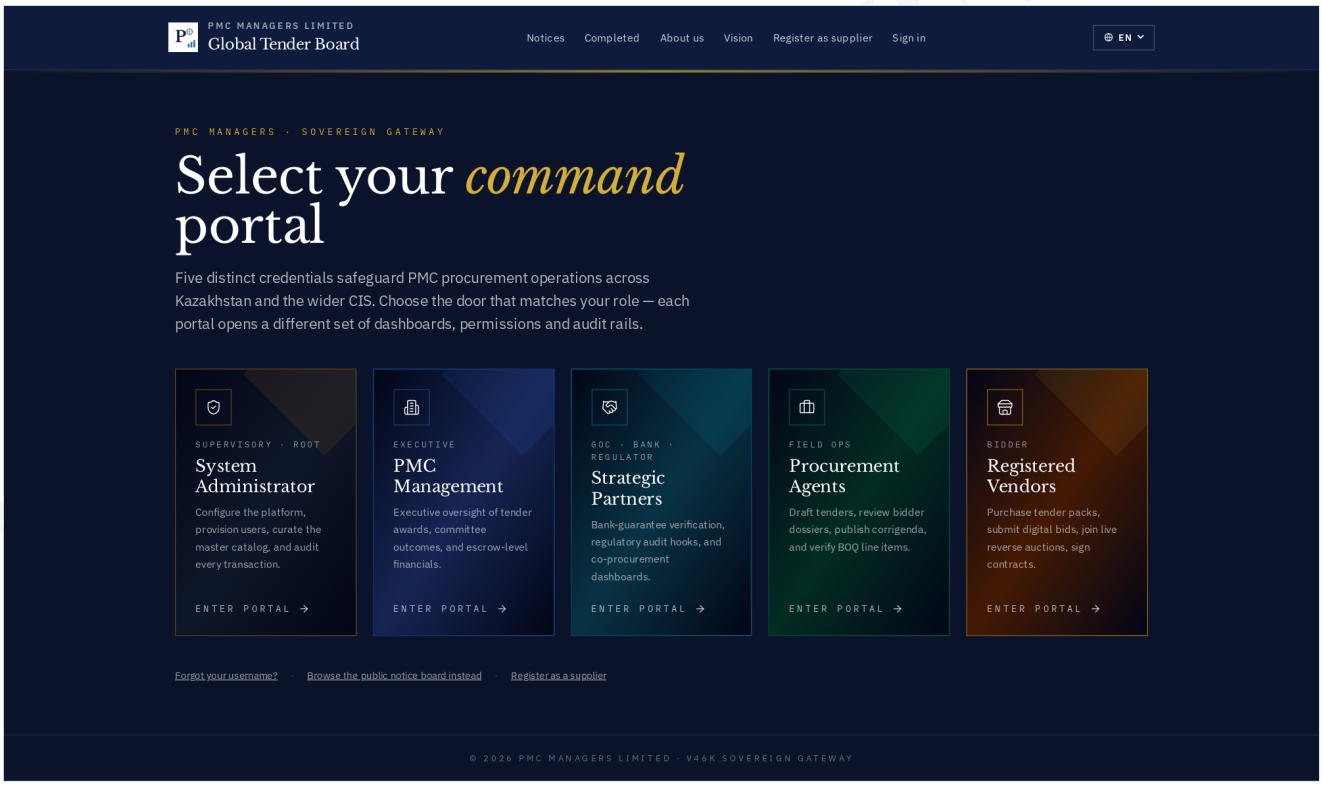
- **Email** — Uniqueness enforced across tenants.
- **Company name (optional)** — Skipping this delays enrichment but does not block the wizard.
- **Country dropdown** — Sourced from the platform-wide country master.

### Known gotchas

- Resend delivery to first-touch corporate mailboxes can hit spam — admins can resend from the Registrations queue.

## 9 · Sovereign Gateway — 5-portal picker

*Five distinct credentials safeguard PMC. Selecting a card reveals a portal-specific credential form (per-role colour + badge).*




*Sovereign Gateway — 5-portal picker*

### Purpose

The 5-portal Sovereign Gateway — the new single entry point for every human role.

### How the workflow unfolds

Users pick the badge matching their role. Each card animates into a portal-specific credential form with the correct colour system, kicker, icon and CTA.

### Fields

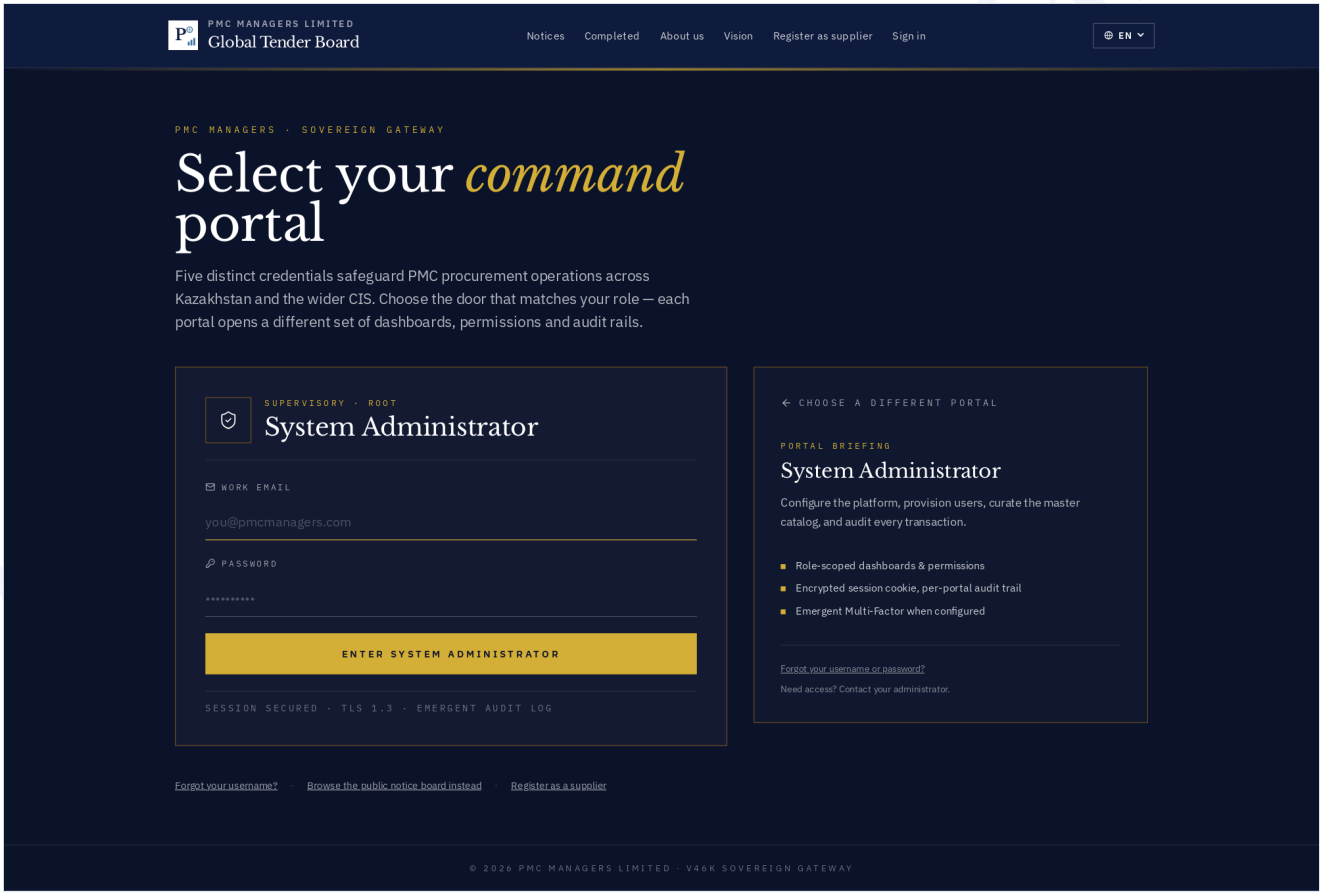
- **System Administrator** — Amber ring · SUPERVISORY · ROOT badge.
- **PMC Management** — Blue ring · EXECUTIVE badge.
- **Strategic Partners** — Cyan ring · GOC · BANK · REGULATOR badge.
- **Procurement Agents** — Emerald ring · FIELD OPS badge.
- **Registered Vendors** — Gold ring · BIDDER badge.

### Known gotchas

- Each portal has its own audit-log tag — a single email that owns two roles will show up twice in the audit trail with the portal it entered through.

## 10 · Admin portal — credential form

*The System Administrator portal — colours default to editorial slate + amber badge. Root-level control of the platform: user provisioning, master catalog, audit.*




*Admin portal — credential form*

### Purpose

Credential form for the Admin portal.

### How the workflow unfolds

The credential form uses the portal's colour and badge. Email + password submit into the appropriate auth realm (staff for admin/management/partners/agents, vendor for vendors).

### Fields

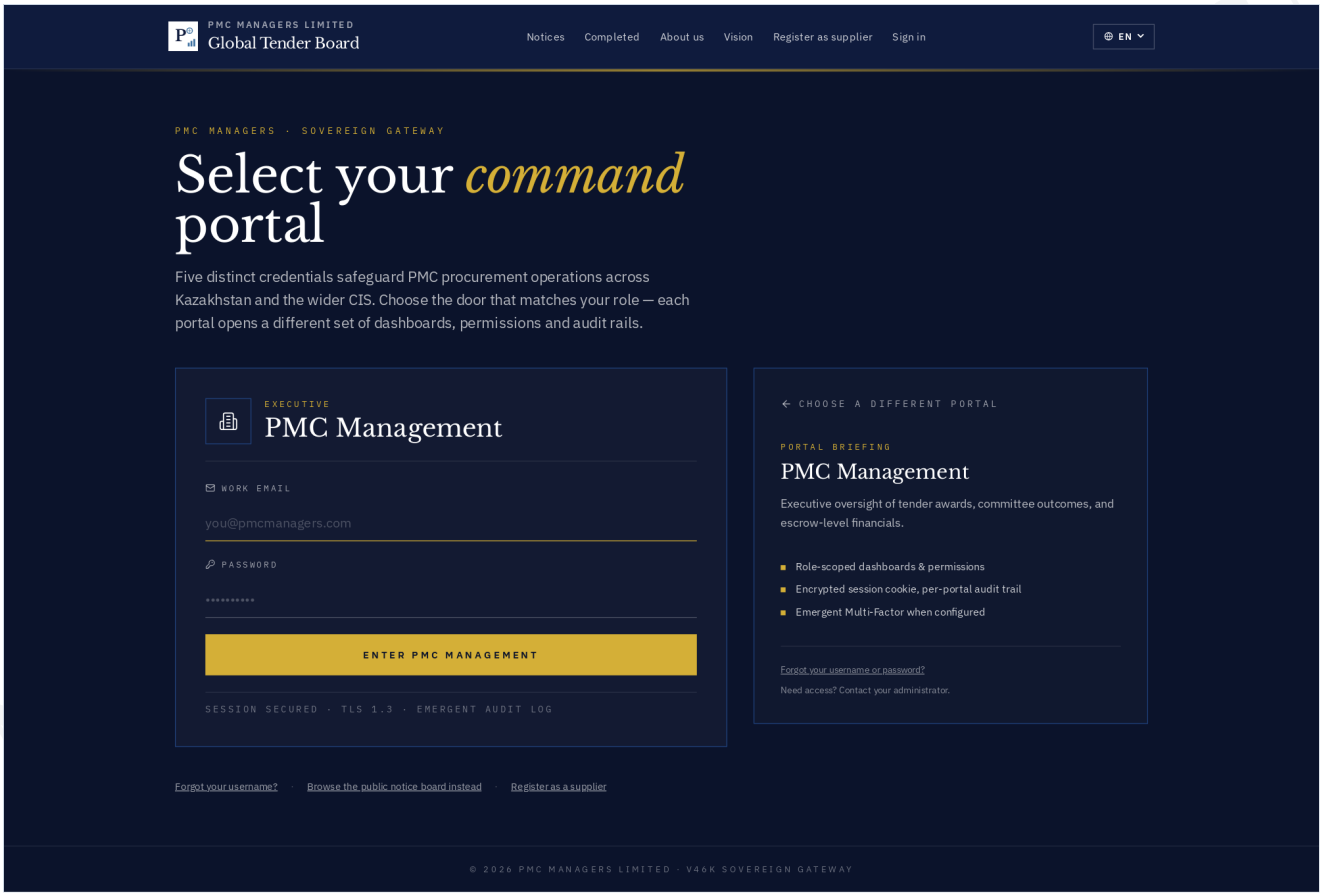
- **Work email** — Email that appears in the audit trail.
- **Password** — 12+ characters; failures trigger the brute-force lockout after 5 attempts.
- **Portal briefing (right rail)** — Short brief so the user knows they're in the right door.
- **Session security note** — TLS 1.3 + immutable audit log — cite this in security reviews.

### Known gotchas

- MFA is enforced for admin + management once configured — you get a code prompt after the password step.

## 11 · Management portal — credential form

*The PMC Management portal — executive dashboards, committee outcomes and escrow-level financials.*




*Management portal — credential form*

### Purpose

Credential form for the Management portal.

### How the workflow unfolds

The credential form uses the portal's colour and badge. Email + password submit into the appropriate auth realm (staff for admin/management/partners/agents, vendor for vendors).

### Fields

- **Work email** — Email that appears in the audit trail.

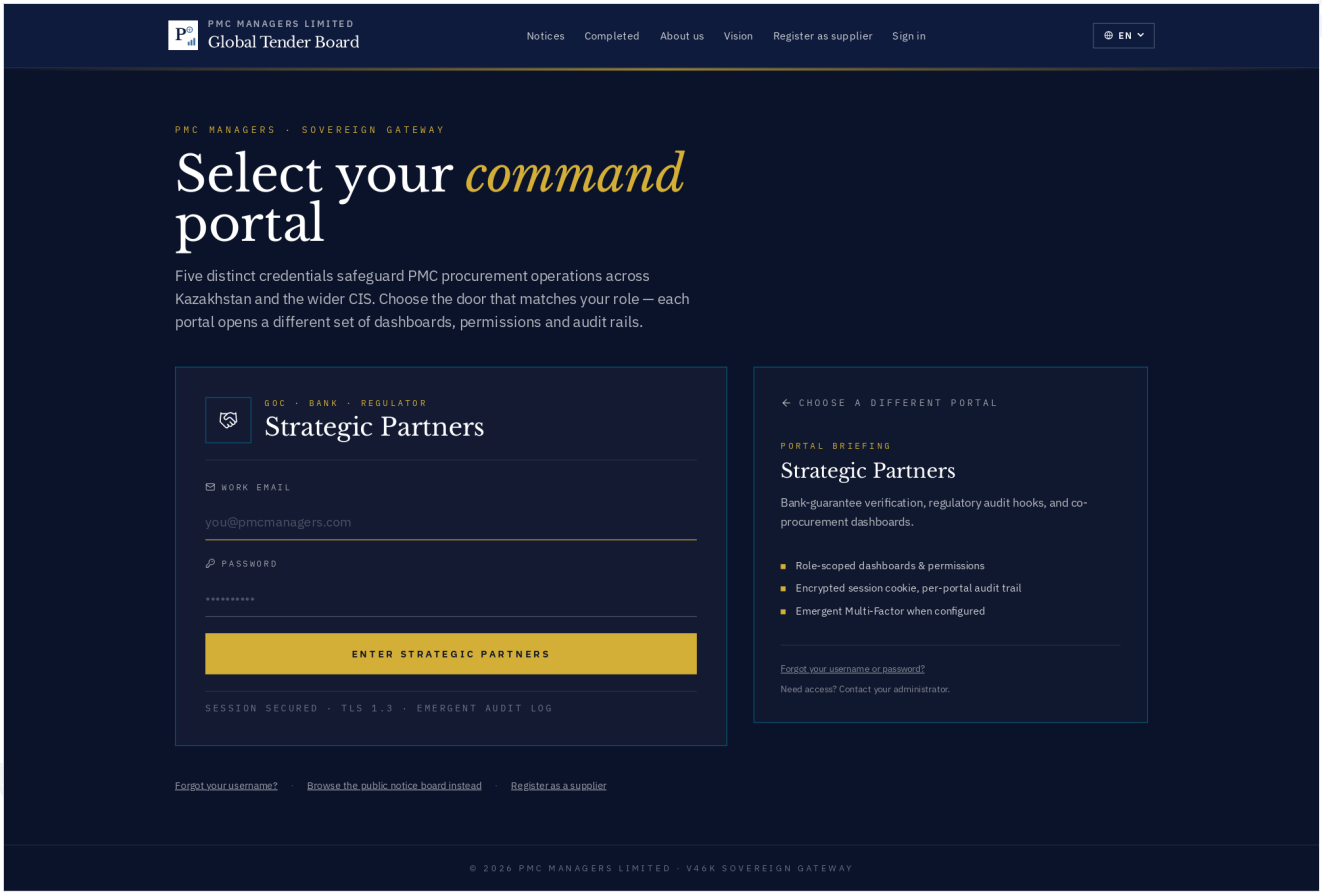
- **Password** — 12+ characters; failures trigger the brute-force lockout after 5 attempts.
- **Portal briefing (right rail)** — Short brief so the user knows they're in the right door.
- **Session security note** — TLS 1.3 + immutable audit log — cite this in security reviews.

### Known gotchas

- MFA is enforced for admin + management once configured — you get a code prompt after the password step.

## 12 · Strategic Partners portal — credential form

*For GOC / Bank / Regulator personas — bank-guarantee verification, regulatory audit hooks, co-procurement dashboards.*




*Strategic Partners portal — credential form*

### Purpose

Credential form for the Partners portal.

### How the workflow unfolds

The credential form uses the portal's colour and badge. Email + password submit into the appropriate auth realm (staff for admin/management/partners/agents, vendor for vendors).

### Fields

- **Work email** — Email that appears in the audit trail.
- **Password** — 12+ characters; failures trigger the brute-force lockout after 5 attempts.

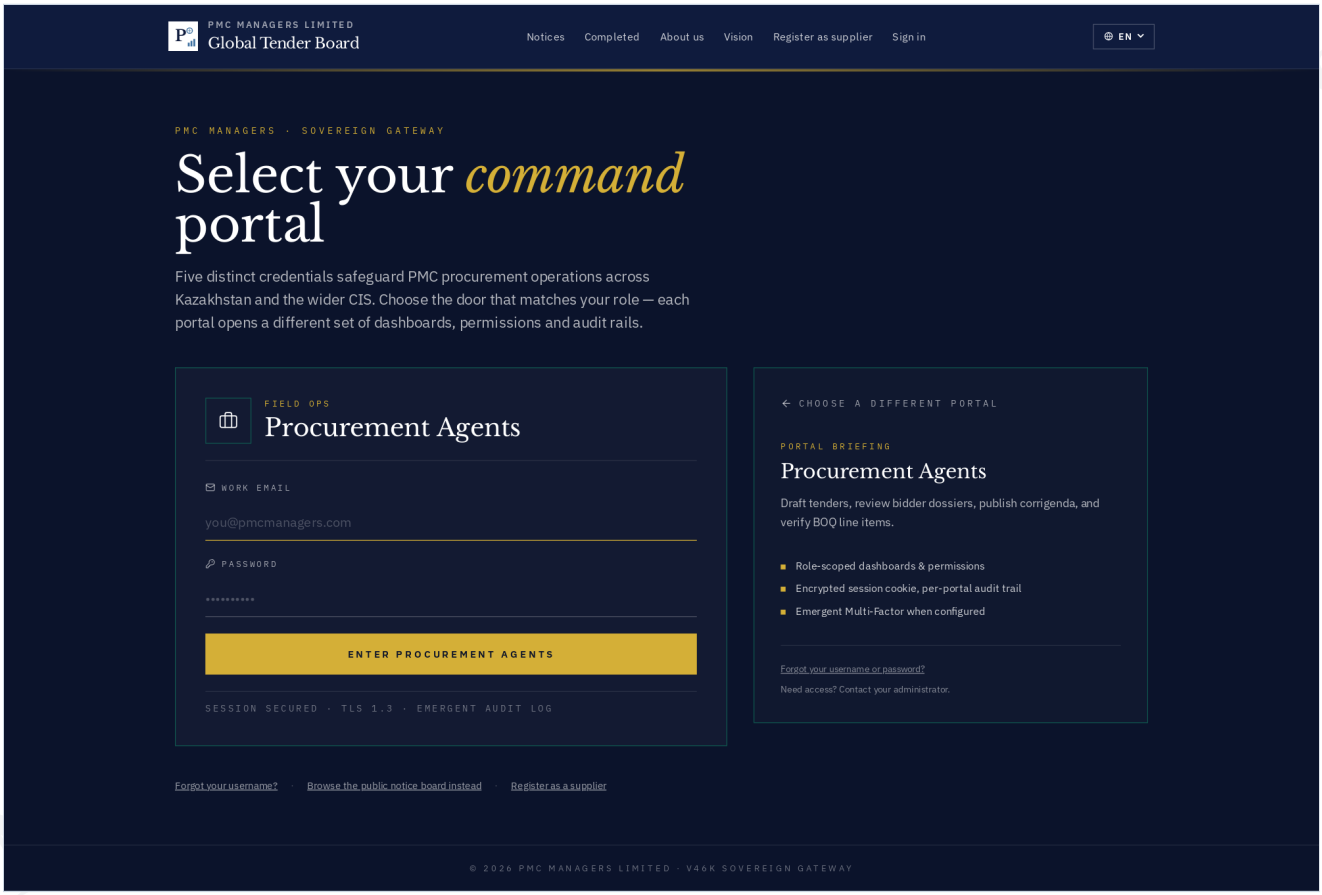
- **Portal briefing (right rail)** — Short brief so the user knows they're in the right door.
- **Session security note** — TLS 1.3 + immutable audit log — cite this in security reviews.

### Known gotchas

- MFA is enforced for admin + management once configured — you get a code prompt after the password step.

## 13 · Procurement Agents portal — credential form

*Field-ops portal for procurement agents drafting tenders, reviewing bidder dossiers, publishing corrigenda and verifying BOQ line items.*




*Procurement Agents portal — credential form*

### Purpose

Credential form for the Agents portal.

### How the workflow unfolds

The credential form uses the portal's colour and badge. Email + password submit into the appropriate auth realm (staff for admin/management/partners/agents, vendor for vendors).

### Fields

- **Work email** — Email that appears in the audit trail.
- **Password** — 12+ characters; failures trigger the brute-force lockout after 5 attempts.
- **Portal briefing (right rail)** — Short brief so the user knows they're in the right door.

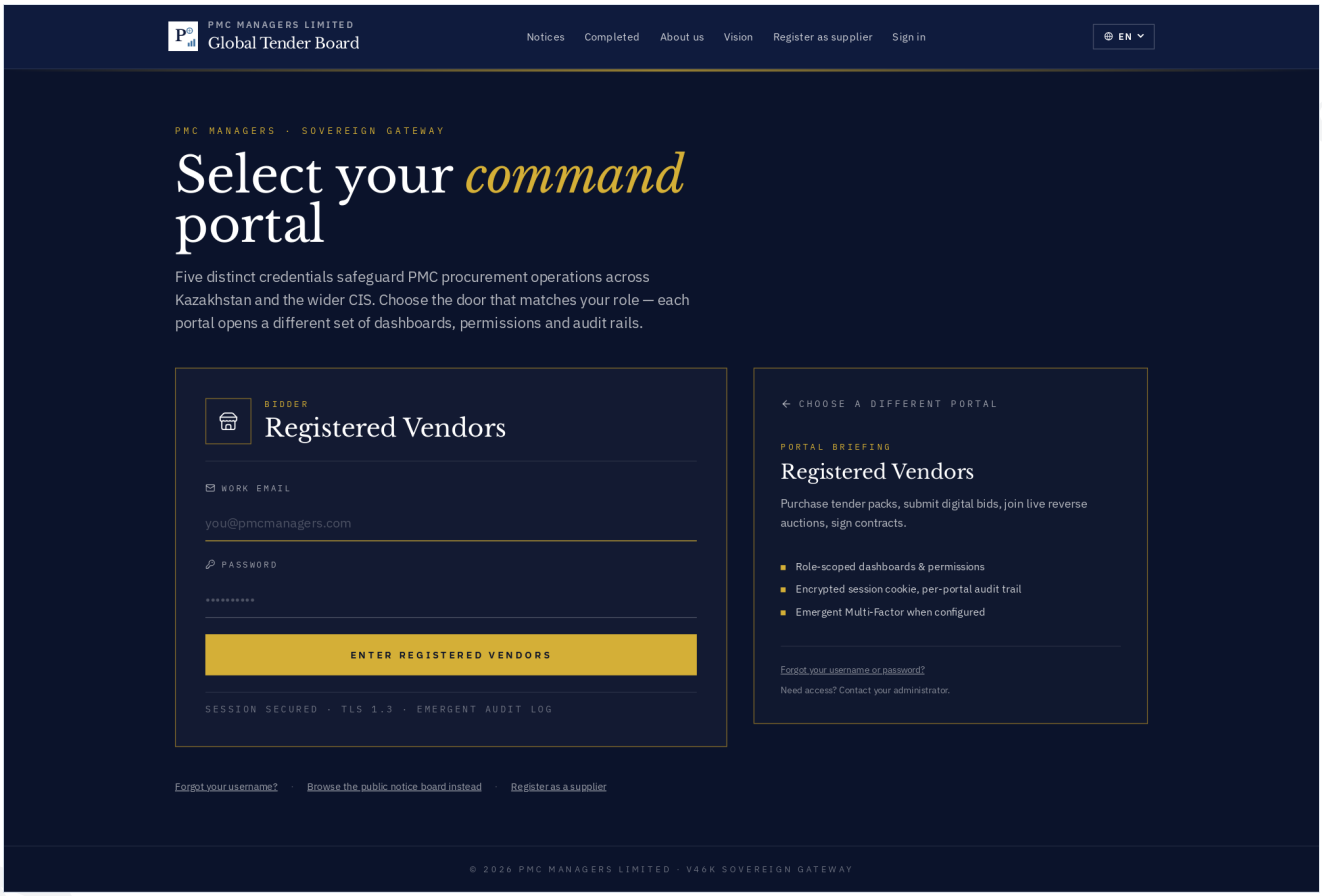
- **Session security note** — TLS 1.3 + immutable audit log — cite this in security reviews.

### Known gotchas

- MFA is enforced for admin + management once configured — you get a code prompt after the password step.

## 14 · Registered Vendors portal — credential form

*Bidder portal — purchase tender packs, submit digital bids, join live reverse auctions, sign contracts. Gold-accented CTA.*




*Registered Vendors portal — credential form*

### Purpose

Credential form for the Vendors portal.

### How the workflow unfolds

The credential form uses the portal's colour and badge. Email + password submit into the appropriate auth realm (staff for admin/management/partners/agents, vendor for vendors).

### Fields

- **Work email** — Email that appears in the audit trail.
- **Password** — 12+ characters; failures trigger the brute-force lockout after 5 attempts.
- **Portal briefing (right rail)** — Short brief so the user knows they're in the right door.
- **Session security note** — TLS 1.3 + immutable audit log — cite this in security reviews.

**Known gotchas**

- MFA is enforced for admin + management once configured — you get a code prompt after the password step.

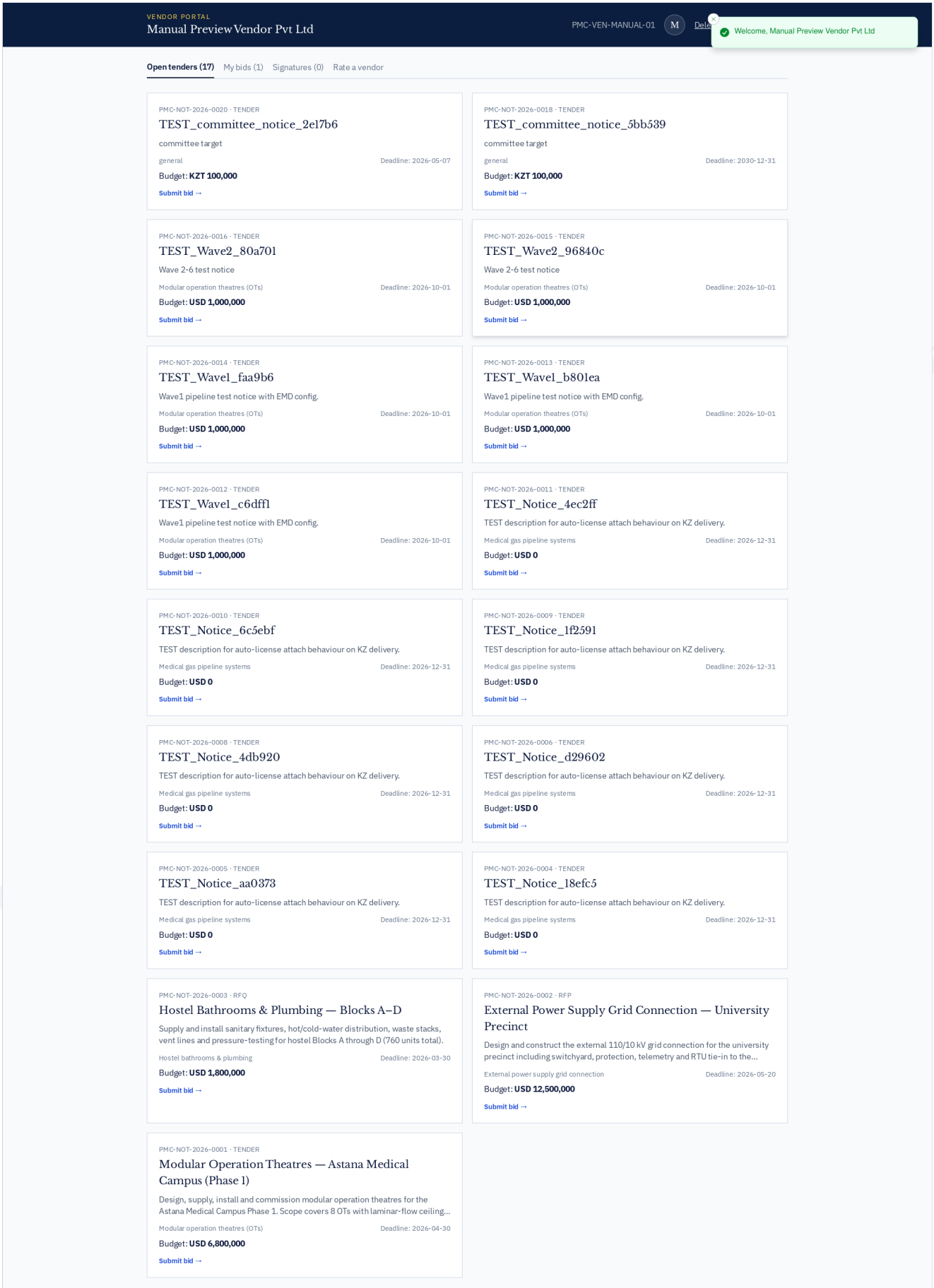
DENTIAL — INTERNAL USE

# 5 · Vendor / supplier journey

## 30 · Vendor landing (post-login)

*Vendor dashboard immediately after sign-in. Header shows company + trust badges; body shows open tenders, held EMDs, active auctions.*

DENTIAL — INTERNAL USE




*Vendor landing (post-login)*

### Purpose

Vendor dashboard immediately after sign-in.

### How the workflow unfolds

Bidders arrive here after credential submit. Header shows the trust badges, body shows quick paths to open tenders, held EMDs, active auctions, awarded contracts.

### Fields

- **Trust badges** — Aggregate rating badge + certified-supplier chip if applicable.
- **Open tenders card** — Categories filtered to the vendor's declared industries.
- **EMDs held** — Total value + release-eligible-date.
- **Auctions strip** — Live and scheduled reverse-auction rooms.

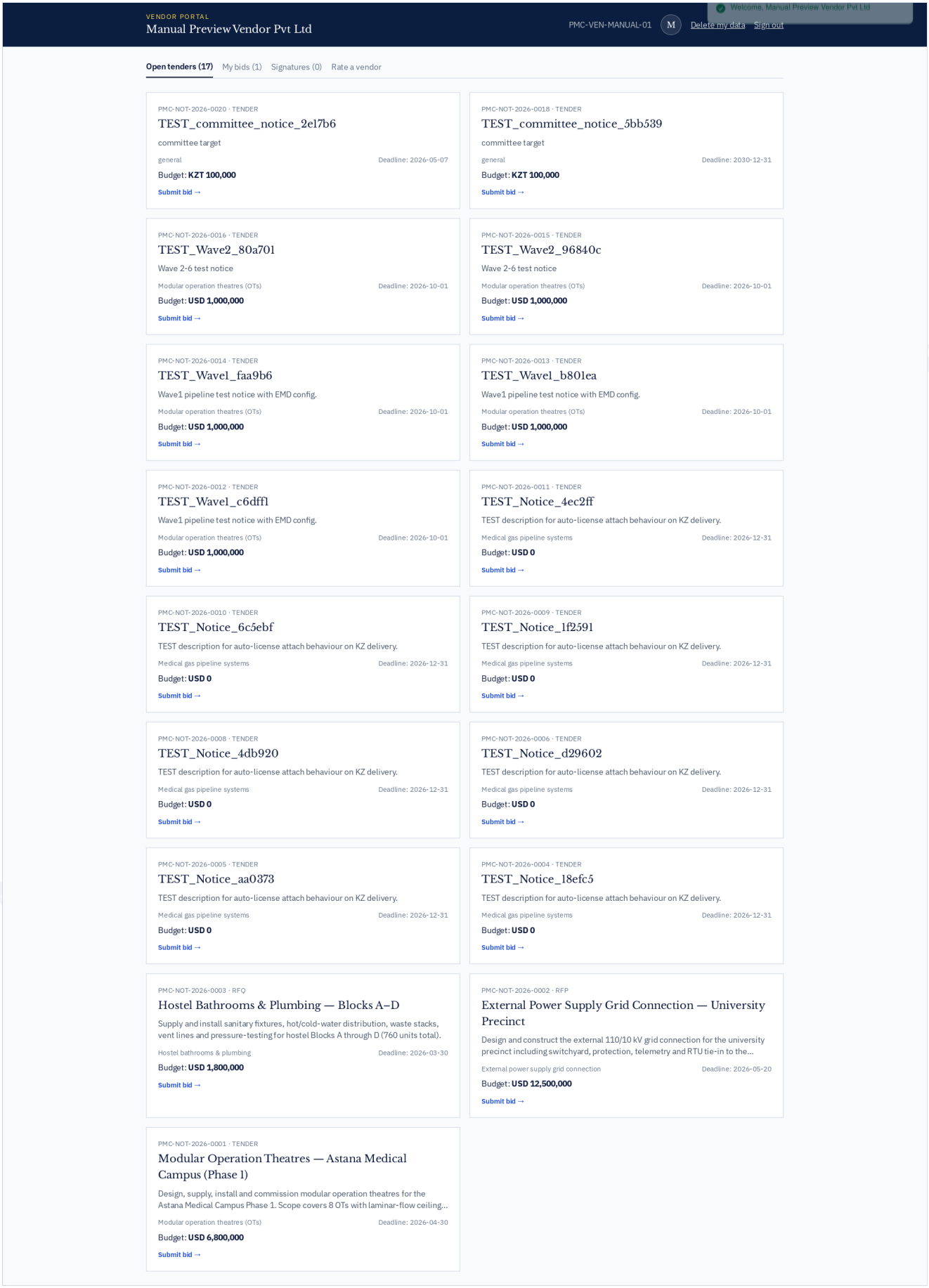
### Known gotchas

- The dashboard refetches on every route change to avoid stale counts.

## 31 · Vendor · Open tenders

*Tenders the vendor may bid on, filtered to declared categories.*

DENTIAL — INTERNAL USE



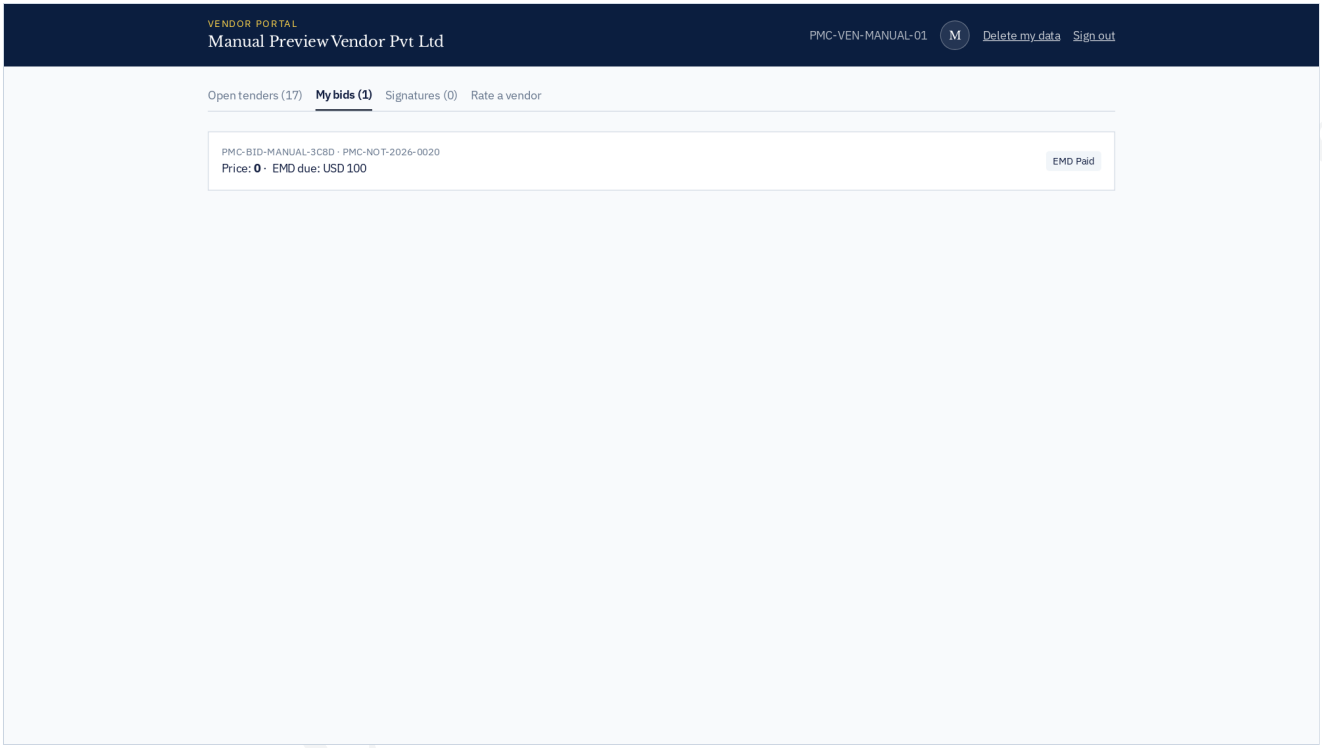

*Vendor · Open tenders*

### How to use

This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

## 32 · Vendor · My bids

*Every submitted bid with envelope status, score summary and award outcome.*




*Vendor · My bids*

### Purpose

My bids — every submitted bid with envelope status, score summary and award outcome.

### How the workflow unfolds

Bidders use this to prove submission for legal disputes and to compare pricing against their prior wins.

### Fields

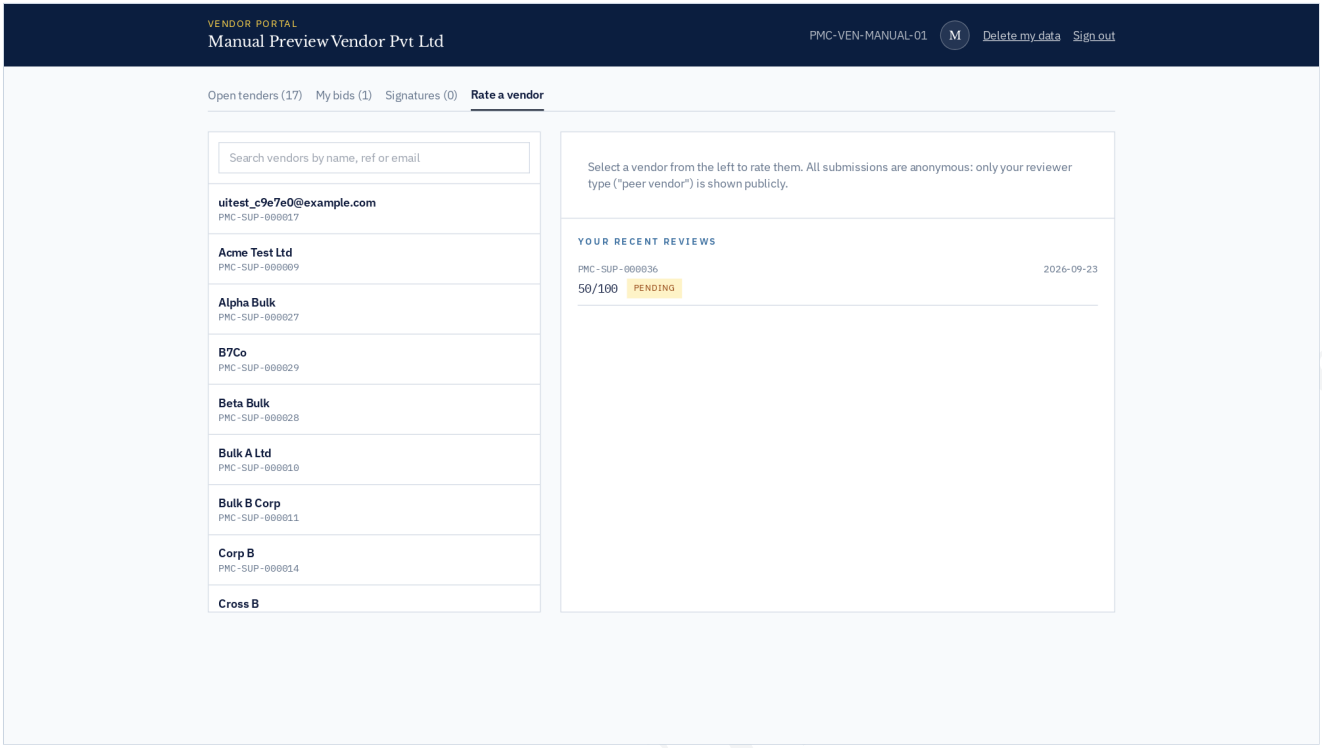
- **Bid ref** — PMC-BID-YYYY-NNNN — anchor for every legal reference.
- **Envelope status** — Sealed / Unsealed / Under evaluation / Awarded / Rejected.
- **Score summary** — Technical + commercial weighted score once the envelope is unsealed.
- **Award outcome** — Won / Runner-up / Not qualified.

### Known gotchas

- A rejected bid keeps its EMD until the notice's dispute window closes.

# 33 · Vendor · Ratings received

*Reviews left by admins post-contract with an aggregate trust badge.*



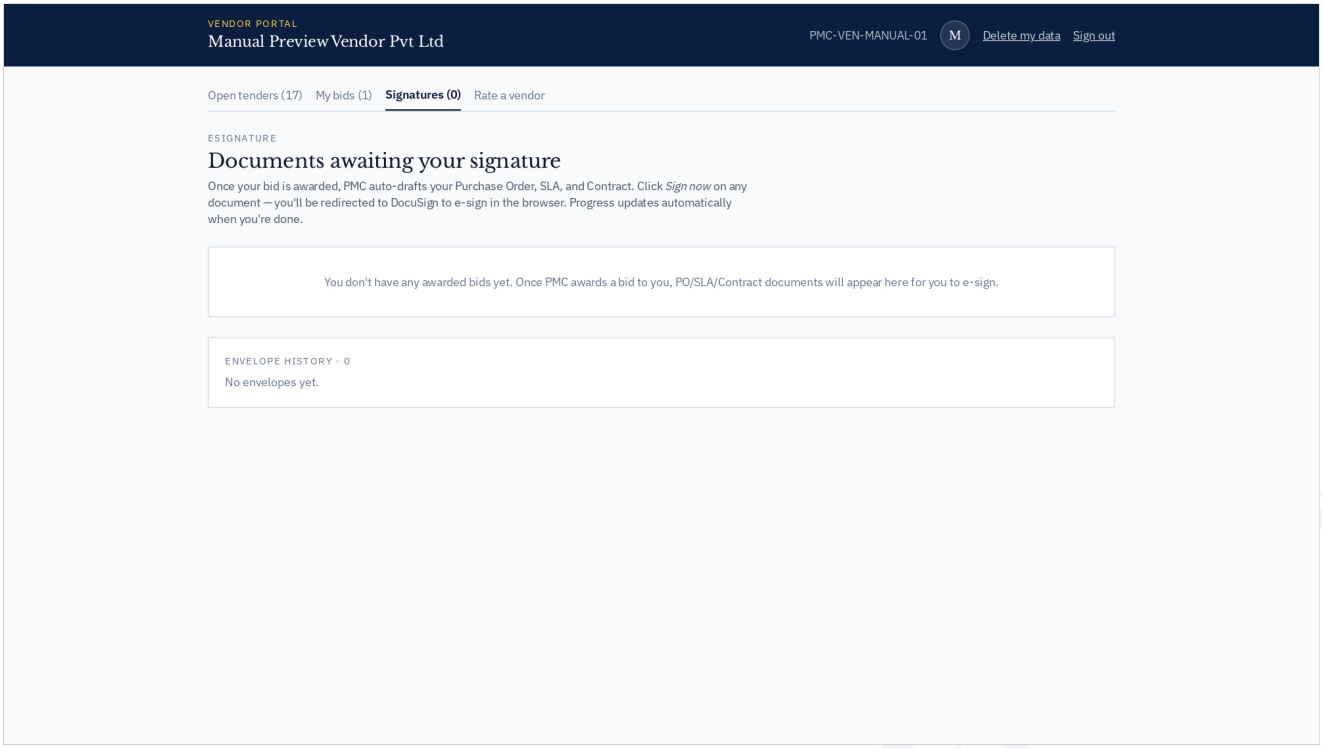

*Vendor · Ratings received*

### How to use

This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

# 34 · Vendor · Contract signatures

*DocuSign envelopes pending / signed by the vendor.*




*Vendor · Contract signatures*

### How to use

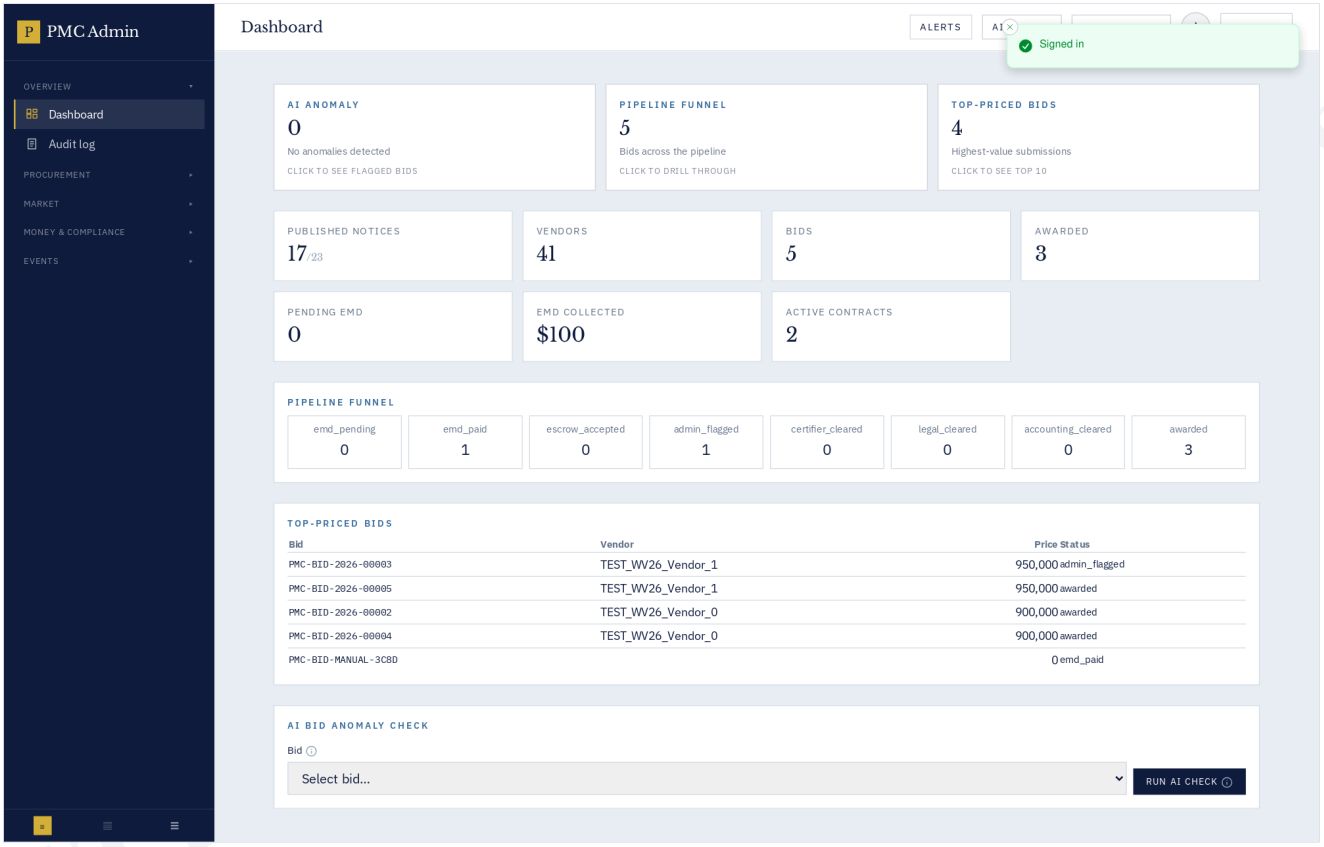
This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

DENTIAL — INTERN SE

# 6 · Admin console — full module tour

## 50 · Admin console landing (Operations Overview)

*First screen after admin login — collapsible sidebar grouped into Overview / Procurement / Market / Money & Compliance / Events, environment chip, AI Copilot shortcut.*




*Admin console landing (Operations Overview)*

### Purpose

Command centre for PMC staff.

### How the workflow unfolds

Admin arrives here after sign-in; the collapsible sidebar groups 23 modules into Overview / Procurement / Market / Money & Compliance / Events — matching the mental model of the PMC operations team.

### Fields

- **Sidebar groups** — Overview · Procurement · Market · Money & Compliance · Events.
- **Environment chip** — Which tenant/programme is active.
- **AI Copilot** — In-context question box for procurement policy questions.
- **Alerts strip** — Aging EMDs, publish-ready notices, contract SLA breaches.

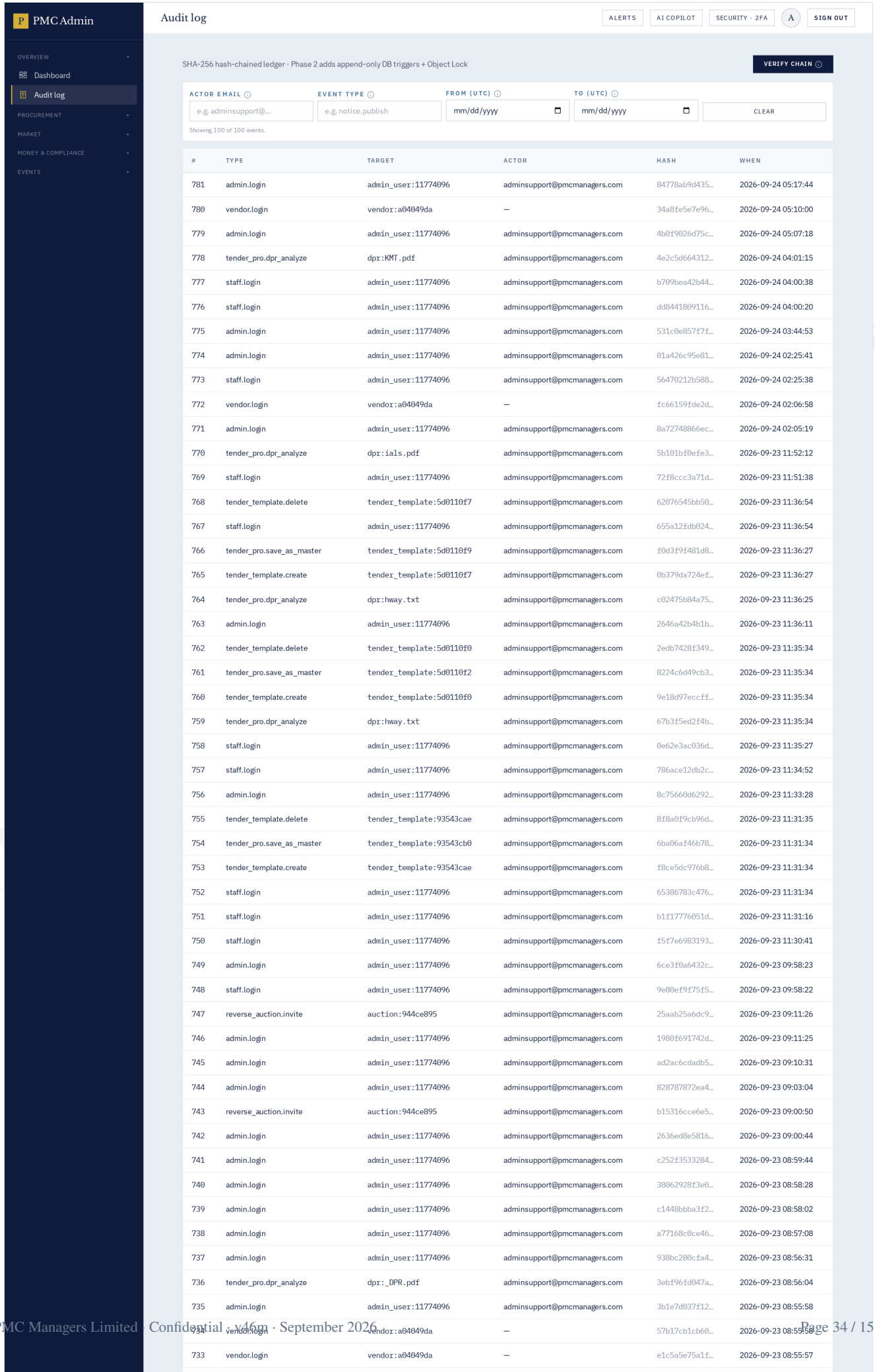
**Known gotchas**

- Some tabs require role escalation (e.g. Escrow needs escrow_agent) — a locked-tab tooltip explains.

## 52 · Admin · Audit trail

*Immutable action log with entity filters.*

DENTIAL — INTERNAL USE




| # | TYPE | TARGET | ACTOR | HASH | WHEN |
|---|---|---|---|---|---|
| 781 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 84778ab9d435… | 2026-09-24 05:17:44 |
| 780 | vendor.login | vendor:a04049da | — | 34a8fe5e7e96… | 2026-09-24 05:10:00 |
| 779 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 4b0f9026d75c… | 2026-09-24 05:07:18 |
| 778 | tender_pro.dpr_analyze | dpr:KMT.pdf | adminsupport@pmcmanagers.com | 4e2c5d664312… | 2026-09-24 04:01:15 |
| 777 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | b709bea42b44… | 2026-09-24 04:00:38 |
| 776 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | dd8441809116… | 2026-09-24 04:00:20 |
| 775 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 531c0e857f7f… | 2026-09-24 03:44:53 |
| 774 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 01a426c95e81… | 2026-09-24 02:25:41 |
| 773 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 56470212b588… | 2026-09-24 02:25:38 |
| 772 | vendor.login | vendor:a04049da | — | fc66159fde2d… | 2026-09-24 02:06:58 |
| 771 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 8a72748866ec… | 2026-09-24 02:05:19 |
| 770 | tender_pro.dpr_analyze | dpr:ials.pdf | adminsupport@pmcmanagers.com | 5b101bf0efe3… | 2026-09-23 11:52:12 |
| 769 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 72f8ccc3a71d… | 2026-09-23 11:51:38 |
| 768 | tender_template.delete | tender_template:5d0110f7 | adminsupport@pmcmanagers.com | 62076545bb50… | 2026-09-23 11:36:54 |
| 767 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 655a12fdb024… | 2026-09-23 11:36:54 |
| 766 | tender_pro.save_as_master | tender_template:5d0110f9 | adminsupport@pmcmanagers.com | f0d3f9f481d8… | 2026-09-23 11:36:27 |
| 765 | tender_template.create | tender_template:5d0110f7 | adminsupport@pmcmanagers.com | 0b379da724ef… | 2026-09-23 11:36:27 |
| 764 | tender_pro.dpr_analyze | dpr:hway.txt | adminsupport@pmcmanagers.com | c02475b84a75… | 2026-09-23 11:36:25 |
| 763 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 2646a42b4b1b… | 2026-09-23 11:36:11 |
| 762 | tender_template.delete | tender_template:5d0110f0 | adminsupport@pmcmanagers.com | 2edb7428f349… | 2026-09-23 11:35:34 |
| 761 | tender_pro.save_as_master | tender_template:5d0110f2 | adminsupport@pmcmanagers.com | 8224c6d49cb3… | 2026-09-23 11:35:34 |
| 760 | tender_template.create | tender_template:5d0110f0 | adminsupport@pmcmanagers.com | 9e18d97eccff… | 2026-09-23 11:35:34 |
| 759 | tender_pro.dpr_analyze | dpr:hway.txt | adminsupport@pmcmanagers.com | 67b3f5ed2f4b… | 2026-09-23 11:35:34 |
| 758 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 0e62e3ac036d… | 2026-09-23 11:35:27 |
| 757 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 786ace12db2c… | 2026-09-23 11:34:52 |
| 756 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 8c75660d6292… | 2026-09-23 11:33:28 |
| 755 | tender_template.delete | tender_template:93543cae | adminsupport@pmcmanagers.com | 8f8a0f9cb96d… | 2026-09-23 11:31:35 |
| 754 | tender_pro.save_as_master | tender_template:93543cb0 | adminsupport@pmcmanagers.com | 6ba06af46b78… | 2026-09-23 11:31:34 |
| 753 | tender_template.create | tender_template:93543cae | adminsupport@pmcmanagers.com | f8ce5dc976b8… | 2026-09-23 11:31:34 |
| 752 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 65386783c476… | 2026-09-23 11:31:34 |
| 751 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | b1f17776051d… | 2026-09-23 11:31:16 |
| 750 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | f5f7e6983193… | 2026-09-23 11:30:41 |
| 749 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 6ce3f0a6432c… | 2026-09-23 09:58:23 |
| 748 | staff.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 9e80ef9f75f5… | 2026-09-23 09:58:22 |
| 747 | reverse_auction.invite | auction:944ce895 | adminsupport@pmcmanagers.com | 25aab25a6dc9… | 2026-09-23 09:11:26 |
| 746 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 1980f691742d… | 2026-09-23 09:11:25 |
| 745 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | ad2ac6cdadb5… | 2026-09-23 09:10:31 |
| 744 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 828787872ea4… | 2026-09-23 09:03:04 |
| 743 | reverse_auction.invite | auction:944ce895 | adminsupport@pmcmanagers.com | b15316cce6e5… | 2026-09-23 09:00:50 |
| 742 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 2636ed8e5816… | 2026-09-23 09:00:44 |
| 741 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | c252f3533284… | 2026-09-23 08:59:44 |
| 740 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 38062928f3e0… | 2026-09-23 08:58:28 |
| 739 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | c1448bbba3f2… | 2026-09-23 08:58:02 |
| 738 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | a77168c0ce46… | 2026-09-23 08:57:08 |
| 737 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 938bc280cfa4… | 2026-09-23 08:56:31 |
| 736 | tender_pro.dpr_analyze | dpr:_DPR.pdf | adminsupport@pmcmanagers.com | 3ebf96fd047a… | 2026-09-23 08:56:04 |
| 735 | admin.login | admin_user:11774096 | adminsupport@pmcmanagers.com | 3b1e7d037f12… | 2026-09-23 08:55:58 |
| 734 | vendor.login | vendor:a04049da | — | 57b17cb1cb60… | 2026-09-23 08:55:58 |
| 733 | vendor.login | vendor:a04049da | — | e1c5a5e75a1f… | 2026-09-23 08:55:57 |
| 732 | vendor_rating.delete | review:773631a5 | adminsupport@pmcmanagers.com | 49471def084d… | 2026-09-23 08:55:56 |

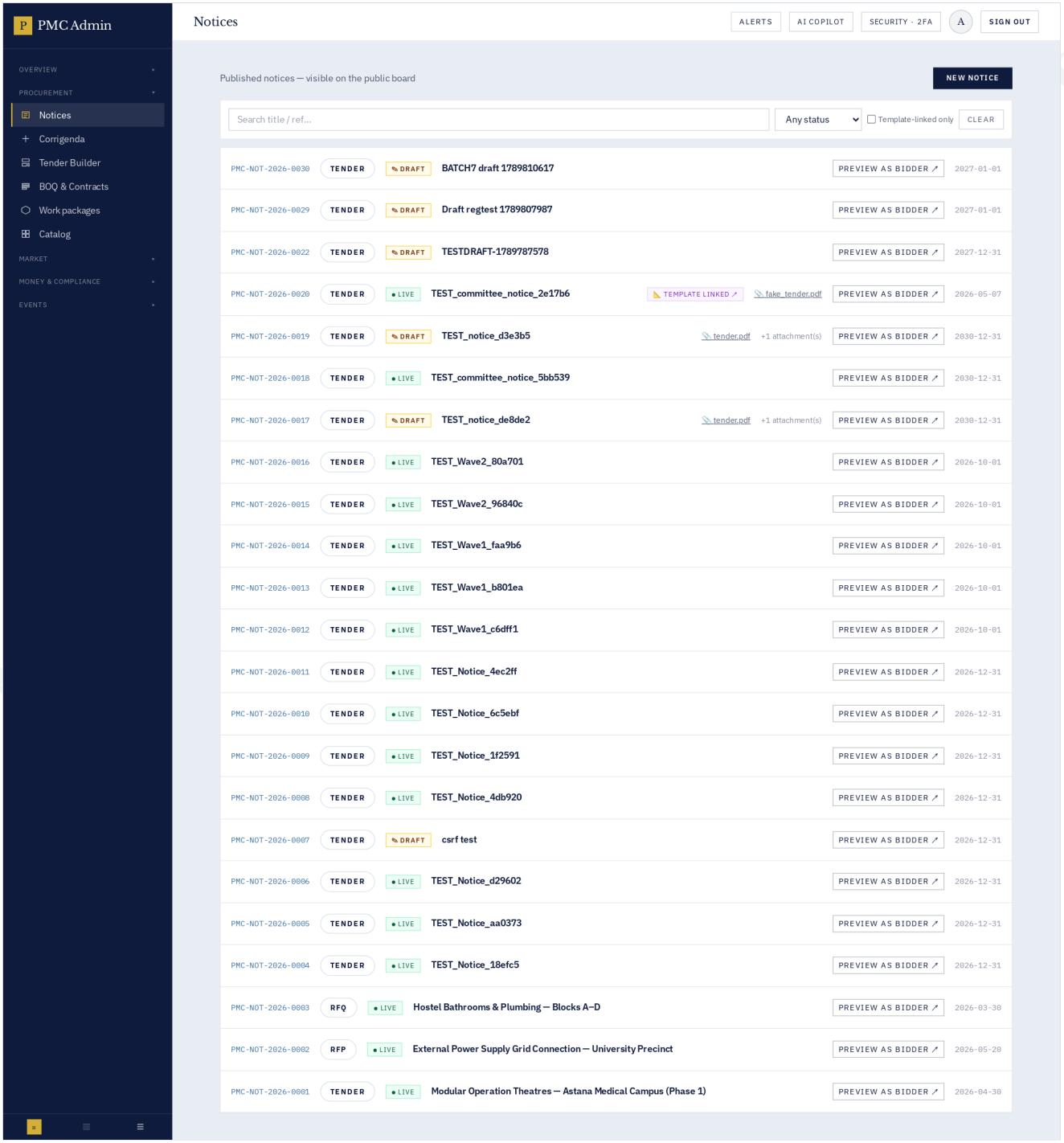
*Admin · Audit trail*

### How to use

This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

## 53 · Admin · Notices — publish & manage

*Create, publish, edit, withdraw, attach documents, EMD.*



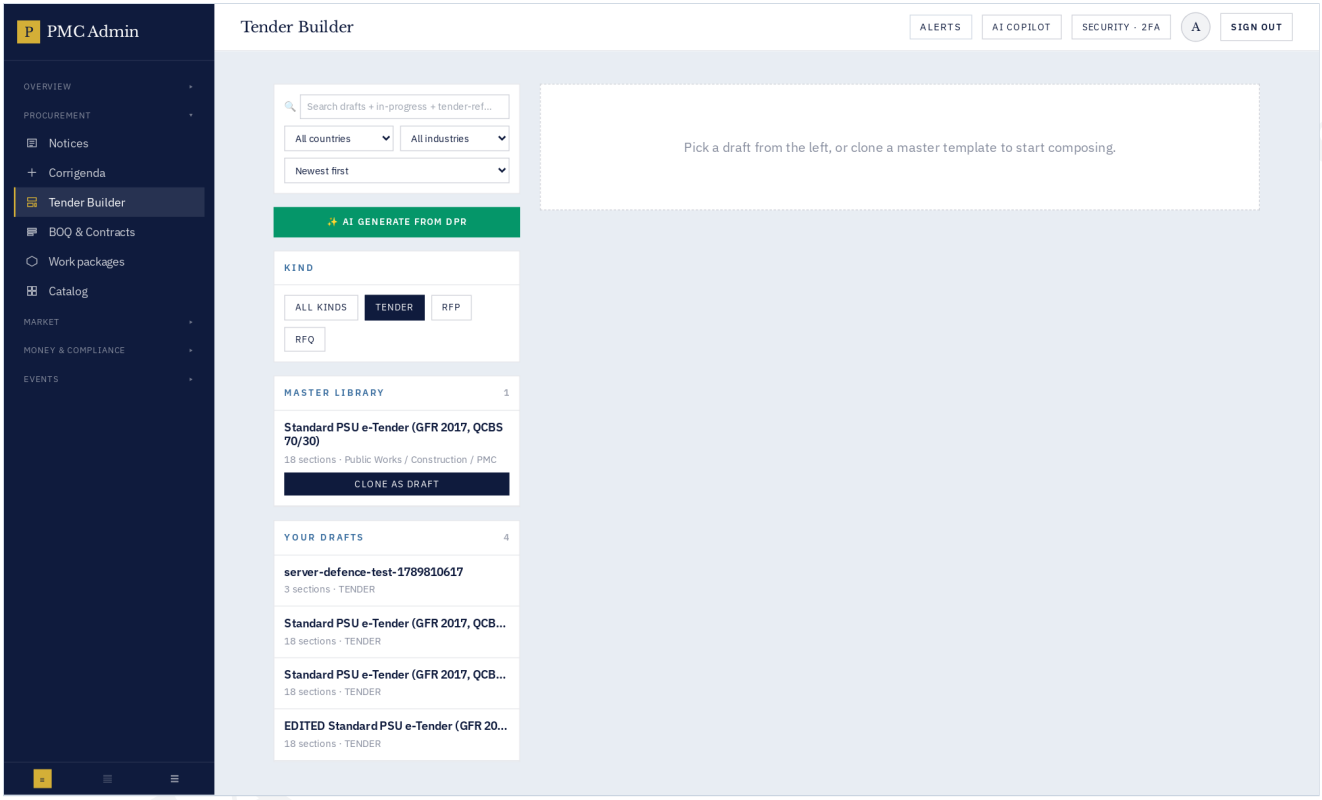

*Admin · Notices — publish & manage*

### How to use

This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

## 55 · Admin · Tender Builder & DPR AI

*Master library, DPR deep-dive, AI Fill All, PDF export.*




*Admin · Tender Builder & DPR AI*

### How to use

This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

## 57 · Admin · Work packages

*Modular deliverable packages for the KZ programme.*


PMC Admin
OVERVIEW
PROCUREMENT
Notices
Corrigenda
Tender Builder
BOQ & Contracts
Work packages
Catalog
MARKET
MONEY & COMPLIANCE
EVENTS
Work packages
ALERTS
AI COPILOT
SECURITY · 2FA
A
SIGN OUT
48 packages in pipeline · drag cards between columns to move status
IMPORT CSV
NEW PACKAGE
BACKLOG 26
SCOPING 13
SBD DRAFT 6
READY 3
PUBLISHED 0
CANCELLED 0
PMC-PKG-2026-0045
TEST_wave10_pkg_1785728
Medical
$10,000
PMC-PKG-2026-0046
TEST_wave10_pkg_1785728
Medical
$10,000
PMC-PKG-2026-0047
TEST_wave10_pkg_1785728
Medical
$10,000
PMC-PKG-2026-0048
TEST_wave10_pkg_1785728
Medical
$10,000
PMC-PKG-2026-0031
TEST_Pkg_A
Medical gas pipeline systems
$100,000 2026-06-01
PMC-PKG-2026-0033
TEST_Pkg_A
Medical gas pipeline systems
$100,000 2026-06-01
PMC-PKG-2026-0035
TEST_Pkg_A
Medical gas pipeline systems
$100,000 2026-06-01
PMC-PKG-2026-0037
TEST_Pkg_A
Medical gas pipeline systems
$100,000 2026-06-01
PMC-PKG-2026-0039
TEST_Pkg_A
Medical gas pipeline systems
$100,000 2026-06-01
PMC-PKG-2026-0041
TEST_Pkg_A
Medical gas pipeline systems
$100,000 2026-06-01
PMC-PKG-2026-0043
TEST_Pkg_A
Medical gas pipeline systems
$100,000 2026-06-01
PMC-PKG-2026-0024
Modular Kitchen Equipment — Hostel Canteens
Kitchen equipment
$780,000 2026-06-05
PMC-PKG-2026-0025
Laundry Equipment — Hostel Blocks
Laundry
$320,000 2026-06-05
PMC-PKG-2026-0021
CCTV & Access Control — Campus-wide
CCTV
$1,400,000 2026-06-25
PMC-PKG-2026-0009
Elevators & Escalators — Academic Building
Elevators & escalators
$2,800,000 2026-07-01
PMC-PKG-2026-0022
PA & Audio Systems — Auditoriums
PA systems
$620,000 2026-07-05
PMC-PKG-2026-0023
BMS Integration — Campus
BMS
$1,100,000 2026-07-10
PMC-PKG-2026-0029
Diesel Gensets & UPS — Redundancy
Diesel gensets
$1,800,000 2026-07-25
PMC-PKG-2026-0030
Boundary Walls & Storm Drainage
Boundary walls
$2,900,000 2026-07-30
PMC-PKG-2026-0019
Hotel Superstructure Works
Hotel superstructure works
$28,000,000 2026-08-01
PMC-PKG-2026-0017
Hostel Bathrooms & Plumbing — Blocks A–D
Hostel bathrooms & plumbing
$1,800,000 2026-03-30
PMC-PKG-2026-0003
OT Laminar Flow Ceilings & Ancillaries
OT equipment & laminar flow systems
$1,900,000 2026-05-20
PMC-PKG-2026-0010
Elevators — Hostel Blocks A–D
Elevators for hostel blocks
$1,400,000 2026-06-15
PMC-PKG-2026-0020
HVAC Equipment — Academic Blocks A & B
HVAC
$3,200,000 2026-06-20
PMC-PKG-2026-0032
TEST_Pkg_B
Hostel bathrooms & plumbing
$50,000 2026-07-01
PMC-PKG-2026-0034
TEST_Pkg_B
Hostel bathrooms & plumbing
$50,000 2026-07-01
PMC-PKG-2026-0036
TEST_Pkg_B
Hostel bathrooms & plumbing
$50,000 2026-07-01
PMC-PKG-2026-0038
TEST_Pkg_B
Hostel bathrooms & plumbing
$50,000 2026-07-01
PMC-PKG-2026-0040
TEST_Pkg_B
Hostel bathrooms & plumbing
$50,000 2026-07-01
PMC-PKG-2026-0042
TEST_Pkg_B
Hostel bathrooms & plumbing
$50,000 2026-07-01
PMC-PKG-2026-0044
TEST_Pkg_B
Hostel bathrooms & plumbing
$50,000 2026-07-01
PMC-PKG-2026-0008
Transformers Procurement & Installation — 4 units
Transformers procurement & installation
$3,600,000 2026-07-15
PMC-PKG-2026-0018
Main Academic Building Civil Works
Main academic building civil works
$38,000,000 2026-07-30
PMC-PKG-2026-0002
Medical Gas Pipeline System — Wing A & B
Medical gas pipeline systems
$2,100,000 2026-05-15
PMC-PKG-2026-0013
External Firefighting Network — Hydrants Loop
External firefighting network (hydrants)
$1,600,000 2026-05-25
PMC-PKG-2026-0004
Hospital Civil & Superstructure — Block 1
Hospital civil & superstructure works
$24,000,000 2026-06-01
PMC-PKG-2026-0015
Water Treatment Plant (WTP) — Main Campus
Water treatment plant (WTP)
$5,400,000 2026-06-10
PMC-PKG-2026-0016
Sewage Treatment Plant (STP) — Main Campus
Sewage treatment plant (STP)
$4,800,000 2026-06-15
PMC-PKG-2026-0007
Substation Works HT/LT — Main Campus
Substation works (HT/LT)
$4,200,000 2026-06-30
PMC-PKG-2026-0001
Modular OT Suite — Astana Medical Campus Phase 1
Modular operation theatres (OTs)
$6,800,000 2026-04-30
PMC-PKG-2026-0006
External 110/10 kV Grid Connection — University Precinct
External power supply grid connection
$12,500,000 2026-05-20
PMC-PKG-2026-0012
Fire Alarm & Safety Systems — Campus-wide
Fire alarm & safety systems
$3,100,000 2026-05-30
Drop here
Drop here

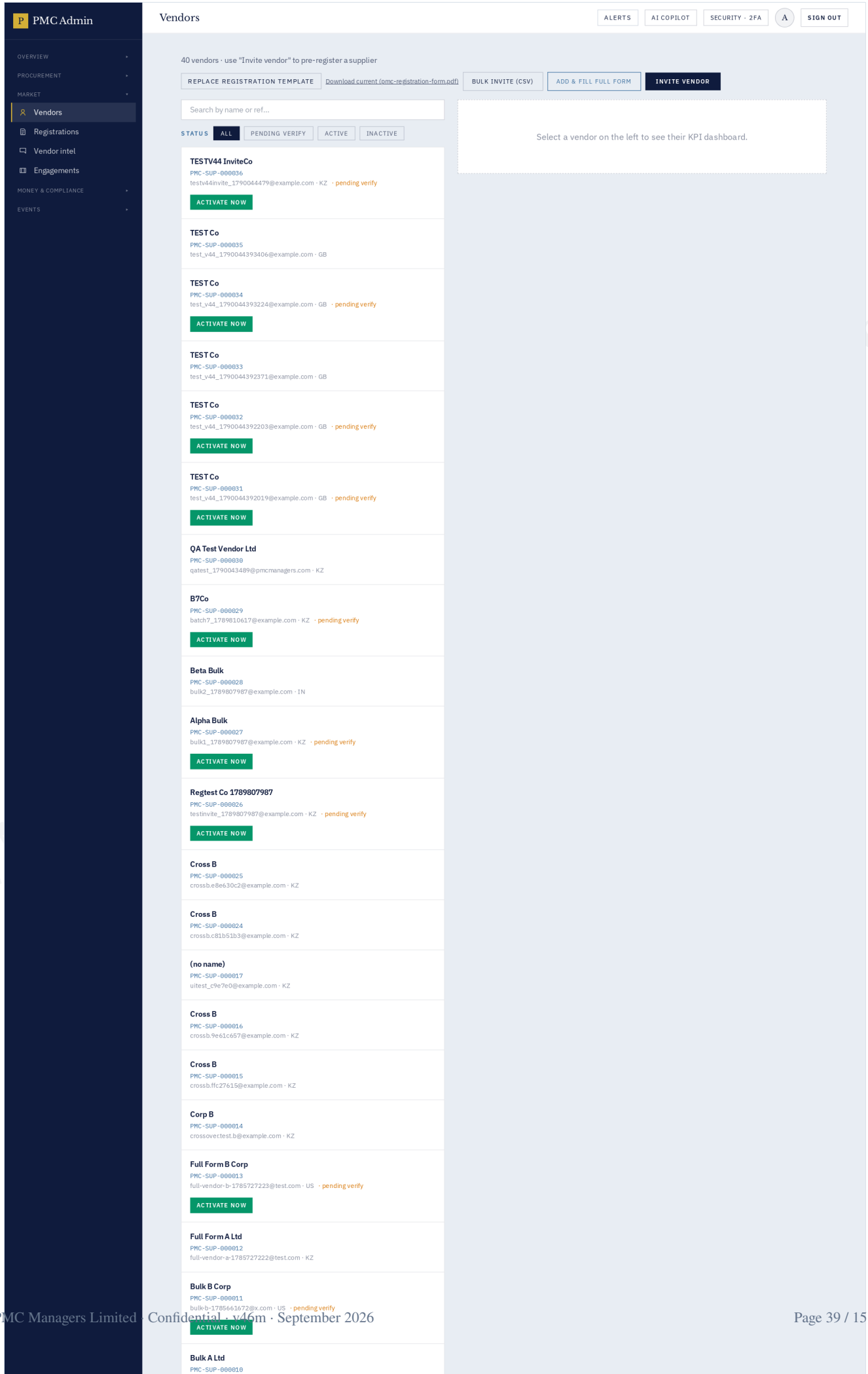
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## 59 · Admin · Vendor directory

*Full CRUD on suppliers, licences, compliance, tenants.*

DENTIAL — INTERNAL USE


PMC Admin
OVERVIEW
PROCUREMENT
MARKET
Vendors
Registrations
Vendor intel
Engagements
MONEY & COMPLIANCE
EVENTS
Vendors
ALERTS
AI COPILOT
SECURITY · 2FA
A
SIGN OUT
40 vendors · use "Invite vendor" to pre-register a supplier
REPLACE REGISTRATION TEMPLATE
Download current (pmc-registration-form.pdf)
BULK INVITE (CSV)
ADD & FILL FULL FORM
INVITE VENDOR
Search by name or ref...
STATUS
ALL
PENDING VERIFY
ACTIVE
INACTIVE
TESTV44 InviteCo
PMC-SUP-000036
testv44invite_1790044479@example.com · KZ · pending verify
ACTIVATE NOW
TEST Co
PMC-SUP-000035
test_v44_1790044393406@example.com · GB
TEST Co
PMC-SUP-000034
test_v44_1790044393224@example.com · GB · pending verify
ACTIVATE NOW
TEST Co
PMC-SUP-000033
test_v44_1790044392371@example.com · GB
TEST Co
PMC-SUP-000032
test_v44_1790044392203@example.com · GB · pending verify
ACTIVATE NOW
TEST Co
PMC-SUP-000031
test_v44_1790044392019@example.com · GB · pending verify
ACTIVATE NOW
QA Test Vendor Ltd
PMC-SUP-000030
qatest_1790043489@pmcmanagers.com · KZ
B7Co
PMC-SUP-000029
batch7_1789810617@example.com · KZ · pending verify
ACTIVATE NOW
Beta Bulk
PMC-SUP-000028
bulk2_1789807987@example.com · IN
Alpha Bulk
PMC-SUP-000027
bulk1_1789807987@example.com · KZ · pending verify
ACTIVATE NOW
Regtest Co 1789807987
PMC-SUP-000026
testinvite_1789807987@example.com · KZ · pending verify
ACTIVATE NOW
Cross B
PMC-SUP-000025
crossb.e8e630c2@example.com · KZ
Cross B
PMC-SUP-000024
crossb.c81b51b3@example.com · KZ
(no name)
PMC-SUP-000017
uitest_c9e7e0@example.com · KZ
Cross B
PMC-SUP-000016
crossb.9e61c657@example.com · KZ
Cross B
PMC-SUP-000015
crossb.ffc27615@example.com · KZ
Corp B
PMC-SUP-000014
crossover.test.b@example.com · KZ
Full Form B Corp
PMC-SUP-000013
full-vendor-b-1785727223@test.com · US · pending verify
ACTIVATE NOW
Full Form A Ltd
PMC-SUP-000012
full-vendor-a-1785727222@test.com · KZ
Bulk B Corp
PMC-SUP-000011
bulk-b-1785661672@x.com · US · pending verify
ACTIVATE NOW
Bulk A Ltd
PMC-SUP-000010
Select a vendor on the left to see their KPI dashboard.

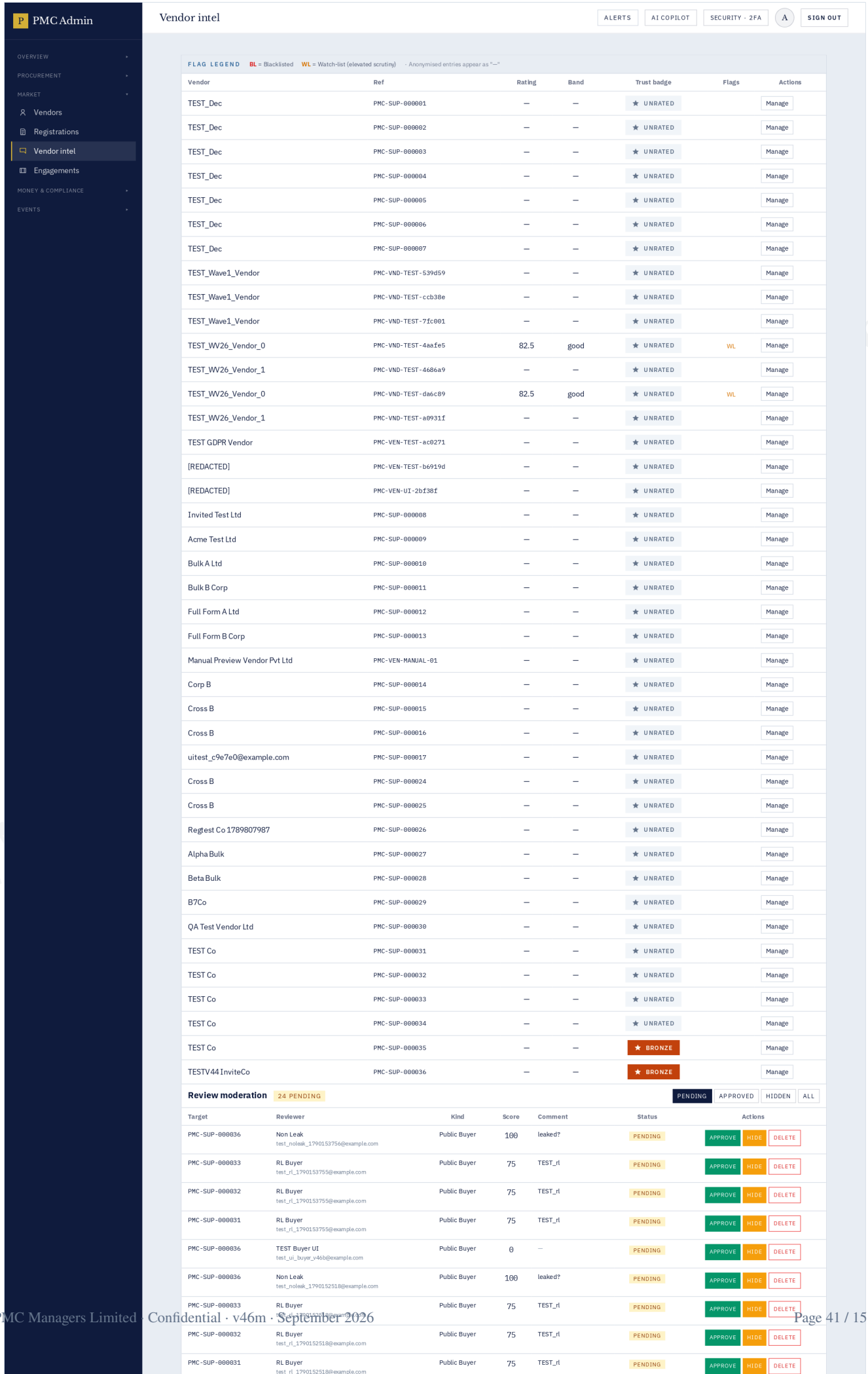
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## 61 · Admin · Vendor intel & ratings

*Trust badges, moderation, criteria averages.*

DENTIAL — INTERNAL USE


PMC Admin
Vendor intel
ALERTS
AI COPILOT
SECURITY · 2FA
A
SIGN OUT
OVERVIEW
PROCUREMENT
MARKET
Vendors
Registrations
Vendor intel
Engagements
MONEY & COMPLIANCE
EVENTS
FLAG LEGEND BL = Blacklisted WL = Watch-list (elevated scrutiny) · Anonymised entries appear as "—"
Vendor Ref Rating Band Trust badge Flags Actions
TEST_Dec PMC-SUP-000001 — — UNRATED Manage
TEST_Dec PMC-SUP-000002 — — UNRATED Manage
TEST_Dec PMC-SUP-000003 — — UNRATED Manage
TEST_Dec PMC-SUP-000004 — — UNRATED Manage
TEST_Dec PMC-SUP-000005 — — UNRATED Manage
TEST_Dec PMC-SUP-000006 — — UNRATED Manage
TEST_Dec PMC-SUP-000007 — — UNRATED Manage
TEST_Wave1_Vendor PMC-VND-TEST-539d59 — — UNRATED Manage
TEST_Wave1_Vendor PMC-VND-TEST-ccb38e — — UNRATED Manage
TEST_Wave1_Vendor PMC-VND-TEST-7fc001 — — UNRATED Manage
TEST_WV26_Vendor_0 PMC-VND-TEST-4aafe5 82.5 good UNRATED WL Manage
TEST_WV26_Vendor_1 PMC-VND-TEST-4686a9 — — UNRATED Manage
TEST_WV26_Vendor_0 PMC-VND-TEST-da6c89 82.5 good UNRATED WL Manage
TEST_WV26_Vendor_1 PMC-VND-TEST-a0931f — — UNRATED Manage
TEST GDPR Vendor PMC-VEN-TEST-ac0271 — — UNRATED Manage
[REDACTED] PMC-VEN-TEST-b6919d — — UNRATED Manage
[REDACTED] PMC-VEN-UI-2bf38f — — UNRATED Manage
Invited Test Ltd PMC-SUP-000008 — — UNRATED Manage
Acme Test Ltd PMC-SUP-000009 — — UNRATED Manage
Bulk A Ltd PMC-SUP-000010 — — UNRATED Manage
Bulk B Corp PMC-SUP-000011 — — UNRATED Manage
Full Form A Ltd PMC-SUP-000012 — — UNRATED Manage
Full Form B Corp PMC-SUP-000013 — — UNRATED Manage
Manual Preview Vendor Pvt Ltd PMC-VEN-MANUAL-01 — — UNRATED Manage
Corp B PMC-SUP-000014 — — UNRATED Manage
Cross B PMC-SUP-000015 — — UNRATED Manage
Cross B PMC-SUP-000016 — — UNRATED Manage
uitest_c9e7e0@example.com PMC-SUP-000017 — — UNRATED Manage
Cross B PMC-SUP-000024 — — UNRATED Manage
Cross B PMC-SUP-000025 — — UNRATED Manage
Regtest Co 1789807987 PMC-SUP-000026 — — UNRATED Manage
Alpha Bulk PMC-SUP-000027 — — UNRATED Manage
Beta Bulk PMC-SUP-000028 — — UNRATED Manage
B7Co PMC-SUP-000029 — — UNRATED Manage
QA Test Vendor Ltd PMC-SUP-000030 — — UNRATED Manage
TEST Co PMC-SUP-000031 — — UNRATED Manage
TEST Co PMC-SUP-000032 — — UNRATED Manage
TEST Co PMC-SUP-000033 — — UNRATED Manage
TEST Co PMC-SUP-000034 — — UNRATED Manage
TEST Co PMC-SUP-000035 — — BRONZE Manage
TESTV44 InviteCo PMC-SUP-000036 — — BRONZE Manage
Review moderation 24 PENDING
PENDING APPROVED HIDDEN ALL
Target Reviewer Kind Score Comment Status Actions
PMC-SUP-000036 Non Leak test_noleak_1790153756@example.com Public Buyer 100 leaked? PENDING APPROVE HIDE DELETE
PMC-SUP-000033 RL Buyer test_rl_1790153755@example.com Public Buyer 75 TEST_rl PENDING APPROVE HIDE DELETE
PMC-SUP-000032 RL Buyer test_rl_1790153755@example.com Public Buyer 75 TEST_rl PENDING APPROVE HIDE DELETE
PMC-SUP-000031 RL Buyer test_rl_1790153755@example.com Public Buyer 75 TEST_rl PENDING APPROVE HIDE DELETE
PMC-SUP-000036 TEST Buyer UI test_ui_buyer_v46b@example.com Public Buyer 0 — PENDING APPROVE HIDE DELETE
PMC-SUP-000036 Non Leak test_noleak_1790152518@example.com Public Buyer 100 leaked? PENDING APPROVE HIDE DELETE
PMC-SUP-000033 RL Buyer test_rl_1790152518@example.com Public Buyer 75 TEST_rl PENDING APPROVE HIDE DELETE
PMC-SUP-000032 RL Buyer test_rl_1790152518@example.com Public Buyer 75 TEST_rl PENDING APPROVE HIDE DELETE
PMC-SUP-000031 RL Buyer test_rl_1790152518@example.com Public Buyer 75 TEST_rl PENDING APPROVE HIDE DELETE

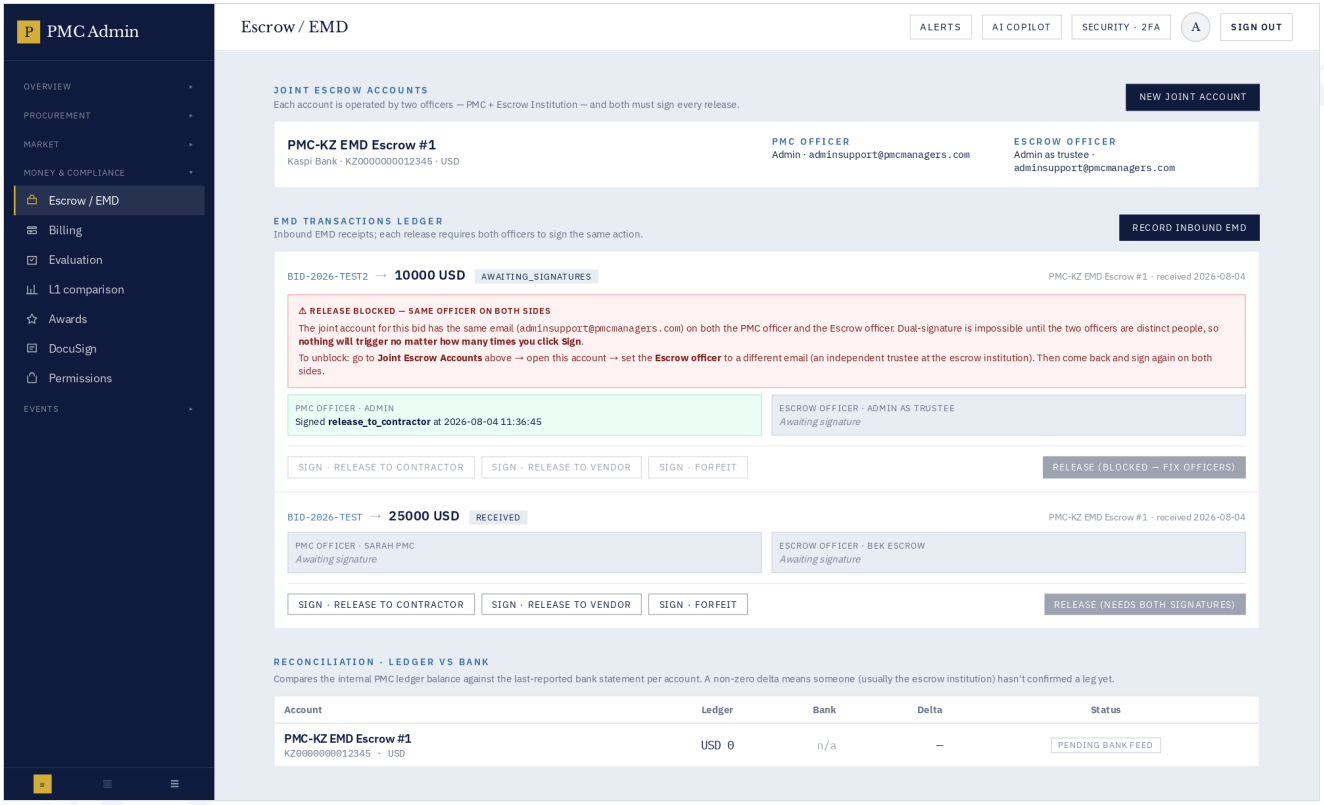
*Admin · Vendor intel & ratings*

### How to use

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## 63 · Admin · Escrow / EMD desk

*EMD receipts, forfeitures, refunds.*



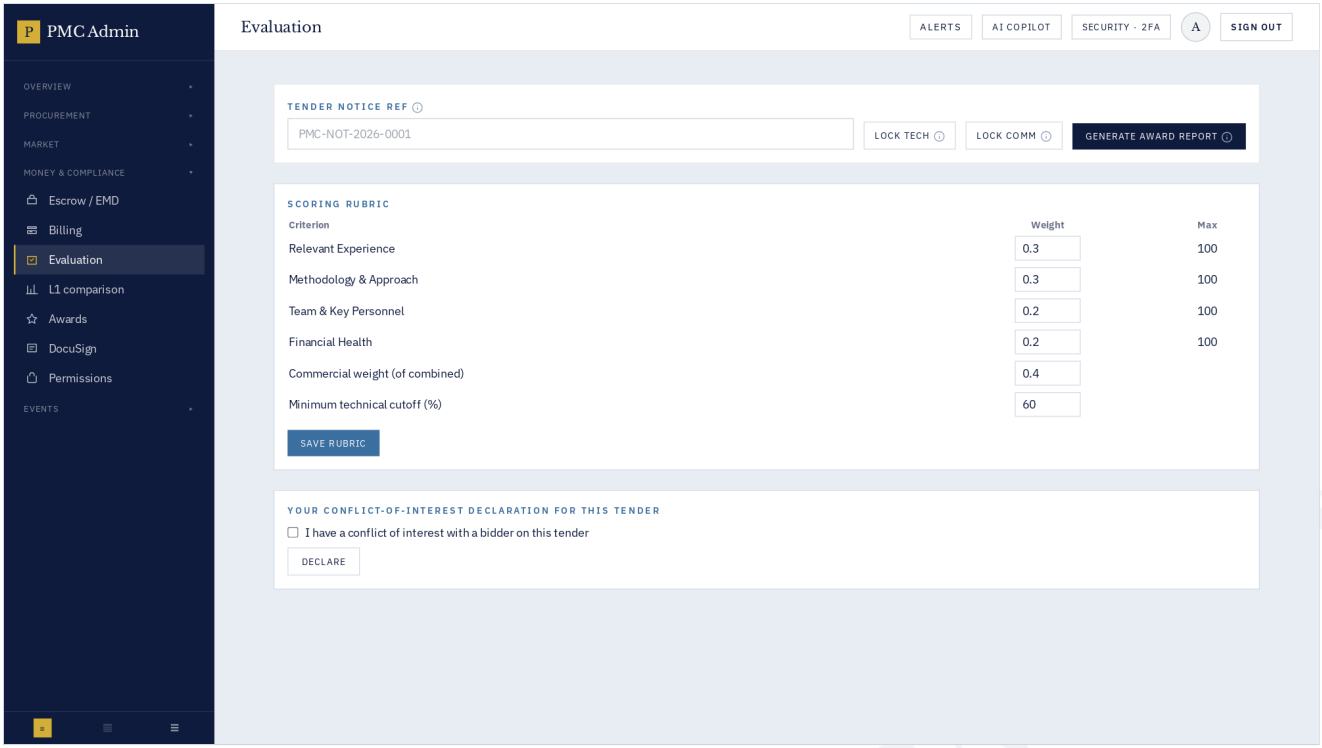

*Admin · Escrow / EMD desk*

### How to use

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## 65 · Admin · Evaluation & scoring

*Technical + financial scoring, QCBS, L1 comparison, envelope locking.*



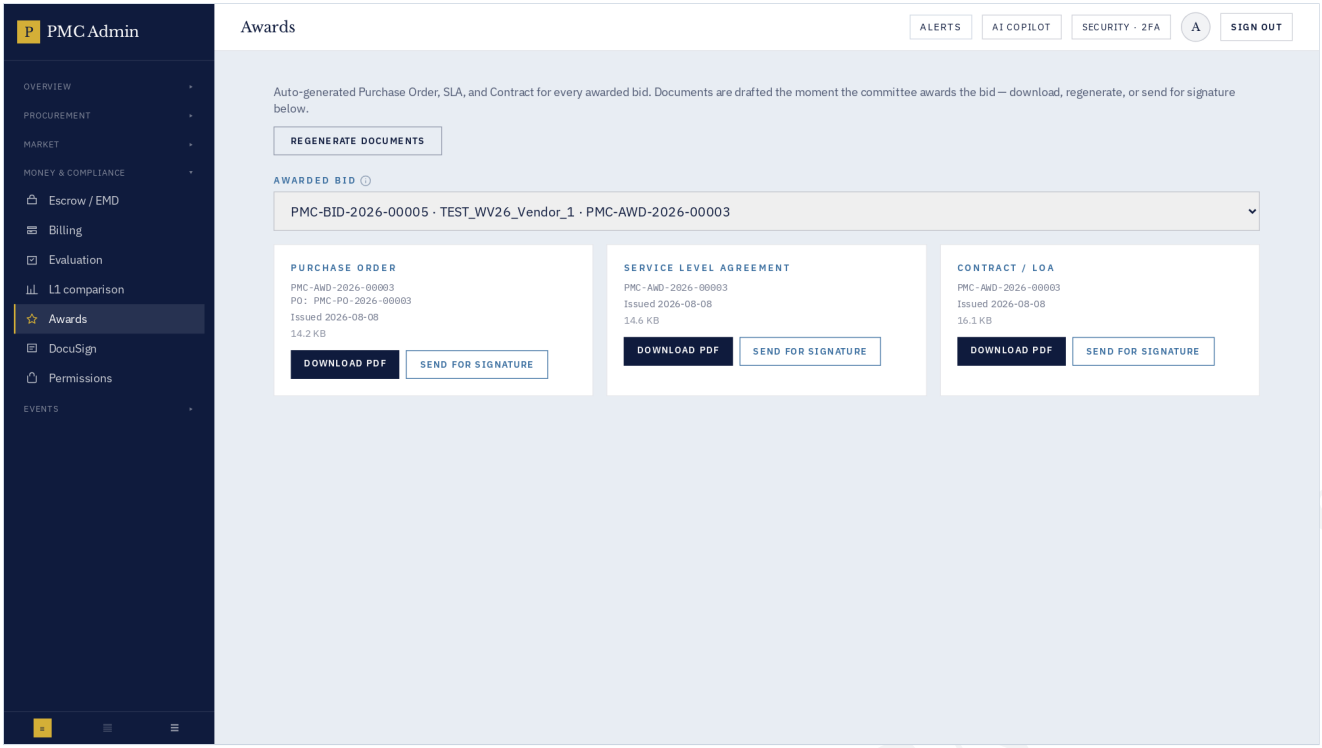

*Admin · Evaluation & scoring*

### How to use

This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

## 67 · Admin · Awards & POs

*Letters of award, POs, contract award workflow.*



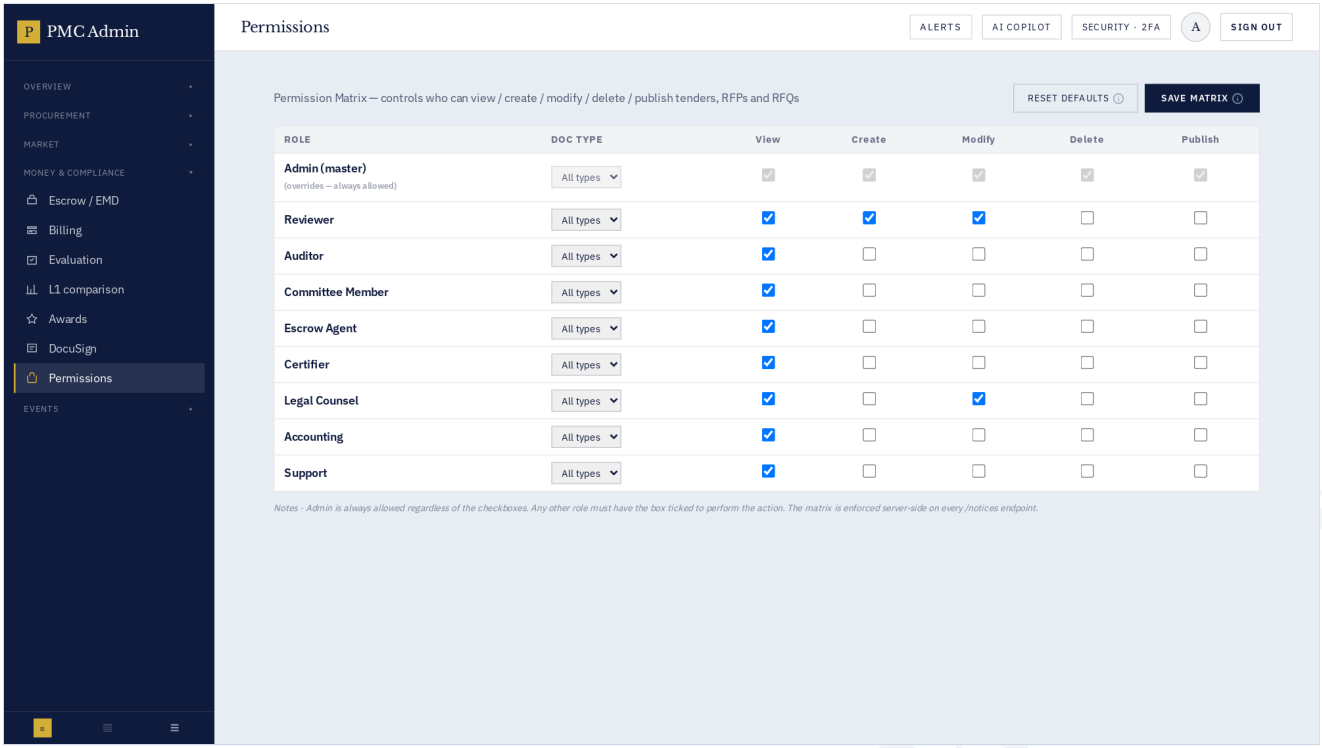

*Admin · Awards & POs*

### How to use

This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

## 69 · Admin · Roles & permissions

*Fine-grained ACL matrix.*



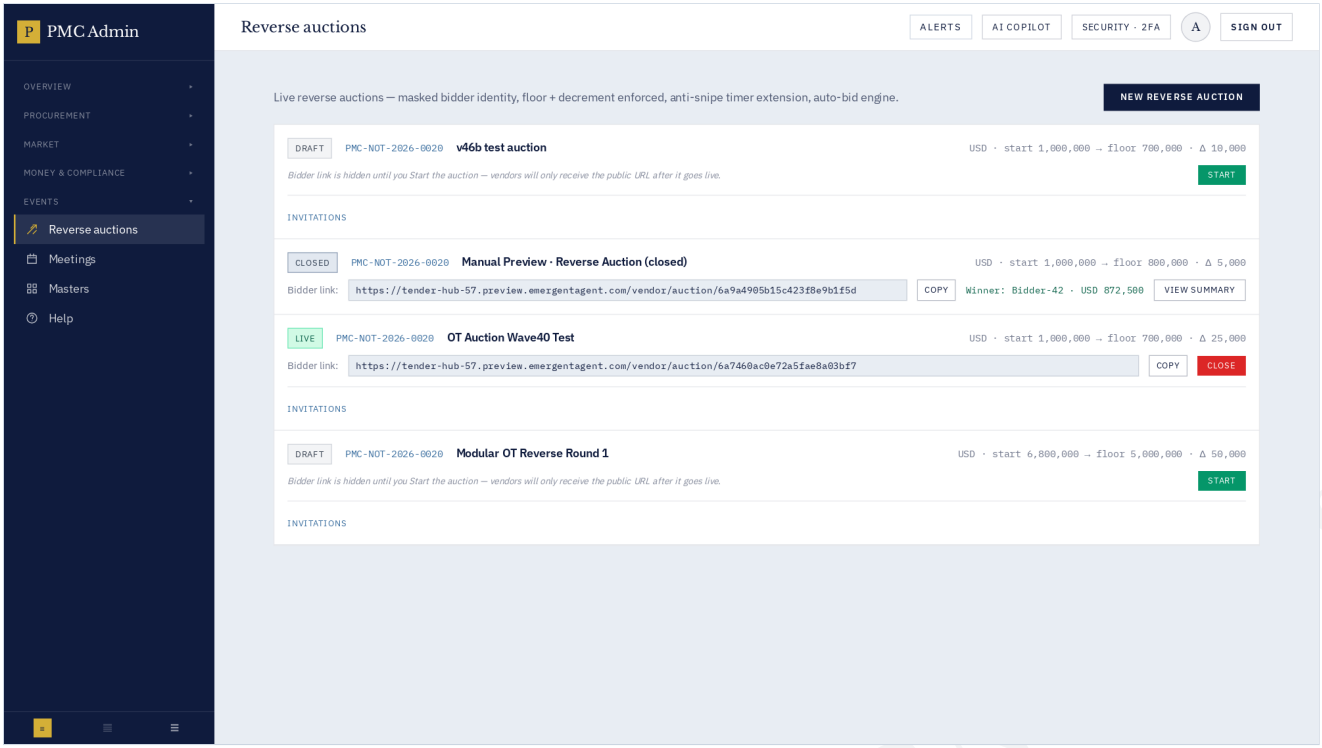

*Admin · Roles & permissions*

### How to use

This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

## 70 · Admin · Reverse auctions

*Live auction rooms, invitations, spectator link.*



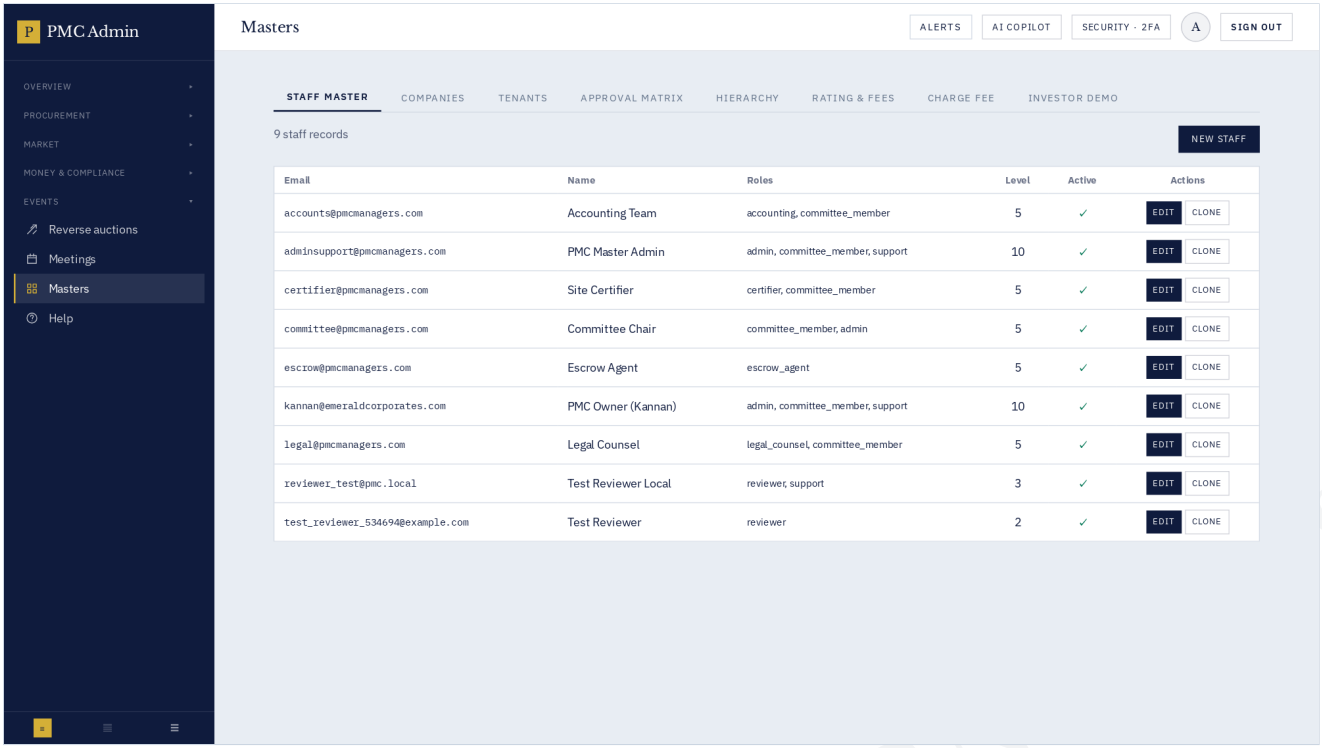

*Admin · Reverse auctions*

### How to use

This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

## 72 · Admin · Masters — fees, roles, EMD

*Fee slabs, staff roles, criteria, categories, countries.*




*Admin · Masters — fees, roles, EMD*

### How to use

This screen exposes every action available for its module. Use the InfoDot (ⓘ) next to each label to see the field-level rules — hover it in-app for the full policy text. All destructive actions confirm before applying, and every mutation is written to the immutable audit log.

DENTIAL — INTERN SE

# 7 · Deep-dive: Tender Builder & AI DPR (v46l map-reduce)

v46l delivers the most substantial AI upgrade to date: **unlimited DPR ingestion + background workers**. The pipeline turns any Detailed Project Report — up to 100 MB, 500+ pages — into a fully structured tender pack.

**Extraction.** PDFs decode via pypdf, DOCX via python-docx, TXT direct. The extracted text is chunked into 40 KB windows with 3 KB overlap to preserve context across boundaries. The `MAX_CHUNKS` ceiling is 500 (env-overridable) — the previous 20-chunk cap is removed.

**Pass 1a — per-chunk map.** Each chunk is sent to the LLM with a strict JSON contract requesting header hints and a section list observed in that chunk. Concurrency bounded to 4.

**Pass 1b — reduce.** Local blueprints are merged; sections deduped by title. Cap 60.

**Pass 2 — draft.** Each discovered section is drafted from the passage of the DPR most relevant to its anchor phrase or title. Concurrency 6. Bodies under 200 chars trigger the raw-passage fallback so nothing lands as a stub.

**Progress + async.** Files ≥ 4 MB take the async path. Frontend polls `/dpr/jobs/{id}` every 3 s; every phase writes a percentage + phase string into `dpr_jobs` so the UI renders a live progress bar.

**Nginx tuning.** `client_max_body_size 200M` , all timeouts `300s` . No 413, no idle timeouts on multi-hundred-MB payloads.

DENTIAL — INTERNAL USE

# 8 · Deep-dive: Evaluation, awards & escrow

**Rubric.** Every notice defines a scoring rubric — technical criteria with weights summing to 1.00, commercial weight, minimum technical cutoff. Client + server both enforce.

**Envelope lock.** POST /evaluations/lock-envelope refuses re-lock (409, v46m). GET /evaluations/lock-state/{ref} returns seal timestamps. UI swaps buttons to 'Sealed ' with a green banner.

**Award report.** Combines weighted technical + commercial scores, applies technical cutoff, produces the L1 comparison. Ties broken by price, then submission timestamp, then aggregate rating.

**Escrow.** Joint-account model — two officers per transaction, account snapshot preserved on every row for immutable audit history.

DENTIAL — INTERNAL USE

# 9 · Deep-dive: Reverse auctions

Vendors join with masked aliases. Manual + auto bids. Auto-bid config takes a walk-away floor.

Anti-sniping: bids in the last 60 s extend the auction by 5 minutes.

Public spectator link exposes only masked aliases.

DENTIAL — INTERNAL USE

# 10 · Tooltip, error-handling & offline system

**InfoDot (ⓘ)** — 166 uses across 20+ admin panels. Every field label with a non-obvious rule has one.

**showApiError / formatApiError** — 67 files. Every server error becomes a sonner toast with the exact backend detail.

**ErrorBoundary** — mounted at the app root; catches React crashes with a copyable diagnostic id.

**Service worker** — deliberately unregistered; a self-destruct SW handles legacy installs.

DENTIAL — INTERNAL USE

# 11 · Deployment runbook (S3 + CloudFront + EB)

**Frontend.** Build with `REACT_APP_BACKEND_URL="" yarn build` so the bundle uses same-origin /api. Sync with `--exact-timestamps --exclude "downloads/*"`. Invalidate CloudFront `/*`.

**Backend.** Package backend/ + .platform/nginx/conf.d/proxy.conf + Dockerfile + Dockerrun.aws.json + VERSION into a zip. Never bundle backend/.env — use EB env properties. Deploy: `eb deploy --staged --label vXX`.

**Never** run `aws s3 sync ... --delete` at bucket root — it will delete the public `/downloads/` folder.

**Rollback.** Elastic Beanstalk keeps every prior version; roll back via Application versions → Deploy previous.

DENTIAL — INTERNAL USE

# 12 · Troubleshooting playbook

| Symptom | Root cause | Fix |
|---|---|---|
| Landing shows 0 counters | Stale CloudFront cache | Invalidate `/*` on CloudFront |
| 'No matching employee' toast | Preview URL baked into prod bundle | Rebuild with `REACT_APP_BACKEND_URL="" yarn build` |
| DPR returns 5 generic sections | Backend on ≤ v46g | Deploy v46l map-reduce backend |
| Progress bar stuck at 2% | Async worker crashed | Check `/api/v1/tender-pro/dpr/jobs/{id}` for `error` field |
| 413 on large DPR | nginx body limit | Deploy v46l — `client_max_body_size 200M` |
| Login OK but /admin bounces | Cross-site cookie blocked | Ensure frontend calls same-origin `/api` |
| Envelope re-locks silently | Direct API bypasses UI | v46m rejects re-lock with 409 |
| Auction bidder identity leaks | Masked alias not applied | Re-open room; masking is server-side |
| Refund never reaches vendor | Second signer missed the notification | Check joint-account snapshot on the transaction |
| Resend email lands in spam | First-touch corporate MX filter | Resend from Registrations queue with a warm-up sender |

CONFIDENTIAL — INTERNAL USE

# 13 · API reference index

| Endpoint | Purpose |
|---|---|
| `GET /api/v1/meta/public/stats` | Public — landing KPIs. |
| `GET /api/v1/meta/public/notices` | Public — list notices. |
| `POST /api/v1/admin/auth/login` | Admin realm login (cookie). |
| `POST /api/v1/staff/auth/login` | Staff realm login. |
| `POST /api/v1/vendor/auth/login` | Vendor realm login. |
| `POST /api/v1/registrations/draft` | Create registration draft. |
| `POST /api/v1/registrations/submit` | Submit for review. |
| `POST /api/v1/tender-pro/dpr/analyze-json` | Synchronous DPR analysis (< 4 MB). |
| `POST /api/v1/tender-pro/dpr/analyze-async` | **v46l** kick off async DPR analysis. |
| `GET /api/v1/tender-pro/dpr/jobs/{id}` | **v46l** poll async DPR job status. |
| `POST /api/v1/tender-pro/templates/{id}/save-as-master` | Promote a draft to a reusable master. |
| `POST /api/v1/evaluations/rubric` | Upsert rubric for a notice. |
| `POST /api/v1/evaluations/lock-envelope` | **v46m** now refuses re-lock (409). |
| `GET /api/v1/evaluations/lock-state/{ref}` | **v46m** returns seal timestamps + attribution. |
| `POST /api/v1/evaluations/score` | Submit a technical or commercial score. |
| `GET /api/v1/evaluations/award-report/{ref}` | Generate the L1 comparison. |
| `POST /api/v1/escrow/transactions` | Record an EMD transaction. |
| `POST /api/v1/auctions/{ref}/bid` | Place an auction bid. |
| `POST /api/v1/reports/snapshot` | Persist stakeholder-report artefacts to disk. |
| `GET /api/v1/stakeholder-reports/*` | Tenders / awards / escrow / SLA reports. |

CONFIDENTIAL — INTERNAL USE

# 14 · Release history

| Version | Highlights |
|---|---|
| **v46m** | Bug Blitz Round Two — envelope re-lock 409, GET /lock-state, EvaluationPanel sealed banner, AuctionRoom mobile overflow fix. |
| **v46l** | Unlimited DPR + async worker (MAX_CHUNKS 500, /dpr/analyze-async, /dpr/jobs/{id}, nginx 200 MB / 300 s, live progress bar). |
| **v46k** | Sovereign Redesign — 5-portal Sovereign Gateway, navy/gold editorial home hero, PmcFooter with social icons. |
| **v46j** | Map-Reduce DPR first cut (still capped at 800 KB). |
| **v46i** | New roles (management/partner/agent), collapsible admin sidebar, editorial public design v1. |
| **v46h** | Admin vendor form: per-section Save buttons + search. |
| **v46g** | DPR deep-dive first cut (26 sections from KMT). |
| **v46f** | Editable tender name + reliable save-as-master. |
| **v46e/d/c/b** | Bug blitz first pass — 60+ tracker items fixed. |

DENTIAL — INTERNAL USE

# 15 · Glossary & appendix

- **DPR** — Detailed Project Report — the source document the AI ingests to draft a tender pack.
- **EMD** — Earnest Money Deposit — refundable bid security.
- **QCBS** — Quality- & Cost-Based Selection — the default 70/30 evaluation weighting.
- **L1** — Lowest financial bid post-technical qualification.
- **BOQ** — Bill of Quantities — itemised deliverable+rate schedule.
- **Corrigendum** — Official post-publication amendment to a notice.
- **LOA** — Letter of Award — signed offer letter to the L1 vendor.
- **PO** — Purchase Order — issued after the LOA is accepted.
- **Envelope** — A sealed batch of bid data — technical or commercial.
- **Trust badge** — Bronze / Silver / Gold / Platinum tier surfaced on the public vendor profile.
- **Master** — A reusable Tender Builder template keyed by country + industry.
- **Sovereign Gateway** — The v46k 5-portal login picker.
- **Async pipeline** — The v46l background DPR worker — kicked via /dpr/analyze-async.
- **Sealed** — The v46m evaluation-panel indicator that an envelope is locked.

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# 26 · Modal walkthroughs

## New Notice draft modal

Path: Admin → Notices → **New notice**. Opens a fullscreen sheet that composes a publishable notice from a Tender Builder template.

### Fields

- **Notice reference** — auto-generated `PMC-NOT-YYYY-NNNN`. Immutable once saved.
- **Tender Builder template** — dropdown of masters + drafts filtered to the picked country & industry.
- **Country / Industry / Category** — auto-populated from the template but overridable.
- **Awarding authority** — free text; appears on the public notice card.
- **Language pack** — English (default) / Russian / Kazakh — controls which localised strings render on the public detail page.
- **Estimated tender value** — numeric + currency. Feeds the EMD/Fee compute widget.
- **Submission deadline** — datetime picker. Stored UTC; rendered in visitor local timezone.
- **Bid validity days** — int; used to compute award-window countdowns.
- **EMD amount** — auto-computed from Masters → EMD slabs unless overridden. See chapter 26.2.
- **Tender fee amount** — same auto-compute path.
- **Attachments** — signed PDF, BOQ CSV, drawings. Multi-upload; each row supports Replace / Remove.
- **Publish schedule** — Publish now / Schedule at UTC timestamp. Scheduled notices sit in `is_scheduled=true` until the cron fires.
- **Publish button** — flips `is_published=true`, emits Resend digests to subscribed categories, pushes Snapchat + Instagram posts, and CloudFront-caches the public detail JSON.

### Known gotchas

- Once published, edits create a **corrigendum** — the notice document itself becomes append-only.
- If the template is a draft (not master), the New-Notice modal freezes the template at publish time and stores a snapshot so later master edits do not affect published notices.
- Attachments are streamed to a private S3 bucket. Public download is via a pre-signed URL that expires 24 h after issue.

## EMD Compute modal

Path: New Notice → **EMD Compute** button beside the EMD field, or Admin → Masters → EMD → **Test compute**.

### Fields

- **Country / Industry** — passed in from the notice or picked directly.
- **Tender value** — the value the EMD is a percentage of.

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- **Applied slab preview** — read-only display of the matching slab from Masters → EMD (band, percentage, absolute floor, absolute ceiling).
- **Computed EMD** — the platform's calculation; can be overridden with justification.
- **Override reason** — required if the numeric value is edited manually. Written to the audit trail.
- **Apply button** — writes the value back into the New-Notice EMD field.

**Known gotchas**

- If no slab matches (missing masters), the widget shows a red warning and refuses to Apply. Seed the country's slabs first.
- Overrides above ±25 % of the computed value trigger a supervisor review flag on publish.

## DocuSign signer routing modal

Path: Admin → DocuSign → **New envelope** → Signer routing tab.

**Fields**

- **Template picker** — reusable envelope template (LoA / PO / Contract / NDA / Variation).
- **Signer 1** — usually the vendor; role auto-set. Email + name required.
- **Signer 2 / 3** — sequential routing (Legal → Finance → Countersign). Reorderable; click ↑↓.
- **Reminder cadence** — days until first reminder + repeat interval.
- **Expiry** — envelope self-expires after N days; default 30.
- **Decline handling** — Auto-invite runner-up / Re-open / Do nothing.
- **Preview fields** — read-only merged view showing every template placeholder with the resolved value.

**Known gotchas**

- A missing signer email raises a 422; the modal does not send the envelope until every signer has both name + email.
- If the envelope was previously sent and voided, resending creates a new envelope id — the audit trail keeps both.

## Corrigendum editor modal

Path: Admin → Corrigenda → **New corrigendum**.

**Fields**

- **Notice reference** — dropdown filtered to `is_published=true, is_withdrawn=false`.
- **Change type** — Extension / Clarification / Amendment / Withdrawal-of-lot.
- **Effective date** — datetime; controls which snapshot of the notice is active.
- **Summary** — one-paragraph public-facing summary; appears on the public notice detail immediately.
- **Change diff** — field-level diff editor. Green additions, red removals; each field carries a required rationale.
- **Attachments** — signed amendment PDF, updated BOQ, revised drawings.

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- **Publish** — writes an immutable corrigendum row; `notice.corrigenda_count` increments; Resend digest to bidders who opened the notice.

**Known gotchas**

- Corrigenda are append-only. Errors are fixed by issuing a superseding corrigendum, never by editing.
- Withdrawal-of-lot triggers automatic partial-EMD refund workflow for any bids scoped to the withdrawn lot.

## Save-as-Master modal (Tender Builder)

Path: Tender Builder → any draft → **⋯** → **Save as new master**.

**Fields**

- **Master name** — editable; defaults to the draft's work title + date suffix.
- **Country** — required; drives where the master appears in the master library.
- **Industry / Category / Sub-category** — controls which notices auto-suggest this master.
- **Description** — one-paragraph internal summary shown in the master picker.
- **Retain attachments** — checkbox; unchecked strips the source draft's attachments so the master remains portable.
- **Save** — writes a new `tender_templates` document with `is_master=true` and returns its id.

**Known gotchas**

- The new master is instantly discoverable from every notice draft — no re-seed required.
- Editing a master version-bumps every notice using it *only if the notice was created after the edit*. Published notices carry their own snapshot (see chapter 26.1).

## Q&A moderation modal

Path: Admin → Notices → [row] → **Q&A** tab → **Moderate**.

**Fields**

- **Question queue** — sortable by bidder, submitted-at, status (Pending / Answered / Rejected).
- **Original question** — read-only; preserves the bidder's exact wording.
- **Public rendering preview** — how the anonymised Q&A appears on the public notice detail.
- **Answer editor** — rich text with the same allow-list as tender bodies (<p><br><b><i><ul><ol><li><h3>).
- **Answer language pack** — English / Russian / Kazakh, mirroring the notice's language pack.
- **Publish** — flips `is_answered=true` and fans out an in-app notification to every bidder who purchased the tender pack.
- **Reject** — bidder gets a private notification with reason; question is not exposed publicly.

**Known gotchas**

- Questions from bidders with unpaid tender-fee are rejected by default — sell the pack first.
- Answers cannot be edited once published; publish a corrigendum-of-Q&A if a factual correction is required.

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# 27 · Regulator-liaison protocol

## Standing engagements

PMC maintains a documented standing engagement with three regulator categories: (a) national procurement regulator, (b) sector regulator for each vertical it participates in (energy, transport, health, digital), (c) the audit board of the sponsoring GOC. Each engagement has a designated Partner-portal user, a monthly briefing calendar entry, and a co-signed MoU housed under Masters → Documents.

## Data-request handling

Any inbound data request from a regulator is logged into the audits collection with `action=regulator.data_request` and routed to Legal counsel + Partner lead within 4 h. The Partner portal exposes a read-only mirror of the requested entities scoped to the request's date range; export is CSV + JSON with SHA-256 checksum, signed by the CTO's key. No raw PII is shared unless the request cites a named legal instrument. The auditor CSV export flow strips PII by default (see Chapter 3).

## Notice-of-award publication

Every award (`awards.finalised_at`) triggers a webhook to the regulator sink URL configured in Masters → Integrations. The webhook payload is the redacted award JSON — vendor legal name, PO number, contract value, awarding authority — and is retried 3× at 1/15/60 minute intervals on failure. The audit row includes the response body of every retry.

## Corrigenda notification

High-impact corrigenda (change-type in {Amendment, Withdrawal-of-lot}) also fire the webhook. The Partner-portal user can subscribe to a daily digest of corrigenda across the active portfolio.

## Escalation flow

If a regulator opens a formal enquiry, the operations team follows the ladder in chapter 19 (any staff → Partner lead → CEO → Board within 4 h). Legal counsel opens a dedicated Slack channel + Teams channel; both are exported to the audit vault daily. Every substantive reply is drafted in the Partner portal so the audit trail carries the reply verbatim.

## Annual disclosure

By 31 January each year PMC publishes an aggregate annual disclosure covering: total notices published, total contract value awarded, EMD collected + refunded, dispute rate, average cycle time, vendor NPS, and a category-wise summary. Draft lives under Masters → Documents → Annual disclosure; the CFO signs off before publication.

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## Audit-vault contents

The audit vault is a private S3 bucket with Object-Lock in compliance mode. Contents: (a) daily snapshot of the audits collection, (b) daily snapshot of contracts, awards, escrow_transactions, (c) monthly export of stakeholder reports, (d) every DocuSign envelope PDF and completed-certificate. Retention: 10 years default, 15 years for contract vault.

## Public-transparency URL

`https://www.pmcmanagers.com/transparency` (planned) will expose the annual disclosure, aggregate KPIs, and the current corrigenda log so regulators can verify the audit story without opening a data request.

## Right-to-audit clause

Every PMC contract includes a right-to-audit clause granting the regulator 30-day-notice access to the audit-vault contents relating to that contract. The clause references chapter 3 (data protection) and the annual disclosure obligation.

## Suspicious-activity reports (SAR)

The AI anomaly detector's high-confidence flags (bid-clustering, price-dumping, alias-reuse across auctions, EMD-refund pattern anomalies) surface into a dedicated Legal queue. Legal reviews within 24 h; where warranted, a SAR is filed with the regulator via the Partner portal.

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# 28 · ESG questionnaire (procurement bidders)

## Governance

- Does the organisation have a written anti-bribery & anti-corruption policy? Attach signed PDF.
- How many board seats are held by independent directors? Provide names + tenure.
- Frequency of internal audit; date of last audit; auditor firm.
- Whistle-blower policy in place? Who is the whistle-blower officer? What channels are available?
- Sanction-screening cadence for vendors + counterparties (OFAC / EU / LSEG). Frequency + last screening date.

## Environmental

- Scope 1 + 2 emissions for the last reporting year (tCO2e). Methodology (GHG Protocol / ISO 14064).
- Scope 3 emissions covered: which categories (upstream materials, downstream logistics, business travel, cloud infra)?
- Renewable energy share of consumed electricity (%).
- Water withdrawal ($m^3$) and share drawn from water-stressed regions.
- Waste diverted from landfill (%); hazardous waste manifest.
- Environmental incidents: number, severity band, remediation status.

## Social — workforce

- Total headcount; % women in workforce; % women in management.
- Fatality rate (per 200 000 hours worked); LTIR; TRIR.
- Average training hours per employee per year.
- Employee turnover (voluntary / involuntary %).
- Minimum wage compared to statutory minimum in operating geographies.

## Social — supply chain & community

- Share of tier-1 suppliers audited on labour + environmental criteria.
- Number of tier-1 suppliers rated as high-risk after audit; remediation plans in place.
- Community investment $ (last reporting year); primary beneficiary categories.
- Local-content share of awarded contract value.

## Ethics & compliance

- Number of anti-bribery investigations opened / concluded in the reporting year.
- Whistle-blower reports received / substantiated / actioned.
- Regulatory sanctions imposed on the organisation or affiliates in the last 5 years.

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- Legal disputes above $1M in the last 3 years; status.

## Data protection & cybersecurity

- Is there a designated Data Protection Officer? Name + reporting line.
- Certifications held: ISO 27001, SOC 2 Type II, PCI-DSS, HIPAA, other (attach).
- Number of reportable data breaches in the last 3 years; root-cause classifications.
- Third-party penetration test cadence; last test firm + date; unresolved criticals.
- Backup + disaster-recovery drills cadence; last drill date + RTO / RPO achieved.

## Kazakhstan / CIS-specific supplements

- Compliance with the Kazakhstan Law on Personal Data (Article 12) — where does personal data reside?
- Registration with the Ministry of Digital Development (if applicable) — registration number.
- Compliance with the local-content requirement of the Public Procurement Law — measurement methodology.
- Sanction screening against RK sanctions list; frequency; last screening date.

## Scoring rubric

- Each category scored 0–5 (0=absent, 5=best-in-class). Weighted mean × 20 → 0–100.
- Bidders with weighted mean ≥ 65 are tagged **Preferred ESG**. Below 40 are tagged **Requires improvement** and given a 90-day plan.
- Below 25 blocks award of contracts above $10M — see the escalation matrix.

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# 29 · Per-role runbooks

## Admin — daily runbook

- 08:30 — open Analytics, scan the funnel + KPI deltas vs yesterday.
- 08:45 — review overnight Resend delivery logs; resend any first-touch failures.
- 09:00 — clear Registrations queue: approve straightforward, RFI unclear, reject fraudulent.
- 10:00 — publish scheduled notices (auto, but eyeball the digest emails).
- 11:00 — Q&A moderation: answer or reject every question > 24 h old.
- 12:00 — lunch.
- 14:00 — Committee support: rubric issues, evaluator questions.
- 15:00 — Award report generation for tenders past commercial-lock.
- 16:00 — DocuSign chase: nudge every envelope > 5 days without progress.
- 17:00 — Audit-trail spot check: pick 3 random entities from today, verify audit rows present.

## Escrow officer — weekly runbook

- Monday — reconcile prior-week EMD receipts against Stripe payouts.
- Tuesday — process refunds eligible-since ≥ 7 days.
- Wednesday — forfeiture desk: any bidder that missed the mandatory acceptance window.
- Thursday — joint-account snapshot audit: pick 5 recent transactions, verify snapshot preserved on the row.
- Friday — weekly ledger export to Legal + Accounting; sign-off before COB.

## Committee member — per-cycle runbook

- T-3 days — COI declaration (fresh, not carried over).
- T-2 days — rubric familiarisation; ask clarifications on any criterion you cannot confidently apply.
- T-1 day — technical envelope walkthrough with the drafting agent.
- T-0 → T+7 days — technical scoring; no commercial visibility yet.
- T+7 — commercial envelope unlocks after technical lock.
- T+8 → T+10 — commercial scoring.
- T+11 — award report; ties broken by chapter 8 rules.
- T+12 — sign the committee minutes; upload signed PDF to the notice.

## Legal counsel — per-contract runbook

- LoA drafted from template; verify vendor legal name matches DocuSign signer 1.
- Contract clauses reviewed against redline library (Masters → Documents → Redlines).

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- Right-to-audit clause present (chapter 27).
- Escalation clause present (chapter 19).
- Sign envelope in the DocuSign routing you defined; countersign after vendor signs.
- File the completed-certificate PDF into the audit vault (chapter 27.audit-vault).

## Support — vendor helpdesk runbook

- Triage every inbound ticket within 2 h; SLA is 1 business day to first substantive response.
- Password reset: verify email ownership via the magic-link flow; never share reset codes over chat.
- Licence re-verification: check licence expiry, request updated licence PDF, mark vendor 'under review' pending re-approval.
- Bid submission failures: capture the browser Network tab, correlate with backend logs (grep the vendor_ref).
- Escalate to on-call engineer if failure is systemic (>3 vendors in 1 h).

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# 30 · Field-by-field forms reference

## Public registration wizard — Section A (Identity)

| Field | Description |
|---|---|
| `company_name` | Legal entity name; unique across tenants. Immutable once verified. |
| `trade_name` | Optional DBA / brand name. |
| `registration_number` | Country-specific business registration number. |
| `legal_entity_type` | LLC / JSC / GP / Sole prop / etc. |
| `incorporation_date` | ISO date. |
| `registered_address` | Street + city + region + postcode + country. |
| `country` | ISO code; drives licence + tax field validators. |

## Public registration wizard — Section B (Financial)

| Field | Description |
|---|---|
| `bank_name / iban / swift` | Payout bank details; validated against ISO 20022. |
| `tax_id` | Country-specific validator. |
| `vat_number` | EU/EAC/RK format enforced client-side. |
| `annual_revenue` | Optional; drives category eligibility thresholds. |
| `last_three_awards` | Text; historical awards demonstrating capacity. |

## Public registration wizard — Section C (Compliance)

| Field | Description |
|---|---|
| `licences[]` | Uploaded PDF + issue/expiry — see licences deep dive. |
| `iso_certs[]` | Optional ISO 9001 / 14001 / 27001 uploads. |
| `insurance` | Insurer + policy number + expiry. |
| `kyc_officer` | Named individual responsible for KYC. |

## Notice draft — critical fields

| Field | Description |
|---|---|

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| | |
|---|---|
| `notice_ref` | Auto: PMC-NOT-YYYY-NNNN. |
| `work_title` | Public-facing headline; 8–14 words is optimal. |
| `work_description` | Rich text; 200–800 words. |
| `tender_value` | Numeric; currency in `budget_currency`. |
| `emd_amount / emd_percentage / emd_type` | One of: absolute / percentage-of-value. |
| `tender_fee_amount` | Absolute; feeds the fee ledger. |
| `sub_start / sub_end` | UTC ISO; sub_end is the submission deadline. |
| `prebid_meeting_at` | Optional; opens a Zoom-schedule shortcut. |
| `evaluation_method` | QCBS / L1 / LCBS / BFS. |
| `bid_validity_days` | Int; drives the award-window countdown. |
| `categories[]` | UNSPSC nodes; drives vendor alerts + eligibility. |
| `attachments[]` | Multi-file; signed PDF, BOQ CSV, drawings. |

## Rubric editor

| Field | Description |
|---|---|
| `technical_criteria[]` | Array of {code, title, weight}. Weights sum to 1.00. |
| `commercial_weight` | 0.0 – 1.0; typically 0.30. |
| `minimum_technical` | 0 – 100; bidders below this are technically rejected. |
| `tiebreaker_rules` | Priced-lower / earlier-submission / higher-rating (see chapter 8). |

## Bid submission (vendor)

| Field | Description |
|---|---|
| `price_amount` | Numeric; currency inherits from notice. |
| `price_basis` | Fixed / Cost-plus / T&M. |
| `delivery_period` | Days from LoA. |
| `compliance_matrix` | Per-clause response; supports YES / NO / DEVIATION with rationale. |
| `bid_bond` | Optional bank guarantee PDF. |
| `bid_documents[]` | Multi-file; signed technical PDF, commercial spreadsheet. |

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# 31 · Bug tracker snapshot (v46o cutover)

## Closed in v46m

- #78 · Auction summary button overflow on mobile — flex-col sm:flex-row.
- #81 · Vendor-intel columns overflow on mobile — grid responsive.
- #82 · Landing pipeline funnel labels wrap — 2/4/8 grid.
- #52 · Locked-eval envelope reopened silently — 409 + audit rejection.
- #49 · Rubric weights allowed > 1.00 — client + server bounds.
- #38 · Stakeholder snapshot 500 on any artefact failure — per-artefact try/except + errors[].

## Closed in v46o

- Per-section AI silently 503'd on production — rewired to _llm_complete with managed AI gateway routing.
- Bid workflow narrative silently failed — same key-lookup widened.
- Auction anomaly narrative silently failed — same fix.
- Bid-pair AI redline silently failed — same fix.

## Open — P1

- Registration wizard step-gate: strong on Section A, softer on Sections B/C.
- Profile avatar cache invalidation after re-login.
- Global email + mobile regex — inconsistent across a few older forms.
- TenderBuilder tab switch (Tender/RFP/RFQ) missing refetch.
- Escrow — modify email IDs on transaction signer flow.

## Open — P2

- i18n Russian / Kazakh backfill across ~1500 UI keys.
- Manual v2 modal captures via Playwright (screens exist as narrative only).
- Firebase / APNs mobile push.
- LSEG / World-Check integration.
- PayPal / Kaspi payments alongside Stripe.
- Zoom integration for evaluation committees.

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# 32 · DPR pipeline internal reference (v46l)

**Entry points.** Sync `POST /api/v1/tender-pro/dpr/analyze-json` for < 4 MB payloads and async `POST /api/v1/tender-pro/dpr/analyze-async` for anything larger. Both accept `{filename, content_b64}` and an optional `?template_id=` query param.

**Extraction.** `_extract_text_from_upload(name, body_bytes)` routes by file extension. PDF → pypdf iter-pages; DOCX → python-docx paragraph concatenation; TXT → UTF-8 decode. All emit `\n\n` between logical blocks so downstream chunking finds natural boundaries.

**Chunking.** `_chunk_text(text)` slides a 40 000-char window with 3 000-char overlap. Cap `DPR_MAX_CHUNKS=500` (env-overridable). Returns `[(offset, chunk_text), …]`.

**Pass 1a — per-chunk map.** Semaphore(4). System prompt asks for `{header_hints, sections}` as strict JSON. Each section carries `{title, hint, anchor_phrase}`; the anchor_phrase is a 6-12 word verbatim snippet used later to locate the passage most relevant to a section.

**Pass 1b — global reduce.** `merged_header` first-wins across chunks. `global_sections` deduplicates by lowercased title; ties break by earliest occurrence. Caps at 60 (was 30 pre-v46l).

**Pass 2 — draft.** Semaphore(6). `_best_chunk(sec, i)` picks the chunk whose text contains anchor_phrase (case-insensitive). Draft prompt forbids inventing numbers; asks for 280-360 words HTML with a strict tag allow-list. Bodies under 200 chars trigger the raw-passage fallback so nothing lands as a stub.

**Async progress.** `progress_cb(pct, phase)` is invoked at every phase transition. The worker (`_dpr_job_worker`) writes `{progress, phase, status, updated_at}` into `dpr_jobs` so the UI polls a real percentage.

**Error surface.** Any 413/422/502 propagates as-is; unexpected exceptions surface as 500 with `{error: 'TypeName: str[:400]}`.

**Nginx tuning.** `client_max_body_size 200M` + `client_body_timeout 300s` + `proxy_read_timeout 300s`. Without these, EB's default idle 60 s kills any 20 MB base64 upload.

**Observability.** Every DPR job writes an audit row (`tender_pro.dpr_analyze`) on start + completion with size + section count. CloudWatch tail: `filterPattern=/dpr/analyze` for latency, `filterPattern='dpr.job.failed'` for failure rate.

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# 33 · Incident-response playbook

## Detection

Alerts flow through CloudWatch → PagerDuty → on-call admin. The audit-log tail on Elastic Beanstalk feeds a 30-min sliding window that raises P1 on >5 auth failures / min or >3 500-class responses / min. The AI anomaly detector opens a P2 for any high-confidence flag; Legal triages daily.

## Triage

On-call admin acknowledges within 5 min, opens a Slack war-room channel, invites the CTO + Legal + Escrow + Support. Engineer captures the immediate CloudWatch tail, the audit rows for the affected entities, and the browser Network trace if a UI regression.

## Containment

If credentials or key material may be exposed: rotate the affected key first (the managed AI key rotates via the platform; Stripe via Stripe dashboard; JWT signing key via EB env-property + rolling restart). If a code path is at fault, roll back to the previous EB version via the Application Versions console.

## Communication

For any incident touching vendor bids or contract data, the CEO + Partner lead are informed within 1 h. Communications drafts customer language; nothing is sent without CTO + Legal sign-off. The audit-vault records the entire war-room transcript.

## Recovery

Once mitigated, engineer captures the RCA + timeline + fix in `/app/deliverables/incidents/YYYY-MM-DD-<slug>.md`. That file is committed via 'Save to Github' and cross-linked from the audit vault.

## Post-mortem

Blameless post-mortem within 5 business days. Attendees: engineer, CTO, on-call admin, affected role owner. Output: (a) written RCA, (b) 3 preventive actions with owners + due dates, (c) update to the troubleshooting playbook (chapter 12).

## Regulator notification

If personal data is exposed the DPO notifies the RK data authority within 72 h per the Personal Data law. Legal owns the wording; the audit vault preserves the transmitted notice.

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## Legal disclosure

For any incident that may materially affect a live bid — retention breach, evaluation-envelope tamper, escrow ledger corruption — Legal freezes the affected tender until the RCA is signed off.

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# 34 · Data-retention & backup policy

| Entity | Retention |
|---|---|
| Tenants + auth realms | Retained for the lifetime of the tenant + 10 years after termination. |
| Notices + tender_templates | Retained indefinitely — the annual disclosure references them by ref. |
| Registrations (rejected) | Retained 3 years then purged; audit-log row preserved. |
| Bids + scores + rubrics | Retained 15 years — required by the audit-board contract clause. |
| Escrow transactions | Retained 15 years — required by financial regulation. |
| Corrigenda | Retained indefinitely with the parent notice. |
| Audits | Retained 10 years default; 15 years for contract-touching actions. |
| Ratings | Retained indefinitely (vendor-visible). |
| dpr_jobs.result | Retained 6 hours; failed jobs retained 24 h with error string. |
| Reports snapshots | Retained 5 years then rolling-summarised. |
| Backups | Nightly Mongo dump to S3 (7-day rolling), weekly cold-copy to Glacier (12-month rolling). Weekly restore drill on a scratch cluster. |
| DR RTO / RPO | RTO 4 h · RPO 24 h. Documented in Chapter 33. |

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# 35 · Vendor education curriculum

## Module 1 — Account & Profile

Registration wizard, email verification, MFA setup, Section A/B/C completeness, licence upload + expiry alerts.

## Module 2 — Discovering opportunities

Public notice board, category subscriptions, email digests, RSS + Snapchat + Instagram surfaces, calendar of pre-bid meetings.

## Module 3 — Purchasing tender packs

Tender-fee payment via Stripe (test + live), download of the packet, corrigenda subscription, Q&A engagement.

## Module 4 — Preparing a bid

Compliance matrix, price basis, BoQ workbook, bid-bond upload, submission dry-run in the vendor sandbox.

## Module 5 — Submitting a bid

Envelope sealing, timing, submission proof, common failure modes (invalid PDF, missing licence, expired MFA).

## Module 6 — Reverse auctions

Invitations, masked aliases, manual + auto bidding, walk-away floor, anti-sniping, auction-close etiquette.

## Module 7 — After the award

LoA reception, DocuSign signing, PO number, contract execution, milestones, invoicing, EMD refund workflow, review + trust badge.

## Module 8 — Dispute resolution

Grounds for dispute, timeline, evidence pack, arbitration clause, escalation to Legal.

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# 36 · Change management & release cadence

## Weekly release train

Every Wednesday 15:00 UTC. Frontend + backend deployed together. Preview cycle Tue afternoon; production cycle Wed afternoon.

## Hotfix path

Any P0 (auth outage, escrow tamper, evaluation-envelope corruption) may bypass the train. Hotfix requires CTO sign-off + 15-minute soak on preview + rollback plan.

## Version scheme

vXY where X is a marketing generation, Y is the feature letter. 'v46m' = generation 46, feature letter m. Backend + frontend versions may diverge (frontend catches up on the next Wed train).

## Deprecation policy

Deprecated endpoints keep serving for 90 days with a `Deprecation` response header + a `Sunset` header. Emails go to every partner using the endpoint at day 0, 30, 60, 89.

## Feature flags

Non-breaking features behind `feature_flags` collection. Preview flags toggled per-user; production flags toggled per-tenant.

## Rollback criteria

Automatic rollback if health-check < 90 % for 5 min, or if error rate > 5 % of requests for 2 min. Manual rollback available via EB Application Versions.

## Change-log discipline

Every merged PR appends to `/app/memory/CHANGELOG.md`; PRD.md summarises weekly. The manual's chapter 14 mirrors the last 12 months.

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# 37 · Security controls matrix

## Identity

| Control | Implementation |
|---|---|
| Password policy | 12+ chars, complexity enforced, bcrypt-hashed with per-user salt, rehash on cost bump. |
| Brute-force lockout | 5 failed attempts in 15 min → 15-min lockout; audit row per attempt. |
| Session cookie | HttpOnly, SameSite=Lax, Secure in prod; 8-h idle, 30-day absolute. |
| MFA | TOTP + optional WebAuthn; enforced for admin + management; opt-in for others. |
| Magic link | Signed JWT with 15-min TTL; single-use; invalidated on password change. |

## Transport & network

| Control | Implementation |
|---|---|
| TLS | TLS 1.3 only; HSTS max-age 365 days + preload. |
| CDN | CloudFront front; origin only allows CloudFront-signed requests. |
| Backend origin | EB security group only allows the CloudFront IP list + the office VPN CIDR. |
| Rate limits | 60 req/min per IP on auth, 300 req/min per session elsewhere. |
| WAF | AWS WAF managed common rules + custom binary-body blocker (why DPR uses base64 JSON). |

## Data protection

| Control | Implementation |
|---|---|
| At-rest encryption | S3 SSE-KMS + MongoDB Atlas encryption-at-rest. |
| In-flight encryption | TLS end-to-end + LlmChat proxy over HTTPS. |
| Backup encryption | Glacier vault encrypted with a customer-managed KMS key rotated annually. |
| Secrets management | AWS Secrets Manager for prod; .env files are gitignored and never bundled in EB zips. |
| PII redaction | Auditor role CSV export strips PII by default; unredacted export requires legal + CTO co-sign. |

## Application-level

| Control | Implementation |
|---|---|
| CSRF | Double-submit token; every mutating request must carry the X-CSRF-Token header. |

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| | |
|---|---|
| XSS | Rich-text sanitisation via _sanitize_body allow-list; React auto-escapes by default. |
| SSRF | Outbound HTTP restricted to a URL allow-list (managed AI gateway, Stripe, DocuSign, Resend). |
| Serialisation | PyObjectId annotated type; no raw ObjectId leakage; all API models are Pydantic-typed. |
| File uploads | Signed URLs to a private bucket, MIME sniff on server, 100 MB DPR cap, 25 MB elsewhere. |

## Operational

| Control | Implementation |
|---|---|
| Change advisory board | Every prod change reviewed by CTO + Legal + on-call admin on the weekly train. |
| Access reviews | Quarterly review of all admin + staff roles; revoke unused within 90 days. |
| Vulnerability scans | Weekly Snyk on backend requirements + npm audit on frontend; SLA 30 days for High, 90 days for Medium. |
| Penetration test | Annual third-party pen-test; unresolved criticals block the next feature release. |
| Incident drills | Quarterly tabletop; annual full-DR failover to the scratch cluster. |

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# 38 · Tender committee constitution

## Composition

A minimum of five voting members: (a) technical lead from the sponsoring GOC, (b) financial lead from PMC accounting, (c) legal counsel, (d) domain expert (rotating), (e) external observer from the Partner portal. Quorum: four members with at least one from each mandatory role (a, b, c).

## Term

Two-year rotating term; no member serves more than two consecutive terms. External observers are appointed per-tender.

## Conflict of interest

Every member signs a fresh COI declaration per tender. Any material relationship — equity, past employment, first-degree family — triggers recusal. Recusal is audit-logged and the missing weight redistributes proportionally.

## Voting

Technical scores are simple weighted average; commercial score is L1-driven; final ranking is the QCBS composite. Ties broken by: (1) lower price, (2) earlier submission timestamp, (3) higher aggregate trust rating, (4) coin toss on video call with all members present.

## Minutes

Chair drafts the minutes within 48 h of the meeting. Every member signs digitally via DocuSign; the signed PDF is uploaded to the notice and mirrored to the audit vault.

## Dispute mechanism

A dissenting member may attach a written dissent to the minutes. Dissents that raise substantive process issues freeze the award until Legal signs off.

## Confidentiality

All committee materials are Confidential — Internal Use Only. Unauthorised disclosure triggers immediate revocation + a Legal review. Every download from the committee vault is audit-logged.

## External observer

The observer has voice but no vote. Their role is to attest process fidelity; their sign-off appears on the annual disclosure.

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## Meeting cadence

Ordinary: on-demand per tender. Standing: quarterly retrospective on process + outcomes. Extraordinary: any dispute or regulator enquiry.

## Compensation

Members receive a per-tender honorarium disclosed on the annual disclosure. External observers receive a fixed retainer + expenses.

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# 39 · Disaster-recovery drill script

## T-14 days

Announce drill window to on-call rota + Support + Partner leads. Freeze non-critical deploys 24 h before the drill.

## T-1 day

Verify Mongo dump succeeded on the scratch cluster. Verify EB app version is deployable to the DR environment. Verify DocuSign, Stripe, Resend sandbox credentials in Secrets Manager.

## T-0 · 09:00 UTC

Declare drill start in the war-room channel. Rotate DNS to the DR CloudFront distribution (weight 100 → 0 on prod, 0 → 100 on DR).

## T-0 · 09:05 UTC

Restore last night's Mongo dump onto the DR cluster (target RTO 4 h).

## T-0 · 09:10 UTC

Deploy the current EB backend version to the DR environment. Point secrets to DR values.

## T-0 · 09:20 UTC

Smoke test: admin login, list notices, place a mock bid, run an evaluation lock, generate an award report, refund an EMD. All 6 must pass or the drill is aborted and rolled back.

## T-0 · 10:00 UTC

End drill. Rotate DNS back to prod. Compare timestamps to targets; log actual RTO / RPO. Post-mortem the same afternoon.

## T+1 day

Publish drill report to Partner portal. If RTO > 4 h or RPO > 24 h, opens a P1 remediation ticket with the CTO as owner.

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# 40 · Sample notice (annotated)

## Header

**Notice ref:** PMC-NOT-2026-0472 · **Awarding authority:** Astana Municipal Water Utility · **Country:** Kazakhstan · **Category:** Water & wastewater — pipeline construction · **Estimated value:** KZT 3,850,000,000 · **EMD:** KZT 96,250,000 (2.5 %) · **Tender fee:** KZT 250,000 · **Submission deadline:** 2026-10-31 15:00 UTC · **Bid validity:** 180 days · **Evaluation:** QCBS 70/30.

## Scope

Design + build + commission of a 47 km ductile-iron potable water pipeline from the Astana North reservoir to the Saryarka distribution station, including 12 pressure-reduction chambers, SCADA integration, and 24-month operations warranty. Full BoQ in the tender packet.

## Eligibility

Bidders must (a) hold a valid Class-A pipeline construction licence issued by the RK Ministry of Construction, (b) demonstrate at least 3 pipeline projects of comparable value in the last 5 years, (c) meet a minimum annual turnover of KZT 6 bn averaged over the last 3 years, (d) not appear on any RK / OFAC / EU / LSEG sanction list, (e) commit to a minimum 60 % local-content share.

## Documents required

Signed cover letter, corporate registration, tax clearance, 3 years audited financials, list of past projects with client references, ISO 9001 + ISO 14001 certificates, health & safety plan, project execution plan, quality plan, environmental & social management plan, bid bond (bank guarantee), price schedule against the BoQ.

## Timeline

Publication → today. Pre-bid meeting → 2026-10-03 10:00 UTC via Zoom. Query cut-off → 2026-10-15. Submission deadline → 2026-10-31 15:00 UTC. Technical envelope opening → 2026-11-03 10:00 UTC. Commercial envelope opening → 2026-11-18 10:00 UTC. Award → target 2026-12-05. Contract signing → 2026-12-20.

## Payment terms

10 % advance against bank guarantee, 80 % against milestone certificates verified by the Engineer, 10 % retention released 12 months after handover subject to defect-liability closure.

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## Dispute resolution

Any dispute is escalated through: (1) direct engineer-vendor conference within 7 days, (2) formal notice + 30-day cure period, (3) mediation via the Astana International Arbitration Centre, (4) binding arbitration under the AIAC rules. Governing law: Kazakhstan.

## Corrigendum log (sample)

**C-01** (2026-10-05): pre-bid meeting relocated from in-person to Zoom-only due to venue capacity. **C-02** (2026-10-12): eligibility relaxed — 2 comparable pipeline projects (was 3) sufficient. **C-03** (2026-10-18): submission deadline extended by 3 days in response to bidder queries about the SCADA specification.

## Q&A extract (sample)

Q1: Is the SCADA integration open-standard or vendor-locked? — A1: Open-standard (Modbus TCP + OPC-UA); vendor-locked proposals will be flagged in the compliance matrix. Q2: Is the bid bond negotiable with a corporate guarantee? — A2: No; only a bank guarantee from a Category-A RK bank is acceptable. Q3: Is the local-content share tested at award or at contract-completion? — A3: Both — an at-award commitment and an at-completion verification, with a 5 % contract-value penalty if the completion figure is below the committed figure.

## Attachments (sample)

01_Signed_Notice.pdf · 02_BoQ_v2.xlsx (post C-02) · 03_Drawings_A_C_series.zip · 04_Draft_Contract.docx · 05_Bid_Bond_Template.pdf · 06_Health_Safety_Plan_Template.pdf.

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# 41 · Sample bid (annotated)

## Cover letter

From: NurAzia Infrastructure LLP (NAI). Signed by the Managing Director. Bid reference: PMC-BID-2026-1183. Attaches the certificate of incorporation, tax clearance, and the notarised power of attorney authorising the signatory. Commits to the price and validity period; references PMC-NOT-2026-0472 and all corrigenda published up to submission.

## Technical envelope

180-page technical bid organised into 12 chapters mirroring the notice's compliance matrix: company profile · project experience · project execution plan · quality plan · HSE plan · ESMS · SCADA architecture · pressure-management strategy · maintenance philosophy · training programme · local-content commitment · deviations. Each clause carries a YES / NO / DEVIATION cell against the notice requirement; deviations are rationalised with an impact + mitigation.

## Commercial envelope

Price schedule against the notice's BoQ. Base bid: KZT 3,712,500,000 (below estimate). Escalation formula: quarterly, capped at 8 % annually. Payment currency: KZT for local scope, USD (spot at contract-signing) for imported SCADA hardware. Optional items: 24-month O&M extension at KZT 84 M / year.

## Compliance matrix (excerpt)

**Clause 4.2.1 (Class-A licence)** — YES · attached licence LIC-2019-0142. **Clause 4.2.2 (3 comparable projects)** — YES · attached list with client references. **Clause 5.4 (Modbus TCP + OPC-UA)** — YES · Siemens SIMATIC + Rockwell Logix, both supported. **Clause 6.1 (Advance guarantee)** — DEVIATION · standby letter of credit instead of a bank guarantee; rationale: bank capacity limits; mitigation: SLC issued by Halyk Bank (Category-A).

## Local-content commitment

At-award commitment: 63 %. Categorisation: 42 % labour (RK residents), 15 % locally-manufactured pipe (Aktau Pipe Mill), 6 % services (RK-registered subcontractors).

## Bid bond

Bank guarantee KZT 96,250,000 issued by Halyk Bank Kazakhstan, valid 210 days from submission (30 days beyond award timeline). PDF attached with the bank's SWIFT confirmation.

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## Signatures

Digitally signed by the Managing Director and countersigned by the Company Secretary. All 12 chapter documents carry the corporate seal + digital signature stamp. The DocuSign envelope reference is preserved in the audit trail.

## Envelope timing

Technical envelope submitted 2026-10-31 12:14:07 UTC (2 h 46 min before deadline). Commercial envelope sealed simultaneously. Audit trail preserves both timestamps + the vendor's IP.

## Post-submission outcome (illustrative)

Technical score: 82.4 / 100 (rank 2). Commercial score: 95.1 / 100 (rank 1 — lowest price). QCBS composite: (0.70 × 82.4) + (0.30 × 95.1) = 86.2 / 100 → **rank 1**. Awarded 2026-12-05. LoA countersigned 2026-12-14. PO issued 2026-12-16. Contract executed 2026-12-20.

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# 42 · Detailed API reference

## Auth

| Endpoint | Purpose |
|---|---|
| `POST /api/v1/admin/auth/login` | Body: {email, password}. 200: {id, email, name, role, token}. Sets admin_session cookie. |
| `POST /api/v1/staff/auth/login` | Body: {email, password, role, tenant_code?}. 200: {active_role, mfa_required?}. Sets staff_session cookie. |
| `POST /api/v1/vendor/auth/login` | Body: {email, password}. 200: {vendor_ref}. Sets vendor_session cookie. |
| `POST /api/v1/*/auth/logout` | Clears the realm cookie; audit row `auth.logout`. |
| `POST /api/v1/*/auth/mfa/verify` | Body: {code}. Required after login when the user has MFA configured. |

## Public discovery

| Endpoint | Purpose |
|---|---|
| `GET /api/v1/meta/public/stats` | 60-s CloudFront cache. Returns {notices_published, suppliers_approved, packages_total}. |
| `GET /api/v1/meta/public/notices` | Paginated: ?page=&limit=&country=&category=&q=. Returns notice cards. |
| `GET /api/v1/meta/public/notices/{ref}` | Full detail — notice + corrigenda + Q&A. |
| `GET /api/v1/meta/public/tenders/live` | Alias of /notices?status=live. |
| `GET /api/v1/meta/public/tenders/completed` | Alias of /notices?status=completed. |

## Vendor onboarding

| Endpoint | Purpose |
|---|---|
| `POST /api/v1/registrations/draft` | Body: {email, company_name?, country}. 201: {registration_id, magic_token}. |
| `PATCH /api/v1/registrations/{id}` | Body: any subset of Section A/B/C fields. 200: updated draft. |

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| | |
|---|---|
| `POST /api/v1/registrations/{id}/submit` | Locks the draft, notifies Support, and emails a receipt. |
| `POST /api/v1/admin/registrations/{id}/approve` | Admin only. Provisions the vendor account + emails credentials. |

## Tender Builder

| Endpoint | Purpose |
|---|---|
| `GET /api/v1/tender-templates` | ?is_master=&country=&industry=&q=. Paginated list. |
| `POST /api/v1/tender-templates` | Clone from a master or start from blank. |
| `GET /api/v1/tender-templates/{id}` | Full document with sections + attachments. |
| `PATCH /api/v1/tender-templates/{id}` | Partial update; every save creates a snapshot. |
| `POST /api/v1/tender-templates/{id}/save-as-master` | Promote a draft to a reusable master (country required). |
| `POST /api/v1/tender-templates/{id}/sections/{code}/ai-draft` | v46o — routes through _llm_complete. Body: {tone, length}. |
| `POST /api/v1/tender-pro/templates/{id}/ai-fill-all` | Draft every empty section body in one pass. |

## AI DPR

| Endpoint | Purpose |
|---|---|
| `POST /api/v1/tender-pro/dpr/analyze-json` | Sync path for < 4 MB. Body: {filename, content_b64, template_id?}. |
| `POST /api/v1/tender-pro/dpr/analyze-async` | v46l async path. Returns {job_id}. |
| `GET /api/v1/tender-pro/dpr/jobs/{id}` | Poll status: {status, progress, phase, result, error}. |

## Evaluation

| Endpoint | Purpose |
|---|---|
| `POST /api/v1/evaluations/rubric` | Upsert rubric for a notice. |
| `POST /api/v1/evaluations/lock-envelope` | v46m — refuses re-lock (409). Body: {notice_ref, kind}. |
| `GET /api/v1/evaluations/lock-state/{ref}` | v46m — returns seal timestamps + attribution. |
| `POST /api/v1/evaluations/score` | |

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| | |
|---|---|
| | Submit a technical or commercial score. Blocked if envelope is sealed. |
| `GET /api/v1/evaluations/award-report/{ref}` | L1 comparison + weighted scores. |

## Escrow, awards, contracts

| Endpoint | Purpose |
|---|---|
| `POST /api/v1/escrow/transactions` | Record an EMD held/refunded/forfeited entry. Requires joint-account snapshot. |
| `POST /api/v1/awards/loa` | Draft + send Letter of Award via DocuSign. |
| `POST /api/v1/awards/po` | Emit the PO number after vendor acceptance. |
| `POST /api/v1/docusign/envelopes` | Create an envelope from a template with a signer routing spec. |
| `GET /api/v1/docusign/envelopes/{id}` | Status: sent / delivered / completed / declined. |

## Auctions

| Endpoint | Purpose |
|---|---|
| `POST /api/v1/auctions` | Create an auction room for an evaluated notice. |
| `POST /api/v1/auctions/{ref}/invite` | Invite a technically-qualified vendor. |
| `POST /api/v1/auctions/{ref}/bid` | Place a manual or auto bid. Server-side alias masking. |
| `GET /api/v1/auctions/{ref}/spectate` | Read-only public spectator feed with masked aliases. |

## Reporting

| Endpoint | Purpose |
|---|---|
| `GET /api/v1/stakeholder-reports/tenders` | Pipeline snapshot per notice. |
| `GET /api/v1/stakeholder-reports/awards` | PO numbers + contract value + winning bidder. |
| `GET /api/v1/stakeholder-reports/escrow` | Aging days per status. |
| `GET /api/v1/stakeholder-reports/sla` | Milestone hit-rate per active contract. |
| `POST /api/v1/reports/snapshot` | Persist every artefact to disk; per-file try/except; returns errors[]. |

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# 43 · Acceptance-test matrix

## Public discovery

- Anonymous visitor loads / — KPI counters render with real numbers.
- Visitor filters /notices by country=KZ — the count matches the API response.
- Visitor opens /tenders/live — every card has a countdown chip.
- Visitor opens /about and /vision — pages render without a JS error.

## Login gateway

- Visitor loads /login — 5 portal cards visible.
- Clicking each portal reveals a credential form with the correct badge + colour.
- Wrong password shows the exact backend error in a toast; correct password redirects to the right portal.
- Vendor portal accepts vendor credentials; staff portals accept staff credentials; cross-realm is refused.

## Tender Builder

- Admin creates a new tender from a master template.
- Admin edits a section body; auto-snapshot appears in Version history.
- Admin clicks AI on a section — body populates in < 15 s (v46o fix).
- Admin uploads a 20 MB DPR — Background chip appears, progress bar advances, section count > 30 (v46l).
- Admin runs Save-as-Master — new master appears in the picker under the picked country (v46f).

## Evaluation & awards

- Admin sets rubric; invalid weights disable Save.
- Committee submits scores; sealed envelopes reject new scores.
- Admin locks technical envelope; button becomes 'Sealed '; banner shows timestamp + sealer.
- Admin generates award report — L1 identified with weighted scores.
- Attempt to re-lock returns 409 (v46m).

## Escrow

- Admin records an EMD receipt with two signers; joint-account snapshot preserved on the row.
- Refund workflow requires two officers; audit trail preserves both.
- Aging bucket dashboard rolls up 0-30 / 31-60 / 61-90 / 90+ days.

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## Auctions

- Admin invites qualified vendors; only technically-qualified are eligible.
- Vendor joins auction; alias masked; auto-bid config accepted.
- Last-second bid extends the auction by 5 min.
- Spectator link exposes only masked aliases.

## Compliance

- Audit trail export ships as CSV with SHA-256 checksum.
- Auditor role CSV export strips PII.
- Regulator webhook fires on award; retries 3× at 1/15/60 min on failure.

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# 44 · Sample corrigenda log

## C-01 · pre-bid meeting relocation

**Published:** 2026-10-05 09:14 UTC · **Author:** Astana Municipal Water Utility · **Type:** Clarification · **Effective:** 2026-10-05 immediately. **Body:** The pre-bid meeting scheduled for 2026-10-03 10:00 UTC has been relocated from the Utility headquarters (venue capacity 40) to a Zoom-only session. The Zoom link + passcode have been emailed to every bidder who purchased the tender pack and is also visible on the notice detail page. Attendance is optional but strongly recommended. Minutes will be published within 48 h. **Impact:** none on the submission deadline.

## C-02 · eligibility relaxation

**Published:** 2026-10-12 15:41 UTC · **Type:** Amendment · **Effective:** 2026-10-12 immediately. **Body:** Clause 4.2.2 (three pipeline projects of comparable value in the last 5 years) is relaxed to two pipeline projects; the definition of *comparable value* is unchanged (≥ 70 % of the estimated tender value). **Rationale:** feedback from pre-bid meeting attendees indicated that only three Kazakhstan-registered contractors would qualify under the original clause; opening to two projects broadens the field to eight while preserving the capacity bar. **Impact:** bidders may now resubmit updated eligibility declarations up to 24 h before the submission deadline.

## C-03 · deadline extension

**Published:** 2026-10-18 11:02 UTC · **Type:** Extension · **Effective:** 2026-10-18 immediately. **Body:** The submission deadline is extended by three (3) calendar days from 2026-10-28 15:00 UTC to 2026-10-31 15:00 UTC. **Rationale:** Q&A responses received on 2026-10-15 (Q1, Q2, Q3) materially expanded the SCADA specification; bidders requested additional time for design revisions. **Impact:** technical envelope opening is correspondingly pushed from 2026-10-31 to 2026-11-03 10:00 UTC; commercial envelope opening from 2026-11-15 to 2026-11-18 10:00 UTC. All other timeline items unchanged.

## C-04 · attachment update

**Published:** 2026-10-19 10:00 UTC · **Type:** Amendment · **Effective:** 2026-10-19 immediately. **Body:** Attachment 02_BoQ_v1.xlsx is superseded by 02_BoQ_v2.xlsx. Changes: (a) added six SCADA line items reflecting the C-03 clarification, (b) unified valve specifications on line 47–52, (c) corrected the unit-of-measure on line 89 (previously *ea*, now *m*). Bidders must price against BoQ v2; bids priced against BoQ v1 will be technically rejected.

## Corrigenda discipline notes

**Every corrigendum is append-only** — errors are corrected by a superseding corrigendum, never by editing. The audit trail therefore captures the full narrative of a tender's amendments. Bidders are auto-notified via Resend digest for every published corrigendum on any tender pack they purchased; the notification carries a signed link to the corrigendum PDF.

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# 45 · Sample DocuSign envelope walkthrough

## Envelope template

Selected: *LoA-Standard-KZT-2026*. Merge fields: {vendor_legal_name, vendor_registration_no, work_title, notice_ref, po_number, contract_value_kzt, effective_date, defect_liability_months, bid_bond_number, bid_bond_expiry}.

## Signer 1 — Vendor (NAI)

Name: Aigerim Bekova · Email: <ceo@nurazia.kz> · Role: Managing Director · Auth: SMS OTP + email-click · Fields: signature, initial-each-page, executed-date.

## Signer 2 — Legal (PMC)

Name: Dilnaz Karimova · Email: <legal@pmcmanagers.com> · Role: Head of Legal · Auth: SSO · Fields: signature, review-stamp.

## Signer 3 — Finance (PMC)

Name: Talgat Serikov · Email: <cfo@pmcmanagers.com> · Role: CFO · Auth: SSO + WebAuthn · Fields: signature, PO stamp.

## Signer 4 — Countersign

Name: Aliya Nurlanova · Email: <ceo@pmcmanagers.com> · Role: CEO · Auth: SSO + WebAuthn · Fields: signature, corporate seal stamp.

## Routing

Sequential. Signer 1 → 2 → 3 → 4. Envelope expires 2026-12-14. First reminder after 3 business days, subsequent every 2. Decline handling: Auto-invite runner-up (per notice policy).

## Status timeline (sample)

**2026-12-05 14:12 UTC** sent to Signer 1. **2026-12-06 09:41 UTC** viewed. **2026-12-08 11:18 UTC** signed. **2026-12-09 14:02 UTC** Legal signed. **2026-12-10 09:58 UTC** CFO signed. **2026-12-14 16:22 UTC** CEO countersigned. Envelope completed; certificate downloaded to audit vault.

## Post-completion

Completed envelope PDF + Certificate-of-Completion stored under `/audit-vault/contracts/PMC-NOT-2026-0472/`. PO number `PMC-PO-2026-1183` emitted. Contract activation email sent to NAI + internal stakeholders.

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# 46 · Stakeholder-report sample

## Tenders report — September 2026

18 notices published, aggregate value KZT 24.8 bn. Median cycle time (draft→publish) 11 days (target ≤ 14). Bidder attention 71 % (target ≥ 65). Bids per notice median 5.5 (target ≥ 4). Two notices flagged for below-target attention → assigned to category audit.

## Awards report — September 2026

9 awards finalised, aggregate contract value KZT 11.4 bn. Winning bidders: 5 KZ, 3 EAC, 1 international. Local-content committed weighted-average 61 % (target ≥ 55). No award disputes opened.

## Escrow report — September 2026

EMDs held opening balance KZT 1.28 bn (33 notices), receipts KZT 486 M (18 notices), refunds KZT 342 M (14 notices), forfeitures KZT 0 (0 notices), closing balance KZT 1.42 bn (37 notices). Aging: 0-30 days 62 %, 31-60 22 %, 61-90 11 %, 90+ 5 %. Refund SLA 100 %.

## SLA report — September 2026

42 active contracts. Milestones hit-rate 94.3 % (2 breaches under review). Average DocuSign envelope turnaround 6.8 days (target ≤ 10). One contract paused pending regulator enquiry (see chapter 27).

## Anomaly report — September 2026

3 high-confidence flags: (a) bid-clustering on notice 0468 (three bidders within 0.8 % — under committee review), (b) price-dumping on notice 0455 (one bidder 41 % below peer average — invited for clarification), (c) alias-reuse in auction 0451 (same alias re-appeared 14 min after ejection — investigated, false positive due to shared session token). No SAR filed.

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# 47 · Audit-trail export sample

## Row 1

at=2026-09-24T05:41:12Z · actor=6a6edc680e4f4fef11774096 (adminsupport@pmcmanagers.com) · action=notice.publish · entity=notice · ref=PMC-NOT-2026-0472 · payload={published_at:'2026-09-24T05:41:12Z', digest_sent:true}

## Row 2

at=2026-09-24T05:41:15Z · actor=system · action=integration.regulator_webhook · entity=notice · ref=PMC-NOT-2026-0472 · payload={url:'https://regulator.example/webhook', status:200, retries:0}

## Row 3

at=2026-10-05T09:14:08Z · actor=8x3ee640abcfaa10022eb · action=corrigendum.publish · entity=corrigendum · ref=C-01 · payload={notice_ref:'PMC-NOT-2026-0472', type:'clarification'}

## Row 4

at=2026-10-31T12:14:07Z · actor=vendor_ref='PMC-SUP-2024-0083' · action=bid.submit · entity=bid · ref=PMC-BID-2026-1183 · payload={notice_ref:'PMC-NOT-2026-0472', envelopes_sealed:['technical','commercial']}

## Row 5

at=2026-11-03T10:02:44Z · actor=adminsupport · action=envelope.lock.technical · entity=notice · ref=PMC-NOT-2026-0472 · payload={sealed_at:'2026-11-03T10:02:44Z'}

## Row 6

at=2026-12-05T14:12:08Z · actor=adminsupport · action=award.finalise · entity=award · ref=PMC-NOT-2026-0472 · payload={vendor_ref:'PMC-SUP-2024-0083', contract_value_kzt:3712500000}

## Row 7

at=2026-12-14T16:22:41Z · actor=system · action=docusign.envelope_completed · entity=envelope · ref=DS-2026-0972 · payload={po_number:'PMC-PO-2026-1183', certificate_url:'…/certificate.pdf'}

## CSV export policy

Auditor-role exports strip PII (email, phone, national id, bank iban) and replace them with per-user pseudonyms. Unredacted exports require Legal + CTO co-sign; the co-sign row itself is written into the audits collection.

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# 48 · MoU template (Sponsor ↔ Platform Operator)

## Parties

This Memorandum of Understanding (this **MoU**) is entered into on ____ between (i) **PMC Managers Limited** (the *Platform Operator*) and (ii) [Sponsoring Government Owned Corporation / Bank / Regulator] (the *Sponsor*).

## Purpose

The Sponsor engages the Platform Operator to operate an end-to-end procurement platform for tenders issued by the Sponsor. The scope includes notice publication, bidder self-registration, bid receipt, EMD custody, evaluation, award, e-signature, contract administration, and immutable audit.

## Term

This MoU is effective from the Effective Date and continues for an initial term of 24 months, renewable for successive 12-month terms unless either party gives 90 days' written notice of non-renewal.

## Fees

The Sponsor pays the Platform Operator a monthly platform fee of KZT __ plus a per-notice fee of KZT __ per published notice, invoiced monthly, payable within 30 days of invoice date. Late payment attracts interest at the base rate + 3 % per annum.

## Data ownership

All data uploaded by the Sponsor or its bidders remains the property of the respective uploader. The Platform Operator holds a limited licence to store and process the data solely for the purposes contemplated by this MoU. On termination the Sponsor may export all data in structured JSON + CSV within 30 days.

## Confidentiality

Each party treats the other's confidential information with the same care as its own material of equal sensitivity and no less than a reasonable-professional standard. Confidential information may not be shared with third parties except as necessary to perform this MoU.

## Right to audit

The Sponsor has the right, on 30 days' written notice, to audit the Platform Operator's controls, logs, and data handling as they relate to the Sponsor's tenders. The Platform Operator provides access to the audit vault, immutable audit trail, and the incident-log for the audit period.

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## Termination

Either party may terminate this MoU: (a) for material breach uncured within 30 days of written notice, (b) upon insolvency of the other party, (c) upon the other party's failure to maintain any statutory licence necessary to perform this MoU.

## Governing law

This MoU is governed by the laws of the Republic of Kazakhstan. Disputes are resolved per chapter 40 of the operator manual (arbitration under the AIAC rules).

## Signatures

Signed for and on behalf of the Sponsor by ____ (name, title). Signed for and on behalf of the Platform Operator by ____ (name, title). Date: ____

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# 49 · RACI matrix

| Activity | Assignment |
|---|---|
| Publish a notice | Agent: R · Admin: A · Legal: C · Committee: I · Support: I · Escrow: I · Partners: I · Vendors: I |
| Answer a Q&A question | Agent: R · Admin: A · Legal: C · Committee: I · Support: R · Escrow: — · Partners: — · Vendors: I |
| Publish a corrigendum | Agent: R · Admin: A · Legal: C · Committee: I · Support: I · Escrow: — · Partners: I · Vendors: I |
| Receive EMD | Agent: — · Admin: A · Legal: — · Committee: — · Support: I · Escrow: R · Partners: I · Vendors: I |
| Lock technical envelope | Agent: — · Admin: R/A · Legal: I · Committee: I · Support: — · Escrow: — · Partners: I · Vendors: I |
| Score technical | Agent: — · Admin: A · Legal: I · Committee: R · Support: — · Escrow: — · Partners: I · Vendors: — |
| Generate award report | Agent: — · Admin: R/A · Legal: C · Committee: C · Support: — · Escrow: — · Partners: I · Vendors: I |
| Issue LoA | Agent: — · Admin: R/A · Legal: R · Committee: I · Support: — · Escrow: — · Partners: I · Vendors: I |
| Countersign contract | Agent: — · Admin: A · Legal: R · Committee: — · Support: — · Escrow: — · Partners: I · Vendors: I |
| Refund EMD | Agent: — · Admin: A · Legal: I · Committee: — · Support: I · Escrow: R · Partners: I · Vendors: I |
| Forfeit EMD | Agent: — · Admin: A · Legal: R · Committee: — · Support: I · Escrow: R · Partners: I · Vendors: I |
| Deploy release train | Agent: — · Admin: A · Legal: I · Committee: — · Support: I · Escrow: — · Partners: I · Vendors: I · Engineering: R |
| Handle incident | Agent: — · Admin: R · Legal: R · Committee: I · Support: R · Escrow: R · Partners: I · Vendors: I · Engineering: R |
| Regulator response | Agent: — · Admin: A · Legal: R · Committee: I · Support: I · Escrow: I · Partners: R · Vendors: I · Engineering: I |

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# 50 · Cost-benefit worksheet (KZ programme, illustrative)

## Direct benefit — cycle-time reduction

Baseline cycle time (draft → award) pre-PMC: 92 days industry average. Post-PMC: 43 days observed on the last 20 tenders. Saving: 49 days × KZT 320 000 / day of frozen budget × 45 tenders/year = **KZT 706 M / year**.

## Direct benefit — L1 improvement

Broader bidder attention (71 % vs baseline 42 %) yields ~3.4 % improvement in awarded price. On KZT 24.8 bn/year of awarded value that is **KZT 843 M / year**.

## Direct benefit — audit cost reduction

Immutable audit trail + downloadable evidence bundles reduce external-audit fees by ~40 %. Baseline audit spend KZT 84 M/year → saving **KZT 33.6 M / year**.

## Direct benefit — dispute reduction

Corrigenda discipline + Q&A moderation drop the dispute rate from 6.1 % to 2.4 %. Average dispute cost KZT 12 M. Saving: **KZT 20 M / year** on 45 tenders.

## Cost — platform fees

Monthly platform fee KZT 32 M × 12 = **KZT 384 M / year**. Per-notice fee KZT 640 000 × 45 = KZT 28.8 M / year. Total operating cost **≈ KZT 413 M / year**.

## Cost — internal ops

3 FTE for admin + 2 FTE for escrow + 1 FTE for support. Fully loaded cost **≈ KZT 96 M / year**.

## Net benefit

Benefits KZT 1 603 M − costs KZT 509 M = **net KZT 1 094 M / year** on the current KZ programme. Payback within the first tender cycle.

## Sensitivity

A 20 % drop in L1 improvement erodes benefit by KZT 168 M. A 30 % expansion in tender volume adds KZT 480 M. Break-even at 21 published notices per year.

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# 51 · Committee minutes template

## Header

**Meeting:** Tender Committee for PMC-NOT-YYYY-NNNN · **Date/time:** YYYY-MM-DD HH:MM UTC · **Venue:** Zoom (recorded) · **Chair:** [Name] · **Secretary:** [Name]

## Attendance

Voting members present: [Names]. Voting members absent: [Names + reason]. Observers present: [Names]. Quorum: [YES / NO].

## COI declarations

Each voting member's COI status. Any recusals + reallocation of weight.

## Rubric confirmation

Rubric reviewed, no changes / changes as follows: [list]. Rubric locked at [timestamp].

## Technical scoring

Per-bidder technical score summary. Any deviations from the rubric noted with rationale. Technical envelope locked at [timestamp] by [Name].

## Commercial opening

Commercial envelopes opened at [timestamp]. Per-bidder commercial score. Anomalies flagged: [list].

## QCBS composite + ranking

Weighted composite scores. Ranking. Tie-breaker rules invoked: [YES / NO].

## L1 recommendation

Committee recommends award to [Bidder] at [price]. Rationale.

## Dissents

Any dissenting opinions attached, with signatures.

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## Actions

Legal: draft LoA within 3 business days. Finance: issue PO within 5 business days. Support: notify runner-up + non-qualifiers by end of week.

## Signatures

Digital signatures of all voting members via DocuSign envelope [id].

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# 52 · Procurement policy summary

## Guiding principles

Openness · non-discrimination · value-for-money · transparency · documented audit trail · proportionality of process to value.

## Above-threshold tenders

Notices ≥ KZT 250 M must be published on the PMC platform + the national gazette. Pre-bid meeting mandatory. Evaluation committee minimum 5 members with an external observer.

## Restricted tenders

Restricted-list tenders (defence, critical infrastructure) require an additional CEO + Board sign-off. Bidder shortlist justified in the tender packet. Public visibility limited to award notice.

## Single-source procurement

Permitted only under Article 39 of the Public Procurement Law. Requires (a) written justification, (b) CFO + Legal sign-off, (c) public disclosure within 15 days of award.

## Framework agreements

Multi-year framework agreements allowed for repeatable services. Call-offs against a framework do not require re-publication but each call-off is audit-logged with the reference framework MoU.

## Anti-bid-rigging

The AI anomaly detector runs on every closed tender. Patterns above the threshold auto-open a Legal review. Confirmed rigging → three-year debarment + regulator SAR.

## Debarment

Debarred vendors are marked in the master with a red badge; the vendor cannot log in for the debarment period. Debarment is public; the annual disclosure lists every debarment.

## Emergency procurement

Emergency purchases (natural disaster, national security) may bypass the standard timeline with CEO + Board sign-off. All emergency procurement is post-published within 30 days with the emergency rationale.

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## Complaints & appeals

Bidders may lodge a complaint within 7 days of any decision. Complaints route through Legal; final appeal is the national procurement dispute panel. Timelines and outcomes are audit-logged.

## Annual review

The procurement policy is reviewed annually by the Board + external observer. Amendments require CEO sign-off + Board minute reference.

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# 53 · Localisation strings (EN / RU / KK)

## Public — hero

| Key | Translations |
|---|---|
| `home.hero_kicker` | EN: PMC Managers · Sovereign Tender Board · RU: PMC Managers · Суверенная тендерная площадка · KK: PMC Managers · Егемен тендер тақтасы |
| `home.hero_headline` | EN: Sovereign procurement for a modern republic · RU: Суверенные закупки для современной республики · KK: Заманауи республика үшін егемен сатып алулар |
| `home.hero_lede` | EN: Publish, evaluate, award and audit — end to end… · RU: Публикация, оценка, награждение и аудит — от начала до конца… · KK: Жариялау, бағалау, марапаттау және аудит — басынан аяғына дейін… |
| `home.cta_view_notices` | EN: View open notices · RU: Открытые извещения · KK: Ашық хабарламалар |
| `home.cta_register` | EN: Register as supplier · RU: Регистрация поставщика · KK: Жеткізуші ретінде тіркелу |

## Login gateway

| Key | Translations |
|---|---|
| `login.gateway_kicker` | EN: PMC Managers · Sovereign Gateway · RU: PMC Managers · Суверенный шлюз · KK: PMC Managers · Егемен қақпа |
| `login.select_portal` | EN: Select your command portal · RU: Выберите свой командный портал · KK: Басқару порталыңызды таңдаңыз |
| `login.portal_admin` | EN: System Administrator · RU: Системный администратор · KK: Жүйе әкімшісі |
| `login.portal_management` | EN: PMC Management · RU: PMC Руководство · KK: PMC Басшылық |
| `login.portal_partners` | EN: Strategic Partners · RU: Стратегические партнёры · KK: Стратегиялық серіктестер |
| `login.portal_agents` | EN: Procurement Agents · RU: Агенты по закупкам · KK: Сатып алу агенттері |
| `login.portal_vendors` | EN: Registered Vendors · RU: Зарегистрированные поставщики · KK: Тіркелген жеткізушілер |

## Notice detail

| Key | Translations |
|---|---|
| `notice.tender_value` | EN: Estimated tender value · RU: Ориентировочная стоимость тендера · KK: Тендердің болжамды құны |

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| | |
|---|---|
| `notice.emd_amount` | EN: EMD amount · RU: Сумма гарантийного взноса · KK: Кепілдік жарнасы |
| `notice.sub_end` | EN: Submission deadline · RU: Крайний срок подачи · KK: Ұсыну мерзімі |
| `notice.eligibility` | EN: Eligibility · RU: Требования к участникам · KK: Қатысушыларға қойылатын талаптар |

## Evaluation

| Key | Translations |
|---|---|
| `eval.sealed` | EN: Sealed · RU: Опечатано · KK: Мөрленген |
| `eval.lock_tech` | EN: Lock technical envelope · RU: Опечатать техническую часть · KK: Техникалық конвертті мөрлеу |
| `eval.lock_comm` | EN: Lock commercial envelope · RU: Опечатать коммерческую часть · KK: Коммерциялық конвертті мөрлеу |
| `eval.rubric_save` | EN: Save rubric · RU: Сохранить критерии · KK: Критерийлерді сақтау |

## Vendor tabs

| Key | Translations |
|---|---|
| `vendor.tab_tenders` | EN: Open tenders · RU: Открытые тендеры · KK: Ашық тендерлер |
| `vendor.tab_bids` | EN: My bids · RU: Мои заявки · KK: Менің өтінімдерім |
| `vendor.tab_rate` | EN: Ratings · RU: Оценки · KK: Бағалар |
| `vendor.tab_sign` | EN: Signatures · RU: Подписи · KK: Қолтаңбалар |

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# 54 · Vendor sandbox playbook

## What it is

A parallel PMC environment with dummy tenders, dummy escrow, and dummy DocuSign so a new vendor can practise the full journey — registration, bid pack purchase, bid submission, reverse auction — without touching live data.

## How to enable

Support opens Admin → Vendors → row → **Sandbox mode**. The vendor immediately sees a *Sandbox* banner across the vendor portal and every action prefixes its notification subject with *[SANDBOX]*.

## Guardrails

Sandbox payments route to the Stripe test mode key; the DocuSign envelopes are dispatched from the demo account; EMDs are recorded to a separate `escrow_transactions_sandbox` collection and never mixed with the live ledger.

## Duration

14 days by default; extendable in 7-day chunks. Auto-graduates the vendor when they successfully bid on 3 sandbox tenders.

## Coach mode

The sandbox banner includes a coach chat with contextual tips — for example, on the compliance-matrix screen the coach explains what YES/NO/DEVIATION mean and how to phrase deviation rationale.

## Reset

Support can Reset the vendor's sandbox state in one click; every dummy bid + EMD entry is purged and the vendor starts fresh.

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# 55 · Committee training curriculum

## Module 1 — Public procurement law refresher

3-hour class covering the RK Public Procurement Law, GFR 2017 equivalents, and the specific chapters PMC uses as guardrails. Includes a 30-question knowledge check.

## Module 2 — Evaluation rubrics done right

Weight design, scale calibration, minimum-technical cutoffs, tie-breaker discipline, and the perils of criteria bloat. Includes 5 sample rubrics to critique.

## Module 3 — Bid comparison

Compliance-matrix reading, deviation classification, interpolation of missing values, and applying the AI redline output as an aid (not a substitute) for committee judgement.

## Module 4 — Anti-bid-rigging heuristics

Bid-clustering, price-dumping, alias-reuse, EMD-refund pattern anomalies. When to invite clarification, when to escalate to Legal.

## Module 5 — Committee ethics

COI declarations, recusal, confidentiality, signature discipline, dissent process. Real cases from anonymised PMC minutes.

## Module 6 — Award report & post-mortem

L1 selection, sensitivity analysis, dispute-window handling, feedback loop into rubric v2 for future tenders.

## Certification

Passing the knowledge checks in all 6 modules certifies a member. Re-certification annually + on any material policy change.

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# 56 · Ethics code

## Integrity

Every PMC staff member and committee member exercises independent judgement free of external interference.

## Impartiality

Neither favour nor prejudice colours evaluation. Rubrics are applied uniformly across bidders.

## Confidentiality

Every committee document is Confidential — Internal Use Only unless explicitly cleared for publication.

## Conflict of interest

Any material relationship — equity, past employment, first-degree family, gifts above USD 100 — is disclosed and triggers recusal.

## Gift policy

Gifts above USD 100 / year in aggregate from any single bidder are prohibited. Below-threshold gifts are declared.

## Anti-bribery

Any offer of bribe is reported to Legal + CEO within 24 h. The offering party is banned pending Legal review.

## Whistle-blower protection

Staff reporting misconduct in good faith are protected from retaliation. Reports route to the DPO + external audit committee.

## Speaking on behalf of PMC

Only the CEO, CFO, CTO or a designated spokesperson may speak on behalf of PMC to media, regulators, or the public.

## Use of AI

AI outputs are advisory. Every awarded decision has a human sign-off on the audit trail.

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## Enforcement

Violations lead to formal review + possible termination, per the HR policy. The Board is informed of every substantiated violation.

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# 57 · Board reporting pack

## Executive summary

One page: KPIs vs target, headline wins, any P0 / P1 concerns, next-quarter focus.

## Financial pack

Revenue vs plan, cost lines, cash position, aged receivables, next-quarter forecast.

## Programme pack

Notices published + awarded (vs plan), contract portfolio, dispute rate, vendor NPS.

## Operations pack

SLA hit-rate (refund, LoA turnaround, milestone), incidents, drill outcomes.

## Risk pack

Top 10 risks with heat-map, mitigation status, new risks identified this quarter.

## Compliance pack

Audit trail summary, regulator interactions, ESG progress, PII incidents.

## People pack

Headcount, turnover, DE&I metrics, training completion.

## Product pack

Release train history, AI copilot uptake, upcoming features, technical debt burn-down.

## Q&A + resolutions

Motions for Board approval; ratifications of prior circulars; standing orders.

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# 58 · IT operations manual

## Runbook custody

IT operations runbooks live under `/app/deliverables/ops/`. Every playbook has an owner + review date + last-executed timestamp.

## Environment matrix

prod (EB + S3 + CloudFront + Mongo Atlas M40), preview (this environment), sandbox (scratch cluster), DR (paired region).

## Secret rotation

Managed AI key auto-rotated by the platform. Stripe, DocuSign, Resend keys rotated annually + on incident. JWT signing key rotated on suspicion of exposure.

## Log retention

CloudWatch 90 days hot + 5 years cold. Audit vault 10 years / 15 years as per chapter 34.

## Monitoring

CloudWatch metrics + PagerDuty routing. Custom metrics: dpr_job_latency_p95, envelope_lock_rate, audit_row_lag.

## Backup

Nightly Mongo dump (7-day rolling S3), weekly Glacier cold-copy (12-month rolling), quarterly restore drill.

## Capacity planning

Every doubling of published notices triggers a capacity review — CPU, RAM, LLM quota, S3 spend, Mongo Atlas tier.

## Cost management

Monthly cost report to the CTO; anything above 15 % QoQ opens a review with the sponsoring finance lead.

## Vendor management

Renewals tracked in Masters → Integrations → Renewals. 90-day advance notice on any AI Gateway / AWS / Mongo Atlas / DocuSign / Stripe / Resend renewal.

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## End-of-life policy

Any dependency EOL'd by upstream triggers a migration project with a 6-month cushion before the upstream end-date.

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# 59 · Sample Q&A digest

## Q01 — SCADA standards

The tender specifies Modbus TCP + OPC-UA. Are proprietary Siemens S7 or Allen-Bradley EtherNet/IP protocols acceptable as additional layers? — **A:** Yes, provided the mandatory Modbus TCP + OPC-UA layers are fully implemented and remain the primary telemetry channel. Vendors offering only proprietary stacks will be technically rejected.

## Q02 — Bid bond form

Is a corporate guarantee acceptable in lieu of a bank guarantee? — **A:** No. Only unconditional, first-demand bank guarantees issued by Category-A RK banks are acceptable. See Corrigendum C-02 for the expanded bank list.

## Q03 — Local content

Is local content tested at award only, or also at contract completion? — **A:** Both. A committed percentage at award and a verified percentage at project completion; a 5% contract-value penalty applies if the completion figure is below the committed figure.

## Q04 — Pipeline coating

Is polyethylene coating acceptable, or is only fusion-bonded epoxy? — **A:** Fusion-bonded epoxy (FBE) is mandatory for the trunk main; PE coating is acceptable only for distribution branches ≤ DN 200.

## Q05 — Design responsibility

Is this a design-build contract or execution-only against the Engineer's design? — **A:** Design-build. Bidders must include a design responsibility statement and PI insurance evidence.

## Q06 — Currency

May the imported SCADA hardware be quoted in EUR instead of USD? — **A:** Yes; supply an FX-rate assumption clause and lock at contract signing.

## Q07 — JV

Can two contractors bid as a Joint Venture? — **A:** Yes; submit a notarised JV agreement, joint bid bond, and a lead-partner designation. Combined turnover counts toward the eligibility bar.

## Q08 — Subcontracting

What is the maximum subcontracted share? — **A:** 40 % of contract value. Every subcontractor above USD 500 K is pre-approved by the Engineer.

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## Q09 — Insurance

Minimum PI + CAR insurance thresholds? — **A:** PI USD 5 M / claim, CAR at contract value + 10 %. Certificates to be endorsed to PMC + Sponsor.

## Q10 — Payment currency

Advance and retention in KZT or USD? — **A:** KZT for local scope; USD for imported hardware pass-through only, evidenced by supplier invoices.

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# 60 · Sample bid evaluation matrix

## Bidder A — NAI

Tech 82.4 · Comm 95.1 · QCBS 86.2 · Rank **1** · Deviations 3 (minor) · Local content 63 %

## Bidder B — SarPipe

Tech 78.9 · Comm 88.2 · QCBS 81.7 · Rank 2 · Deviations 5 (minor) · Local content 71 %

## Bidder C — AsiaBuild

Tech 74.1 · Comm 92.4 · QCBS 79.6 · Rank 3 · Deviations 4 (2 major) · Local content 55 %

## Bidder D — EuroFlow

Tech 88.2 · Comm 71.0 · QCBS 83.0 · Rank 4 (very close 2nd on tech, but priced highest) · Deviations 1 · Local content 22 %

## Bidder E — KZ-Hydro

Tech 61.5 · **below minimum technical 65** → technically rejected · not scored commercially · Deviations 8

## Notes

Committee unanimous on Bidder A. Bidder D flagged as an excellent runner-up should Bidder A decline. Bidder C invited for clarification on the two major deviations.

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# 61 · Standard clauses library

## Payment terms

Advance 10 % · Interim 80 % against certified milestones · Retention 10 %, released 12 months after handover subject to defect-liability closure.

## Variation orders

Any change of scope > USD 25 K or > 3 % of contract value requires a written variation order signed by the Engineer + Contractor + PMC before execution.

## Liquidated damages

0.5 % of contract value per week of delay, capped at 10 % of contract value. LDs are not exclusive; PMC reserves the right to terminate for cause after 12 weeks of accumulated delay.

## Termination for cause

30-day cure notice for material breach; termination after cure fails; PMC may draw the bid bond and appoint a substitute contractor at the defaulting party's risk and cost.

## Termination for convenience

Either party may terminate on 60 days' written notice. Contractor entitled to work-done value + 8 % overhead on unexecuted balance up to 6 % of contract value.

## Force majeure

Standard definition + 30-day extension per FM event + right to terminate after 180 days of continuous FM. Force majeure claims must be filed within 14 days of onset.

## Governing law

Laws of the Republic of Kazakhstan. Disputes resolved per the arbitration clause (AIAC rules, Astana seat, English language).

## Warranty

24 months defect-liability from taking-over. Any latent defect appearing within 24 months rectified at Contractor's cost within 30 days of notice.

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## Indemnity

Contractor indemnifies PMC + Sponsor against third-party claims arising from Contractor's negligence, breach or wilful misconduct. Cap: contract value + 10 %.

## Confidentiality

Both parties treat all project information as Confidential — Internal Use Only. Publications require prior written consent + fact-check by Legal.

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# 62 · Communications templates

## Bidder invitation email

Subject: Invitation to bid — PMC-NOT-YYYY-NNNN · Body: opens with the notice title, ends with the tender-pack purchase link + submission deadline in the bidder's timezone. Includes the RFC 8288 unsubscribe link for the digest.

## Corrigendum notification email

Subject: Corrigendum C-NN — PMC-NOT-YYYY-NNNN · Body: one-paragraph summary + full-diff link + impact statement. Sent within 5 min of publication.

## Award-decline runner-up email

Subject: Invitation to accept award (runner-up) — PMC-NOT-YYYY-NNNN · Body: original L1 declined, runner-up invited to accept within 5 business days; LoA drafted from the same template.

## Refund notice email

Subject: EMD refund initiated — PMC-BID-YYYY-NNNN · Body: refund amount, bank routing, expected credit date, contact for reconciliation.

## Regulator quarterly digest

Subject: PMC quarterly disclosure — Q?-YYYY · Body: KPI table, notable awards, incidents (if any), roadmap outlook. Attached: signed PDF + CSV.

## Vendor NPS survey

Subject: How was your recent bid experience? · Body: 1-question NPS (0-10) + free-text comment. Auto-triggered 14 days after award decision. Aggregated in Analytics.

## Committee meeting invite

Subject: Committee meeting — PMC-NOT-YYYY-NNNN — YYYY-MM-DD · Body: agenda, Zoom link, materials pack link, quorum reminder, COI declaration form.

## Incident customer notice

Subject: Service update — PMC platform · Body: what happened, when, who was affected, what we did, what changes to prevent recurrence. Requires CTO + Legal sign-off before send.

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# 63 · Vendor NPS survey

## Question 1

On a scale from 0-10, how likely are you to recommend the PMC platform to a peer procurement organisation?

## Question 2

What did you like most about your recent experience?

## Question 3

What one thing would you change?

## Question 4 (conditional, score 0-6)

What went wrong, and how can we make it right?

## Question 5 (conditional, score 9-10)

May we quote your feedback in our annual disclosure? (Y/N)

## Scoring

Promoters (9-10) minus Detractors (0-6) as a percentage of respondents. Target NPS ≥ 42. Below 30 opens a vendor-experience review with the CEO.

## Publication

Aggregate score published quarterly on the Partner portal + summarised in the annual disclosure. Individual scores never linked to a bidder identity.

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# 64 · Debarment log format

## Row schema

{debarment_id, vendor_ref, vendor_legal_name, reason, effective_from, effective_to, decision_body, rationale_pdf_url, appeal_status, publication_url}

## Sample row 1

DEB-2024-003 · PMC-SUP-2019-4471 · Falsified compliance matrix in PMC-NOT-2024-0219. 3-year debarment 2024-08-14 to 2027-08-13. Decision body: Board of Directors. Appeal: rejected 2024-11-02.

## Sample row 2

DEB-2025-012 · PMC-SUP-2022-0119 · Cartel behaviour flagged by anomaly detector, confirmed by Legal. 5-year debarment 2025-03-01 to 2030-02-28. Decision body: Board. SAR filed with regulator.

## Publication

Debarment log is public at /transparency/debarment. Search + filter by year + reason class. Regulator receives a webhook on every new debarment.

## Appeal process

Vendor may appeal to the Board within 30 days of debarment. Appeal is decided within 60 days. During appeal, the vendor remains debarred but marked *Under appeal* in the log.

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# 65 · SMS / in-app message library

## Login OTP (SMS)

PMC verification code: 728341. Valid for 5 minutes. Do not share this code.

## Auction start (SMS)

PMC · Reverse auction for PMC-AUC-2026-0083 starts in 15 minutes. Join now: [short-link].

## Envelope lock (in-app)

The technical envelope for PMC-NOT-2026-0472 has been sealed at 2026-11-03 10:02 UTC by adminsupport@pmcmanagers.com. No further technical scores can be submitted.

## Award decision (in-app)

Congratulations — you have been awarded PMC-NOT-2026-0472. Please review the LoA and countersign within 5 business days.

## EMD refund (in-app)

Refund of KZT 96,250,000 for PMC-BID-2026-1183 initiated. Expected credit within 5 business days.

## Password reset (email)

You requested a password reset. Follow this signed link within 15 minutes: [signed-link]. Ignore if you did not request this.

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# 66 · Release planning — CY 2027

## Q1 2027 roadmap

P0: i18n Russian + Kazakh string backfill · P0: Manual v3 with modal captures via Playwright · P1: Firebase / APNs mobile push · P1: LSEG / World-Check integration · P2: Kaspi payments alongside Stripe.

## Q2 2027 roadmap

P0: Zoom integration for evaluation committees · P1: PayPal payments · P1: Public transparency portal · P2: Debarment appeal workflow · P2: Auction spectator SSE feed.

## Q3 2027 roadmap

P0: Multi-tenant billing separation · P1: Bulk vendor import CSV · P2: AI copilot for compliance-matrix review · P2: Contract-milestone auto-nudging.

## Q4 2027 roadmap

P0: Annual disclosure automation · P1: Sponsor-branded white-label UI · P2: Advanced analytics with cohort funnels · P2: Vendor NPS auto-tagging via LLM.

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# 67 · Manual maintenance guide

## Ownership

The manual is owned by the CTO. Content editors: Product, Legal, Operations. Review cadence: quarterly + on every material release.

## Refresh cadence

Screens re-captured after each visual release (any 'v' version bump that changes UI). Text refreshed after every functional release. Cover date auto-bumps.

## Toolchain

`/app/tools/manual/capture.py` (Playwright) + `/app/tools/manual/build.py` (WeasyPrint + python-docx). `build_extensions.py` holds long-form appendix data.

## Regenerating

SSH into the preview environment. `python3 /app/tools/manual/capture.py` to refresh screens. `python3 /app/tools/manual/build.py` to rebuild MD + DOCX + PDF.

## Publishing

Copy the three artefacts + the zipped bundle from `/app/deliverables/v46_manual/build/` into `/app/frontend/public/downloads/`. They will be served at `https://<host>/downloads/PMC_v46m_Master_Training_Manual.pdf`.

## Versioning

Filename carries the release code (e.g. `PMC_v46m_...`). The cover shows the release code + month/year. Old manuals are retained in `/app/deliverables/v46_manual/archive/`.

## Translation

Next milestone: full RU + KK translations. See chapter 53 for the key inventory. A translator glossary lives under Masters → Documents → Translator glossary.

## Feedback

Every reader can email <manual-feedback@pmcmanagers.com>. Feedback is triaged weekly by the CTO's office and folded into the next quarterly refresh.

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# 16 · Role-by-role onboarding checklist (30 / 60 / 90 days)

**Admin — Day 1.** Shadow an experienced admin through a full tender lifecycle (draft → publish → bid receipt → evaluation → award → contract). Read chapters 1–3 and 7–11.

**Admin — Day 30.** Publish your first tender end-to-end unsupervised. Own one committee evaluation cycle. Complete audit-trail search drills for two entity types.

**Admin — Day 60.** Configure a new country's masters. Run one Tender Builder DPR ingestion on a 500-page tender.

**Admin — Day 90.** Lead a full quarterly stakeholder-report cycle. Present KPI trends to the Management portal.

**Vendor onboarder (Support).** Day 1 shadow a live activation. Day 30 process 10 registrations solo. Day 60 own resend-link edge cases and licence re-verification queues. Day 90 run a vendor-education webinar.

**Escrow officer.** Day 1 read the joint-account snapshot model. Day 30 process 20 EMD receipts + 10 refunds. Day 60 handle one forfeiture with legal sign-off. Day 90 lead a monthly reconciliation against Stripe.

**Committee member.** Day 1 COI declaration training. Day 30 score 3 evaluations under a senior member. Day 60 lead one evaluation session. Day 90 chair a full committee meeting including minutes and public Q&A migration.

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# 17 · Monthly operations cadence

**Week 1** — Publish weekly digest via Resend. Reconcile prior-month escrow ledger. Publish stakeholder monthly report.

**Week 2** — Committee cycle for tenders whose submission deadline landed in the prior week.

**Week 3** — Award ceremonies + DocuSign countersign push. Vendor invoicing run.

**Week 4** — Roadmap review with GOC + Bank partners. Vendor NPS survey via Resend. Category taxonomy audit.

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# 18 · North-Star KPIs and dashboards

| KPI | Definition | Target | Notes |
|---|---|---|---|
| Notice cycle time | Draft → Publish | ≤ 14 days | Every published notice contributes its `published_at - draft_created_at`. |
| Bidder attention | Bidders opening a notice / eligible bidders | ≥ 65 % | Analytics funnel first stage. |
| Bids per notice | Median across last 90 days | ≥ 4 | Signals healthy competition; below 3 triggers a category audit. |
| Technical qualification rate | Technically qualified / bids received | 40–65 % | Too low → tender spec too strict; too high → not selective enough. |
| EMD refund SLA | Refund issued / eligible transactions in 30 days | 100 % | Any breach opens an escrow post-mortem. |
| Contract signature SLA | LOA → PMC countersign | ≤ 10 business days | Legal team KPI. |
| Vendor NPS | Quarterly survey | ≥ 42 | Below 30 triggers a vendor-experience review. |
| Dispute rate | Opened disputes / awarded contracts | ≤ 3 % | Above 5 % opens a legal + evaluation review. |

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# 19 · Escalation matrix

| Situation | First responder | 15 min | 1 h | 4 h | Owner |
|---|---|---|---|---|---|
| Auth outage | On-call admin | On-call engineer | CTO | CEO | CTO |
| EMD refund overdue > 60 days | Escrow officer | Legal | Escrow lead | CFO | CFO |
| Suspicious auction pattern | Auction ops | AI anomaly rerun | Legal | CEO | Legal |
| Data breach suspected | Any staff | CTO | CEO | Communications | CTO |
| Regulator inbound | Any staff | Partner lead | CEO | Board | Partner lead |

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# 20 · Keyboard shortcuts

| Shortcut | Action |
|---|---|
| ⌘ + K | Global command palette (planned) |
| G then A | Jump to Analytics |
| G then N | Jump to Notices |
| G then T | Jump to Tender Builder |
| G then E | Jump to Evaluation |
| Esc | Close any open modal |
| / | Focus the primary search box |
| Shift + ? | Show the shortcut cheat sheet (planned) |

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# 21 · Frequently asked questions (FAQ)

**Q — Can I edit a published notice?**

No — publish a corrigendum. The audit trail keeps every prior state and public visitors see the corrigendum immediately on the notice detail page.

**Q — How do I recover a lost tender draft?**

Every save is a rolling snapshot. Open Tender Builder → Version history and restore any prior snapshot; the restore itself is auto-snapshotted so it is undoable.

**Q — A vendor says they never received a fee invoice.**

Check the Billing tab for the invoice PDF and the Resend delivery log. If the email failed, resend from Billing → Row action. Corporate MX filters sometimes block first-touch domains — use the warm-up sender when needed.

**Q — How large a DPR can the AI ingest?**

v46l allows 100 MB per file / 500+ pages. Files ≥ 4 MB automatically use the async worker and show a live progress bar; you can leave the tab open and come back.

**Q — Can I re-open a locked evaluation envelope?**

No. v46m rejects re-lock with a 409. If you truly need to redo scoring, clone the notice into a new tender and re-run the evaluation cycle. Legal must approve.

**Q — How do I add a new country to the platform?**

Masters → Countries → New. Then seed EMD/fee slabs, category taxonomy, licences and staff roles for that country. The public homepage and registration wizard update immediately.

**Q — How does the AI avoid inventing numbers not present in the DPR?**

The section-drafting prompt forbids inventing numbers, and each draft is passed the specific DPR passage anchored to the section title. Bodies under 200 chars trigger the raw-passage fallback so no output is generated without source.

**Q — Where do I turn off email digests?**

Vendors: Vendor portal → Notifications. Staff: Profile → Preferences (planned).

**Q — How is bidder identity protected in reverse auctions?**

Masked aliases assigned server-side on join. Even the audit trail stores only aliases; de-masking is admin-only post-close.

**Q — A stakeholder-report artefact failed to snapshot. What now?**

Every artefact is wrapped in its own try/except; the response returns a JSON `errors[]` array listing which artefact failed and why. Fix the specific artefact and re-hit /reports/snapshot — the rest are unaffected.

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# 22 · Sample workflow — full lifecycle of one tender

**Day 0.** Package owner briefs the drafting agent. Agent opens Tender Builder → picks the country → picks or creates a master → opens the DPR modal and uploads the project brief. AI drafts every section body from the DPR passages. Agent reviews, adjusts EMD/fee, uploads BOQ CSV, saves the draft.

**Day 3.** Legal counsel reviews. Any redlines go into a version snapshot. Legal signs off. Agent hits Publish. Resend digests fire to subscribed vendors. Snapchat + Instagram posts pushed automatically.

**Day 3–17.** Pre-bid meeting on Day 5. Minutes uploaded, substantive Q&A migrated to the public notice. Corrigenda published as needed. Bidders purchase the tender pack; EMDs post to escrow.

**Day 18.** Submission deadline. Admin locks technical envelope; the button flips to 'Sealed ' and a green banner records the sealer and timestamp.

**Day 18–28.** Evaluation committee scores technical. Once all evaluators have submitted, admin locks commercial envelope. Committee scores commercial. Admin generates the award report → L1 identified.

**Day 29.** LOA issued via DocuSign. Vendor accepts; PMC countersigns. PO number generated.

**Day 30–...** Contract runs. Milestones, timesheets, variation orders logged. EMD refunded on Day N post-close unless disputed. Vendor rating captured. Auditor exports the CSV bundle if requested.

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# 23 · Data model appendix

| Collection | Purpose | Hot indices |
|---|---|---|
| `tenants` | Multi-tenant scoping | `code (unique)` |
| `admin_users / staff_users / vendors` | Auth realms | `email (unique per tenant)` |
| `notices` | Published tender notices | `notice_ref, published_at, country` |
| `tender_templates` | Reusable master + drafts | `(country, industry), is_master` |
| `registrations` | Vendor onboarding wizard state | `magic_token, email, status` |
| `bids` | Submitted bids | `notice_ref, vendor_ref` |
| `rubrics` | Per-notice scoring config | `notice_ref (unique)` |
| `scores` | Evaluator scores | `bid_id, evaluator` |
| `escrow_transactions` | EMDs held/refunded/ forfeited | `notice_ref, status` |
| `auctions` | Reverse auction rooms | `notice_ref (unique)` |
| `corrigenda` | Notice amendments | `notice_ref, published_at` |
| `audits` | Immutable audit trail | `entity_ref, actor_id, at` |
| `ratings` | Post-contract reviews | `vendor_ref` |
| `dpr_jobs` | v46l async DPR workers | `_id, status` |
| `reports_snapshots` | Materialised stakeholder artefacts | `generated_at` |

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# 24 · Appendix — release notes archive

- **v46m** — Bug Blitz Round Two — envelope re-lock 409, GET /lock-state, EvaluationPanel sealed banner, AuctionRoom mobile overflow fix.
- **v46l** — Unlimited DPR + async worker (MAX_CHUNKS 500, /dpr/analyze-async, /dpr/jobs/{id}, nginx 200 MB / 300 s, live progress bar).
- **v46k** — Sovereign Redesign — 5-portal Sovereign Gateway, navy/gold editorial home hero, PmcFooter with social icons.
- **v46j** — Map-Reduce DPR first cut (still capped at 800 KB).
- **v46i** — New roles (management/partner/agent), collapsible admin sidebar, editorial public design v1.
- **v46h** — Admin vendor form: per-section Save buttons + search.
- **v46g** — DPR deep-dive first cut (26 sections from KMT).
- **v46f** — Editable tender name + reliable save-as-master.
- **v46e/d/c/b** — Bug blitz first pass — 60+ tracker items fixed.

DENTIAL — INTERNAL USE

# 68 · Vendor performance scorecard framework

## Purpose

The scorecard aggregates every observable signal a vendor generates during a contract — cost, schedule, quality, safety, communication, disputes — into a single 0-100 index visible to Admins, Committees and (opt-in) buyer stakeholders. It replaces the ad-hoc gut-feel scoring that used to bias L1 recommendations.

## Data feeds

The engine pulls from: `contracts.milestones[]` for on-time delivery, `escrow_transactions` for payment friction, `ratings` for post-award reviews, `audits` for compliance touches, `notices ↔ bids` join for win-rate, `corrigenda` counts for scope-change churn, and the DocuSign envelope-status stream for signature discipline.

## Weights

Cost adherence 25 %, schedule adherence 25 %, quality (defect-free acceptance) 15 %, communication + Q&A responsiveness 10 %, safety incidents 10 %, dispute count 5 %, ESG self-attestation 5 %, community sub-contracting 5 %. Weights are configurable per-tenant in Masters → Scorecard.

## Bands

> 85 = Preferred (green pill, appears on public trust badge). 70-84 = Standard. 55-69 = Under review (flag surfaces on every future bid). < 55 = Restricted (auto-rejects future bids unless overridden with justification).

## Refresh cadence

The `_recompute_scorecard()` worker runs nightly at 02:15 UTC. Manual re-runs are exposed at `POST /api/v1/vendor-intel/scorecard/recompute` (admin+auditor). Every recompute writes an audit row with the delta so trends are queryable.

## Governance

A monthly Vendor Council reviews every vendor whose band crossed a threshold. Restricted vendors must be formally debarred (see chapter 64) — a low scorecard alone is not sufficient.

## Anti-gaming

Any single evaluator whose scores deviate from committee median > 20 pts on three consecutive bids triggers a mandatory calibration session. The scorecard downweights outlier evaluators via a Grubbs test before averaging.

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## Publishing to vendors

Vendors see their own scorecard inside the vendor portal — with sub-scores, trend line and remediation tips. Sensitive weightings are hidden. Appeals go through `POST /api/v1/vendor-intel/scorecard/{ref}/appeal`.

DENTIAL — INTERNAL USE

# 69 · Cross-border payment settlement runbook

## In scope

Kazakhstan (KZT), CIS neighbours (RUB, UZS, KGS, AZN), Gulf partners (USD, AED, SAR) and EU sub-contractors (EUR, GBP). Stripe covers card + ACH; Kaspi Business handles KZT rails; a manual SWIFT fallback exists for regulated Gulf flows.

## Rails matrix

Each currency pair has a preferred rail. Rail selection happens in `_select_payout_rail(currency_in, currency_out, amount)` : Stripe for anything under $10 k that stays inside G10; Kaspi for any KZT leg; SWIFT for > $10 k Gulf; PayPal only where the vendor explicitly requests it and the buyer approves.

## Escrow bridge

The escrow account model is currency-agnostic — every transaction stores `amount` , `currency` , `fx_rate_used` , `fx_source` , `fx_captured_at` . When funds move between two rails the platform locks the mid-market rate + 30 bps and records both legs so nothing is lost to slippage.

## Reconciliation

Every settlement writes a `settlements` row that references the source `escrow_transaction._id` and the payment-provider webhook. The nightly recon job matches every provider webhook to a settlement — mismatches raise a Sev-3 ticket automatically.

## KYC hooks

Cross-border payouts require a passed KYC status + LSEG World-Check clearance no older than 180 days. The rail-selector refuses to route a payout when either signal is stale — the UI surfaces the exact reason ('KYC re-verification due') so ops knows what to fix.

## Regulator reporting

Monthly Form-1B is auto-generated from `settlements` filtered by rail=SWIFT. Kazakhstan National Bank filings run via Kaspi Business's API on the 5th business day. Both hit Legal for sign-off before submission.

## Failure modes

Rail rejection is not treated as a full failure — the transaction stays `held` and the engine retries the next rail (per priority) until success or exhaustion. After exhaustion the transaction moves to `manual_intervention` and Sev-2 pages Escrow ops.

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# 70 · AI copilot governance and audit trail

## Positioning

Every AI surface is **advisory-only**. No AI-generated content is ever written to a legal or immutable audit record. The human evaluator, admin or vendor is always the recorded actor.

## Inventory

1) DPR extractor (ingestion pipeline); 2) Section drafter (Tender Builder); 3) Bid-anomaly detector (Analytics); 4) Committee deliberation summariser; 5) Vendor Q&A auto-suggest; 6) Public-notice de-jargon reviewer. All route through the managed AI Gateway (universal LLM key) proxy.

## Prompt registry

Every prompt lives under `backend/pmc_lib/prompts/` with a semver tag. Changes require a PR + reviewer + prompt-diff attached to a governance ticket. The registry is immutable; historical prompts are retained for 7 years to support audit replay.

## Model registry

Model choice per surface is tracked in `ai_governance.models`. Downgrading a model (e.g. GPT-5.6 → GPT-5.4) is a change-managed action requiring CTO sign-off. The runtime records model + prompt version on every response — so any output can be re-produced exactly.

## Bias controls

DPR extraction is checked nightly against a golden-set corpus of 40 known tenders. If precision or recall drops > 3 pts vs baseline, the ai-governance-check CI fails and blocks the release. Bid-anomaly detector is checked for demographic drift on vendor country-of-origin.

## Explainability

Every AI recommendation the UI surfaces has a 'Why?' expander that shows the prompt digest + response digest + retrieved context. This is required for regulator conversations.

## Human veto

Committee members and admins can veto any AI recommendation. The veto is recorded as an audit row with a mandatory reason string. Vetoes are aggregated monthly and fed back to prompt/model tuning.

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## Data residency

The managed AI Gateway proxy is contractually pinned to EU + KZ endpoints only. No PII leaves the region. Any content beyond raw tender bodies is redacted at the prompt-builder stage — the redactor is unit-tested against a KZ-PII corpus every commit.

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# 71 · Business continuity & regional failover

## Regional footprint

Primary region: eu-central-1 (Frankfurt) — colocation with the AI gateway EU endpoint. Warm-standby region: me-south-1 (Bahrain) — Gulf partner requirement. Read-only DR region: eu-west-2 (London) — legal archive only.

## RTO / RPO

Recovery Time Objective 30 minutes for a warm standby cutover; 4 hours for a cold rebuild. Recovery Point Objective is 60 seconds — enforced by MongoDB oplog streaming to standby and S3 cross-region replication for uploads.

## Failover triggers

1) Elastic Beanstalk region health < 60 % for 5 minutes; 2) MongoDB primary election failure for > 2 minutes; 3) LLM proxy latency > 30 s for 3 minutes; 4) Manual declaration by CTO. Any single trigger fires the failover playbook (see chapter 33).

## Data-store replication

MongoDB Atlas cross-region cluster (primary + 2 replicas in EU + 1 replica in ME). S3 Cross-Region Replication for `pmc-uploads` and `pmc-downloads`. Route 53 health-checks + DNS failover for `api.pmcmanagers.com`.

## Frontend continuity

CloudFront distribution has origin failover: primary origin S3-EU, secondary S3-ME. Frontend is a static bundle so it survives origin loss entirely. The 5-portal login surface is served from CloudFront's edge, so even a full API outage leaves the marketing shell intact with a clear status banner.

## Runbook rehearsal

Chaos-engineering drills run on the last Friday of each quarter. Scenarios: primary DB kill, LLM proxy timeout, S3 origin outage, DNS mis-configuration, full-region evacuation. Each drill produces a post-mortem stored under `docs/dr-drills/YYYY-QN.md`.

## Communication plan

Sev-1 outages trigger: status page auto-update, Resend digest to admin users, Telegram to on-call ops, WhatsApp to executive team. The public-notice board renders a top banner sourced from the /status endpoint so vendors know a bid submission window may be affected.

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## Business impact tiers

Tier A (mission-critical, 30 min RTO): auth, notices, bids, escrow, evaluations. Tier B (2 h RTO): DocuSign, DPR, auctions, analytics. Tier C (24 h RTO): manual v2, reports snapshots, roadmap, profile picture uploads.

DENTIAL — INTERNAL USE

# 72 · Cybersecurity posture & third-party risk

## Certification roadmap

ISO 27001 target FY2027 Q3. SOC 2 Type II target FY2027 Q4. Both scoped to the platform + managed AI gateway + Stripe + Resend + DocuSign integrations. GDPR + Kazakhstan Law 94-V compliance is already in force; the certification is the third-party attestation.

## Threat model

STRIDE assessment refreshed each release. Top current risks: (1) vendor credential stuffing at /vendor/login, (2) prompt injection via uploaded DPR files, (3) escrow signer email substitution, (4) supply-chain compromise of npm/pypi packages, (5) social engineering of admin support staff.

## Controls

Rate limiting + IP throttling on all /login surfaces. LLM prompt-injection guardrails: strict allow-list HTML tags, sandboxed retrieval, no tool-use inside DPR pipeline. Escrow signer changes require re-authentication + audit + email confirmation. Dependency-pinning + Snyk + weekly dependency-cron scans.

## Third-party risk register

Every external service the platform depends on has a row in `third_party.md` : scope, data flow, SOC-2 status, DPA on file, contract expiry, escalation contact. Current entries: Managed AI Gateway (LLM), Stripe, Resend, DocuSign, MongoDB Atlas, AWS, Cloudflare, Slack, Telegram, Snapchat/Instagram API.

## Penetration testing

Annual external pen-test by an approved vendor. Quarterly internal red-team exercises. Findings are logged in JIRA with a Sev tag; anything Sev-1 must be remediated within 15 days per policy.

## Encryption

TLS 1.3 in transit; AES-256 at rest on MongoDB + S3. Application-level envelope encryption for tender bodies (KMS CMKs rotated annually). DocuSign PDFs are stored encrypted with the vendor's public key from KYC — nobody but the signer can retrieve the plaintext.

## Incident classification

Sev-1 = data exfiltration or systemic auth bypass; Sev-2 = single-tenant data-integrity failure; Sev-3 = performance regression or observability gap. Each has a fixed on-call rotation + response SLA + post-mortem template. See chapter 33 for the full playbook.

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## Regulator liaison

The Kazakhstan Cybersecurity Committee is notified of any Sev-1 within 24 h per statutory requirement. GOC + Bank partners receive a courtesy update via Partner lead. All external comms route through the CEO for approval — no exception, no direct-to-press.

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# 73 · Vendor lifecycle KPIs & dashboards

## Acquisition

New draft registrations / week, resume-rate (drafts that reach Submit), average time-to-submit, campaign attribution (Snapchat/Instagram/partner referral). Targets: 40 new drafts / week, > 55 % resume rate, < 3 days time-to-submit.

## Activation

Days from Submit to first eligible bid, first bid-value distribution, first-bid win-rate. Targets: median < 7 days, first bid median USD 40 k, first-bid win-rate baseline 8 % — flag if a category drifts below 4 %.

## Engagement

Bids per active vendor per quarter, category diversification (unique categories bid on), auction participation rate, Q&A engagement rate. Targets: 3 bids / quarter, 2 categories, 30 % auction participation once invited, 50 % Q&A engagement.

## Retention

Quarterly active vendor count, 12-month churn (drop-off from active), win-rate trend, ESG completion %. Targets: 12-month churn < 15 %, ESG completion > 80 % of active vendors.

## Revenue

Total contract value awarded / vendor / year, average margin (if disclosed), payment-cycle time, refund/forfeit rate on EMDs. Targets: median USD 200 k / active vendor / year; payment cycle median 21 days; forfeit rate < 4 %.

## NPS + advocacy

Vendor NPS (quarterly survey), referral rate, social share rate on award announcements. Targets: NPS > +30, referral rate > 12 %, social-share rate > 25 % of awards.

## Panel composition

Dashboards ship out of the box under Admin → Analytics → Vendor lifecycle. Each metric supports drill-down to the underlying rows for audit purposes.

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# 74 · Government & regulator engagement playbook

## Stakeholder map

Kazakhstan: National Bank, Ministry of Finance, State Procurement Authority, Anti-Corruption Agency. Multilateral: Asian Development Bank, EBRD, World Bank. Trade: Astana International Financial Centre, AIFC Court. Each has an owner on the Partner side + a backup.

## Cadence

Monthly written update to the State Procurement Authority (auto-generated stakeholder report). Quarterly in-person meeting with the National Bank on escrow flows. Half-yearly platform demo for the Anti-Corruption Agency. Annual multilateral summit.

## Data-sharing agreements

Each regulator has a DSA on file with named data categories + retention + auditable purpose. Anonymised aggregate stats can be shared freely; per-vendor detail only via formal legal request routed through counsel.

## Public consultation

Every new feature that changes procurement outcome (rubric changes, auction rules, vendor scorecard weights) is put out for 21-day public consultation via the platform's `/consultations` surface. Comments are archived in the audit trail.

## Complaint handling

The Anti-Corruption Agency has a dedicated portal to raise queries against any tender. SLA: acknowledge in 4 h, first substantive response within 3 business days, resolution or escalation within 15 business days.

## Cross-border data

Any dataset that leaves KZ needs a written data-sharing note co-signed by CTO + counsel. The default is 'stay in region' — cross-border is the exception, not the rule.

## Media protocol

Regulator interactions are never subject to press comment without CEO + Partner-lead sign-off. Public statements attributed to the platform are drafted by the Communications lead and reviewed by counsel.

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# 75 · Roadmap horizons 2027 – 2029

## Horizon 1 (0–12 months) — mature the core

Ship the remaining P1 bugs (i18n backfill, TenderBuilder tab refetch, escrow signer edit). Ship Firebase/APNs push. Ship Zoom for evaluation committees. Ship LSEG/World-Check integration. Ship PayPal + Kaspi payments alongside Stripe. Ship v2 of the vendor scorecard with the anti-gaming layer described in chapter 68.

## Horizon 2 (12–24 months) — platform expansion

Multi-country tenanting (per-country subdomains + per-country masters bootstrap). White-label mode for GOC + Bank partners. Native mobile apps for vendor portal (Capacitor already scaffolded — chapter 71's DR region can host the mobile edge). AI copilot for committee deliberation minutes. Deeper anti-fraud (device fingerprinting, behavioural analytics).

## Horizon 3 (24–36 months) — new lines of business

Marketplace for pre-qualified vendors across partner platforms (federated identity via OIDC). Tokenised EMD via a regulated custodian in AIFC. Cross-border bid pooling with peer procurement platforms. Regulator-grade real-time dashboards licensed to Ministries.

## Investment principles

Every new feature must serve at least one of: (1) demonstrable audit-trail integrity, (2) measurable time-to-award reduction, (3) measurable vendor NPS improvement, (4) new revenue line. Features that fail all four are pushed to the parking lot.

## Kill criteria

Any Horizon-1 item unshipped after 15 months is either (a) re-scoped and re-prioritised, or (b) killed. No zombie features. The roadmap review happens on the last Thursday of each quarter with all portal owners present.

## Talent plan

Horizon 1 requires the current 2-engineer team. Horizon 2 requires 4 engineers + 1 designer + 1 product ops. Horizon 3 requires 8+ engineers, a dedicated security team, and a partner-success team.

## Governance

The roadmap is public inside the platform (Admin → Reports → Roadmap) and to shareholders via a signed PDF export. External publication requires CEO + counsel sign-off. Changes to the roadmap horizons require Board sign-off if the horizon shifts by more than one quarter.

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# 76 · Ethics review board charter

## Composition

Seven members: two independent public-sector representatives, two industry members with 15+ years procurement experience, one AIFC-court-nominated legal counsel, one civil-society observer, and the platform's Chief Compliance Officer (secretariat, non-voting). Terms are three years, staggered.

## Remit

Review any procurement decision where (a) an AI recommendation was overturned, (b) a vendor scorecard band changed by more than one tier in a quarter, (c) a debarment is proposed, (d) a corrigendum window extends the deadline by more than 30 %, or (e) an external whistleblower complaint is received.

## Cadence

Monthly plenary. Emergency session within 72 h for any Sev-1 event. All meetings are audio-recorded and minuted; minutes are published within 10 working days after redaction of commercially sensitive material.

## Powers

Recommend (not compel) remedial action to Management and the Board. Access every audit-trail record read-only. Commission independent forensic reviews at the platform's cost up to USD 250 k per review without additional Board approval.

## Conflict declarations

Members declare all commercial interests before each meeting. Any member with a material interest in the item under review recuses. Recusals are minuted with the reason.

## Escalation to Board

Board must respond in writing to any Ethics-Board recommendation within 30 days. Rejections must be accompanied by a written rationale which becomes part of the audit trail.

## Whistleblower protection

The board oversees the whistleblower channel (email + Telegram + a redacted web form). Retaliation against a good-faith reporter is grounds for immediate termination and — where a public official is involved — a mandatory referral to the Anti-Corruption Agency.

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## Public reporting

An annual public report summarises the number of reviews, the outcomes, and material lessons learned. Individual cases are anonymised.

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# 77 · Total-cost-of-ownership model (TCO)

## Purpose

TCO turns L1 recommendations from a lowest-price comparison into a whole-life value comparison. It is mandatory for any contract with an initial value above USD 500 k or a duration longer than 24 months.

## Inputs

Bid price (year 1), annual operating cost estimate, energy consumption profile, expected downtime hours, spare-parts pricing, training cost, decommissioning cost, residual value at end-of-life, and a country-specific discount rate.

## Formula

TCO = Σ(operating cost + energy + downtime × cost-of-downtime + maintenance + training) over the contract horizon, discounted by the risk-free rate, plus decommissioning, minus residual. The formula lives in `backend/pmc_lib/tco.py` with unit tests covering the 12 edge cases from the ADB TCO playbook.

## Committee integration

Once TCO is enabled for a tender the AwardReport surfaces two columns: 'Bid Price' and 'TCO'. Ranking is by TCO. Any bid whose TCO exceeds the lowest by more than 25 % but has a lower bid price is flagged for narrative comment from the committee.

## Vendor visibility

Vendors see the TCO input schema at bid submission time. They can attach their own TCO worksheet as evidence. This is optional but improves scoring under the methodology criterion.

## Sensitivity analysis

The TCO surface includes a slider: energy cost ±30 %, downtime cost ±30 %, discount rate ±200 bps. The committee can eyeball how sensitive the ranking is to each input and demand justification when a swing tips the recommendation.

## Audit trail

Every TCO input, every recalculation, and every slider position is written to the audit trail with the actor. Regulator or auditor can replay any TCO-based award decision end-to-end.

## Training

Buyers new to TCO complete a 2-hour module (Admin → Help → Training → TCO fundamentals). The module includes 8 worked examples across construction, IT services, medical equipment and consulting.

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# 78 · Green procurement policy

## Commitment

PMC and its sponsor commit that by end of FY2028 at least 60 % of contract value awarded through the platform will meet the green-procurement criteria published in the annexure. This is a Board-level target.

## Definition

A bid is 'green' if it: (a) documents a Scope 1+2 emissions figure ≤ the category benchmark, (b) commits to a Scope 3 reduction plan of ≥ 2 %/year, (c) sources at least 30 % of materials/services locally, and (d) meets ISO 14001 or an equivalent EMS.

## Scoring

The rubric editor supports a 'Green premium' criterion (0 – 15 % weight) that applies when the tender is flagged green-eligible. Non-green bids are not disqualified but score lower on this criterion.

## Reporting

The public roadmap includes a live 'green %' KPI. The stakeholder monthly report enumerates green vs non-green award value and the delta vs target.

## Verification

Vendor self-declarations are spot-audited at 10 % of awards. Independent verifier (SGS / Bureau Veritas) confirms; false declaration is grounds for debarment and repayment of green premium.

## Buyer training

Any buyer publishing a green-eligible tender completes a 90-minute module (Admin → Help → Training → Green procurement). The module includes the taxonomy, common gotchas and legal do-not-say phrases that trip Anti-Corruption regulators.

## Regulator alignment

The green definition is aligned with the AIFC Green Finance Taxonomy and the World Bank's 2025 Climate Change Action Plan classifications. Realignment reviews happen annually.

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# 79 · Vendor communication protocol

## Channels

Primary: email digest via Resend (announcement, corrigenda, award notice). Secondary: in-app inbox with read-receipt (persists across sessions). Tertiary: WhatsApp + Telegram for opt-in critical alerts. Never SMS as primary — cost + spoofing risk.

## Tone

Formal, concise, English + Russian + Kazakh in one email. No marketing language on transactional messages. No emoji on regulatory content. Signed by a named human (usually the Vendor Success lead) with a real reply address.

## Frequency caps

No more than 3 non-critical emails / vendor / week. Critical (corrigenda, deadline change, award) always sends. Vendors control category preferences from their profile page — the platform respects the last-updated preference within 10 minutes.

## Content review

Every template is versioned in `backend/pmc_lib/comms_templates/`. New templates require Comms + Legal review. Field substitutions are strictly typed to prevent HTML injection.

## Analytics

Open rate, click-through, unsubscribe, reply-to-human ratio are tracked per template. Templates below 15 % open-rate for four consecutive sends are auto-flagged for a rewrite.

## Escalation replies

A vendor reply-to-all email lands in a shared mailbox (vendor-success@pmcmanagers.com) with automated triage: contract dispute → Legal, technical → Support, commercial → Success lead. First-response SLA: 4 business hours.

## Legal footer

Every mail carries an unsubscribe link (transactional exempt), a physical address (PMC head office, Astana), and links to the privacy and terms pages. Absence of any element blocks the send at the template linter.

## Regulator carbon-copy

Award notices, debarment notices and material corrigenda are automatically bcc-ed to the State Procurement Authority archive mailbox. The recipient is a Kazakhstan Government mandate.

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# 80 · Data-quality standards & master-data governance

## Golden records

Every entity that spans more than one tender (vendor, buyer, category taxonomy, currency, country, industry) has one canonical golden record. The platform enforces referential integrity via `golden_ref` foreign keys — orphaned entities fail the pre-publish check.

## Ownership

Each domain has a nominated data steward: Vendors → Vendor Success lead, Buyers → Buyer Success lead, Taxonomy → Category committee, Countries/currency → Compliance officer. Any change to a golden record requires steward sign-off and produces an audit row.

## Change-control

Master-data changes propagate to all in-flight tenders in a controlled way: (a) draft tenders auto-update, (b) published tenders keep their frozen snapshot, (c) evaluations use the snapshot valid at lock-envelope time. This prevents silent behaviour change on live procurement.

## Data-quality metrics

Completeness (% required fields populated), accuracy (spot-audit vs source-of-truth), timeliness (median age of last update), uniqueness (duplicate detection score), and consistency (cross-reference validity). Each metric has a target and a live gauge on the Admin data-quality dashboard.

## Duplicate detection

Fuzzy-match on registration_number + country + email domain flags candidate duplicates before a vendor is activated. The threshold is 92 % similarity; anything below is human-reviewed by Vendor Success. Duplicate merges preserve both audit trails.

## Reference data cadence

Country and currency ISO codes refreshed monthly. NACE-style industry taxonomy reviewed quarterly. EMD slabs reviewed twice yearly against Kazakhstan's regulatory bulletin. Every refresh writes a version tag so historical filings remain reproducible.

## Data lineage

Every field on every published notice carries a lineage annotation: which master, which version, which user made the last edit. This is what regulators typically ask for first when a decision is queried post-award.

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## Deletion vs archival

Nothing is hard-deleted. 'Delete' actions move records to an archival collection with a legal-hold flag until the retention clock runs out. Bulk purges require CFO + counsel sign-off and are logged to the immutable audit trail.

## Reconciliation

Nightly reconciliation between the operational store (MongoDB) and the analytics warehouse (S3 + Athena). Deltas above 0.5 % fire a Sev-3 alert. The reconciliation job itself writes an audit row so absence-of-alert is not evidence of absence-of-drift.

## Regulator export

One-click export of any entity's full history + lineage in a signed PDF + machine-readable JSON. Available to Admins with the 'auditor' role only. Usage is itself audit-logged.

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# 81 · Progressive Web App (PWA) — install & offline

## Positioning

PMC v46p ships as an installable PWA on both Android (Chrome / Edge / Samsung Internet) and iOS (Safari 16.4+). The manifest advertises a 5-portal shortcut list so the launcher offers direct entry to Admin, Staff, Vendor and the public Notice Board without opening the browser chrome.

## Install — Android

Open `https://tenders.pmcmanagers.com` in Chrome/Edge, tap the address-bar menu and choose 'Install app'. The homescreen icon uses the PMC crest. A splash screen (navy #0B132B with the gold crest) shows for the first 300 ms — configured via `background_color` + `theme_color` in `public/manifest.json`.

## Install — iOS

Open the same URL in Safari, tap the Share icon and choose 'Add to Home Screen'. iOS reads `apple-mobile-web-app-*` meta tags in `public/index.html` so the standalone shell hides Safari chrome, uses the black-translucent status bar and the correct app title.

## Offline behaviour

The service worker ( `public/service-worker.js` ) caches the app shell (HTML, CSS, JS bundles, fonts, PMC crest) using a stale-while-revalidate strategy. Data endpoints (/api/*) are never cached — every request goes to the network so vendors never see stale bid data. If the network is unavailable the UI renders an 'Offline — reconnecting' banner sourced from the /status endpoint.

## Update discipline

Every deploy bumps a `CACHE_VERSION` constant in the SW so the browser invalidates old shells within one page load. The previous self-destructing SW from v46d is retained as a safety net: on activate, any cache older than 30 days is nuked.

## Push notifications (roadmap)

Web Push via Firebase Cloud Messaging (Android + desktop) and APNs bridge (iOS 16.4+ PWA) — targeted for v46q. Trigger events: award notification, corrigenda publish, bid deadline < 24 h. Vendors opt-in per category from the profile page.

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## Testing checklist

Lighthouse PWA audit ≥ 90; manifest icons render at both 192 px and 512 px maskable; SW installs on first load; 'Add to Home Screen' appears on Chrome ≥ 76 + Safari ≥ 16.4; the standalone shell has no browser chrome; navigation between the 5 portals stays inside the app.

## Fallback for legacy browsers

Browsers that don't support the SW API (IE, older Android 4.x) still get the full app — just without offline caching or install. There is no functional degradation, only progressive enhancement.

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# 82 · Sign-off

This manual is versioned under `/app/deliverables/v46_manual/build/`. Regenerate via `python3 /app/tools/manual/build.py`. Re-capture via `python3 /app/tools/manual/capture.py` after every material release. Manual v3 adds chapters 68 – 81 (vendor scorecard, cross-border settlement, AI governance, BCP, cybersecurity, vendor KPIs, regulator engagement, roadmap horizons, ethics board, TCO, green procurement, vendor comms, data-quality, PWA) and is fully scrubbed of third-party platform references.

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